# Introduction to Category Theory for Non-Mathematicians

## LIFE Project

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# Foundations of Mathematics

## A Mathematic or Several Mathematics?

« To present a view of the entire field of mathematical science as it exists, — this is an enterprise which presents, at first sight, almost insurmountable difficulties, on account of the extent and the varied character of the subject. […] Nevertheless it is legitimate to ask whether this exuberant proliferation makes for the development of a strongly constructed organism, acquiring ever greater cohesion and unity with its new growths, or whether it is the external manifestation of a tendency towards a progressive splintering, inherent in the very nature of mathematics, whether the domain of mathematics is not becoming a tower of Babel, in which autonomous disciplines are being more and more widely separated from one another, not only in their aims, but also in their methods and even in their language. In other words, do we have today a mathematic or do we have several mathematics? »

N. Bourbaki, *The Architecture of Mathematics*, American Mathematical Monthly, 57, 1950.

« In a metamathematical sense our theory provides general concepts applicable to all branches of mathematics, and so contributes to the current trend towards uniform treatment of different mathematical disciplines. »

S. Eilenberg and S. Mac Lane, A General Theory of Natural Equivalences, *Trans. Amer. Math. Soc.*, 58(2), 1945, p. 236.

Contemporary attempts

to unify the several « mathematics »

into *a* mathematic:

- **Set theory**

(late 19th and early 20th, centuries; Cantor, Zermelo-Fraenkel, von Neumann, Bourbaki).

- **Category theory**

(1945-; Eilenberg, Mac Lane, Kan, Grothendieck, Lawvere, etc. ).

# What is a Mathematical Object?

- **Substantialist, intrinsic or bottom-up approach (set theory, 1870s-1920s)**:

A mathematical object is intrinsically defined by its ***content,***

namely as a ***set equipped with a structure*** (*fulfilling certain laws*).

- **Structuralist, extrinsic or top-down approach (category theory, 1945-)**:

A mathematical object is extrinsically defined

by its relational ***context***,

namely as a ***node in a network of relations or transformations***.

# Set-Theoretic Foundations

## Set-Theoretic Foundations

(Cantor, Zermelo-Fraenkel, von Neumann, Bourbaki, 1870s-1920s)

Sets are the raw or naked material

out of which mathematical objects can be constructed

by endowing sets with a structure.

Mathematical Object
=
**Set** (raw data)
+
**Structure**
+
**Laws** (conditions/constraints on the structure)

**Every mathematical objects is a kind of *structured dust*.**

The ultimate ontological substratum of mathematical objects

is given by pure multiplicities.

« [...] the decisive break—in which mathematics blindly pronounces on its own essence—is Cantor's creation. It is there alone that it is finally declared that, despite the prodigious variety of mathematical 'objects' and 'structures', they can all be designated as pure multiplicities built, in a regulated manner, on the basis of the void-set alone. »

A. Badiou, *Being and Event*, p. 6.

## Algebraic Example of a Structured Set: Groups (E. Galois, XIX century)

A group $(X, \bullet, e)$ is an algebraic structure defined by

- a set $X$ (***data***)
- enriched with a *closed binary operation* and a *distinguished element* $e$ (***structure***);
- such that (***laws***)

.the distinguished element $e$ is the neutral element for the operation $\bullet$:

$$g \bullet e = e \bullet g = g$$

.for each $g \in X$, there is an element $g^{-1} \in X$ (called the inverse of $g$) such that

$$g^{-1} \bullet g = g \bullet g^{-1} = e.$$

. the operation $\bullet$ is associative:

$$(g \bullet h) \bullet k = g \bullet (h \bullet k).$$

**Example: the entire numbers** $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$

$(\mathbb{Z}, +, 0)$ is a group.

- Neutrality of 0: $n + 0 = 0 + n = n$.
- Existence of inverses: for each $n \in \mathbb{Z}$, there is an inverse denoted $-n$ such that $n + (-n) = 0$.
- Associativity: $(n + m) + k = n + (m + k)$.

**Counterexample: the natural numbers** $\mathbb{N} = \{0, 1, 2, \ldots\}$

- Neutrality of 0 holds.
- Associativity of + holds.
- But there are no inverses !

$(\mathbb{N}, +, 0)$ is not a group, but a ***monoid***.

Other *algebraic structures* that can be defined on top of a set: *rings*, *fields*, etc…

The kind of structure we impose on a set determines whether the resulting mathematical object is algebraic, geometric, or of another kind.

The qualitative diversity of mathematical objects arises from the imposition of different structures upon a common ontological background given by pure multiplicities (sets).

## Geometric Example of a Structured Set: Topology (F. Hausdorff, 1914)

• A set $X$ is not a space (e.g. there is no notion of distance between any two elements $x, y \in X$).

• A *topology* is a first kind of geometric structure that we can define on a set to endow it with a (*sub-metric* or *distance-independent*) geometric nature.

• The fundamental topological notion is the notion of ***continuity***,…

> … that is, changes that do not engage *abrupt breaks, jumps,* or *sharp discontinuities*.

• To do so, one understands each element $x \in X$ as a geometric point and defines which are the ***unbreakable neighborhoods*** of $x$.

• A ***topology*** on a set $X$ is a structure that assigns to each « point » $x \in X$…

.a collection $\mathcal{N}(x)$ of subsets U of $X$

.that will play the role of the unbreakable *neighborhoods of x* (and which satisfy certain axioms).

• A set $X$ endow with a topology $\mathcal{N}$ defines a ***topological space*** $(X, \mathcal{N})$.

• Morphisms between topological spaces are given by ***continuous functions***, which are:

.maps between the underlying sets

.that respects the additional structure, i.e. that does not break what was defined as unbreakable.

More precisely, a function $f : X \to Y$ between topological spaces is said to be *continuous* if for every open subset $U \subset Y$, the preimage $f^{-1}(U) = \{x \in X / f(x) \in U\}$ is an open set of $X$.

## Examples of Topological Spaces

- **Discrete topology.**

A *discrete space* (a space with no continuity at all) is a set $X$ endowed with the so-called *discrete topology*, in which there is a neighborhood $\{x\}$ for each $x \in X$

All the points are isolated from each other.

Here there is no community !

- **Indiscrete topology**.

In the indiscrete topology, there is only one neighborhood for each $x$, namely the whole set $X$.

The most intimate neighborhood of a point $x$ is the whole set $X$ itself.

Here there is no privacy !

# Category-Theoretic Foundations

**Category theory as a *mathematized meta-mathematics.***

« I like to think of category theory

as the mathematics of mathematics […]

Category theory is a theory of mathematics,

but is itself a piece of mathematics. »,

E. Cheng, *The Joy of Abstraction*, pp. 13 & 21.

« Category theory is the study of patterns of mathematical thought. »

E. Riehl, *Infinity Category Theory Offers a Bird's-Eye View of Mathematics prescriptive.*

« [The concept of category] frames a possible template for any mathematical theory […] »

B. Mazur, *When is one thing equal to some other thing?*

# The Threefold Doctrine of Science

## 1. Sciences *qua* theories of « life »

The different sciences

Physics,
Chemistry,
Biology,
Geology
Economy,
Sociology,
etc.

study the different dimensions or « ontic regions » (Heidegger)

of « life » or « nature ».

# The Threefold Doctrine of Science

## II. Mathematics *qua* Theory of Science

Mathematics is not so much a science among others

as a discipline that studies

the formal structures of rational thinking as such.

« […] in any special doctrine of nature there can be only as much *proper* science as there is *mathematics* therein. »

E. Kant, *Metaphysical Foundations of Natural Science.*

« *Τα μαθηματα* means, in Greek, that which, in his observation of beings and interaction with things, man knows in advance. »

M. Heidegger, *The Age of the World Picture*, in *Off The Beaten Track*.

## Mathematics *qua* exact science

Mathematics is a theatre in which the spirit's acts or operations

— rather than being applied to any « ontic » domain » (physics, biology) —

are considered as such, thereby becoming the main characters.

Mathematics is *ex-act* in the precise sense that it exteriorizes or stages

the spirit's rational acts as such

— e.g. counting, computing, deducing, abstracting, etc. —

(Cf.: P. Valery reflections on mathematics)

## Mathematics *qua* Ontology (Badiou)

« Mathematics *is* ontology, the science of being qua being. »

A. Badiou, *Being and Event*, p.4.

« […] in terms of the 'application' of mathematics to the so-called natural sciences […] the clarification is immediately evident if mathematics is the science, in any case, of everything that is, *insofar as it is*. »

A. Badiou, *Being and Event*, p.7.

« A famous query of Eugene Wigner asks for an explanation of the unreasonable effectiveness of mathematics—especially its effectiveness in the various sciences. The emphasis on structure does not really provide a response, except to say that different parts of science frequently exhibit some common structure. One might say that mathematics, too, is a science; namely it is that science whose objects are the forms or structures which are common to different specific scientific realizations. I am inclined to restate this by asserting that mathematics is protean science; its subject matter consists of those structures which appear (unchanged) in different scientific contexts. »

S. Mac Lane, *Structure in Mathematics*, 1996.

## Ontological categories

## correlated to different branches of mathematics

- **Multiplicity** (*being-many*): set theory.
- **Identity** (*being-itself*): group theory.
- **Relation** (*being-with*): set-theoretic relations, category theory.
- **Extension** (*being-there*): geometry.
- **Quality** (*being-thus*): category theory.
- **Quantity** (*being-thus-much*): number theory, algebra.
- **Difference** (*being-different*): category theory, homotopy type theory.
- **Continuity** (*being-unbreakable*): differential and integral calculus.

## Mathematics *qua* speculative construction of rational tools (Frege)

Mathematics is an activity by means of which

we can refine and enhance

the rational faculties and categories

of human understanding in order to perform specialized tasks.

According to Frege's analogies,

mathematical logic is to human language

what a microscope is to the eye

or what a robotic hand is to the human hand.

« I believe that I can best make the relation between my ideography to ordinary language clear if I compare it to that which the microscope has to the eye. Because the range of its possible uses and the versatility with which it can adapt to the most diverse circumstances, the eye is far superior to the microscope. Considered as an optical instrument, to be sure, it exhibits many imperfections, which ordinarily remain unnoticed only on account of its intimate connection with our mental life. But, as soon as scientific goals demand great sharpness of resolution, the eye proves to be insufficient. The microscope, on the other hand, is perfectly suited to precisely such goals, but that is just why it is useless for all others. »

G. Frege, *Begriffsschrift*, Preface.

« Language [...] proves to be deficient in the matter of protecting our thinking from error; it does not even fulfill the first demand made on it in this respect, that of being unequivocal. [...] Language is not in such a way dominated by logical laws that compliance with grammar would of itself guarantee the correctness of thought processes. [...] The shortcomings here stressed are caused by a certain softness and instability of language which, on the other hand, constitute the reason for its many-sided usefulness and potentiality for development. In this respect language can be compared to the hand which, despite its adaptability to the most diverse tasks, is still inadequate. We build ourselves artificial hands-tools for special purposes which function more exactly than the hand is capable of doing. And how is this exactness possible? Through the very rigidity and inflexibility of the parts, the lack of which makes the hand so dexterous. The word language is in the same way also not adequate. We have need of a system of symbols from which every ambiguity has been banned and whose rigorous logical form cannot be escaped by the content. »

G. Frege, *On the Scientific Justification of a Concept-Script*.

To do science means

to try to understand a given « ontic domain »

- by thinking logically (logic);

- by methodically « forgetting » features that are contextually irrelevant (abstraction);

- by quantifying by means of numbers (order structures, algebraic structures, etc.);

- by combining and computing (algebra);

- by predicting (differential and integral calculus);

- by using spatial schematizations (geometry):

- by identifying patterns, symmetries and invariants (group theory).

We can understand the different branches of mathematics

as conceptual and formal technologies

by means of which we can enhance

the faculties of rational cognition as such,

— abstracting, classifying, computing, counting, deducing, identifying patterns

and invariants, operating, ordering, predicting, measuring, schematizing, etc. —

« through [an aprioristic] construction of concepts » (Kant),

that is, independently of the different « ontic domains »

to which they can be applied.

## The Threefold Doctrine of Science

## III. *Theory of the theory of science* or *theory of mathematics*

Category theory can be understood as the *theory of mathematics*.

Since this theory of mathematics is a case of rational thinking,

it has to be itself mathematized.

Category theory can be described as a

***mathematical meta-mathematics***.

## The Categorical Imperative

« We shall thus have to investigate the possibility of a *categorical* imperative entirely *a priori*, since here we do not enjoy the advantage of having its reality given in experience […]

There is therefore only a single categorical imperative and it is this: '*Act only on that maxim through which you can at the same time will that it should become a universal law*.' »

I. Kant, *Groundwork of the Metaphysics of Morals*, pp.87-8

(quoted in M.A. Arbib & E.G. Manes, *Arrows Structures and Functors. The Categorical Imperative*)

The kind of « universality » at stake here

is not only with respect to different « ontic domains » (e.g. physics, biology, etc.)

— which is the universality of mathematics *qua* theory of science—,

but also with respect to different « ontological (i.e. mathematical) domains ».

If category-theoretical metamathematics is itself mathematical,

it should apply to itself, i.e., be its own theory.

« The purpose of category theory is to try to describe certain general aspects of the structure of mathematics. Since category theory is also part of mathematics, this categorical type of description should apply to it as well as to other parts of mathematics. »

J.W. Gray, *Formal category theory: adjointness for 2-categories*, Springer-Verlag, Berlin, 1974, Lecture Notes in Mathematics, Vol. 391.

Does category theory exhibit the kind of reflexivity?

Can category theory

(as mathematical *meta-mathematics*)

provide a theory of itself

(as *mathematical* meta-mathematics)?

Category theory cannot provide such a reflexivity

— that of being its own theory —,

but it can be enhanced

to a fully self-reflective theory called

***∞-category theory***,

the « fixed-point of abstraction » (E. Cheng).

## ∞-Category Theory as a Fixed-Point of Abstraction

« I think of [infinite-dimensional category theory] as a fixed-point of abstraction. If science is the theory of life, and mathematics is the theory of science, and category theory is the theory of mathematics, I think that 2-category theory is the theory of category theory, and 3-category theory is the theory of 2-category theory, and so on. Finally, however, higher-dimensional category theory is the theory of higher-dimensional category theory. We have reached a pinnacle of abstraction. Or perhaps the heart of abstraction, or its deepest roots. Perhaps the pinnacle and the heart and the deepest roots are the same, and that, to me, is the joy of abstraction. »

E. Cheng, *The Joy of Abstraction*, p. 395.

According to category theory

the subject matter of pure mathematics

is given by

***invariant forms*** or ***structures***,

defined by means of (what I will call)

a ***constructive abstraction principle***.

Philosophically, category theory enacts

what we could call ***constructive structuralism***.

These invariant forms or structures

inhabit ***relational ontological contexts***

(the so-called ***categories***)

related by ***functorial bridges***.

These contexts are ***relational***

in the sense that a category

is a network of

***relations***

between

***objects***.

These relational networks

are ***ontological contexts*** for mathematical entities

in the sense that

two fundamental ontological categories

— namely ***identity*** (the *being-identical-to-itself* of any object)

and

***determinacy*** (the *suchness* or *being-so* of any object) —

are context or category-dependent.

Whereas set theory stresses internal ***content***

category theory stresses external ***form*** or ***structure***

defined by relations or transformations.

Category theory can be understood

as the mathematical expression of relational thinking,

in the sense that every set-theoretic content-based construction

is recasted in terms of external relations.

This makes it possible to generalize the corresponding construction

to contexts that are not set-theoretical.

## The Categorical Imperative

I. Think relationally, « […] think with arrows […] express key concepts

in terms of mappings […] rather than in terms of set elements. »

M.A. Arbib & E.G. Manes, *Arrows Structures and Functors. The Categorical Imperative.*

Whereas in set-theoretic foundations,

there is a « fundamental common object »

underlying every mathematical object

— namely, sets —

this is not the case in categorical « foundations ».

# Category Theory

## Category-Theoretic Foundations

(Date of birth: 1945; Eilenberg & Mac Lane, "General theory of natural equivalences")

CT provides a *structuralist*, *relational*, *behavioral*, or s*ociological*

characterization of mathematical objects

that faithfully implements the *Structuralist Credo*:

**The *relations* determine the *relata.***

***Objects are not determined by their content (as in set theory),***

***but rather by their relational context.***

## Category-Theoretic Context Principle (I)

« […] the attitude gradually emerged that the crucial characteristic of mathematical structures is not their internal constitution as set-theoretical entities but rather the relationship among them as embodied in the network of morphisms […] »

J.L. Bell, *Category Theory and the Foundations of Mathematics*, 1981.

## Category-Theoretic Context Principle (II)

« From Dedekind, through Noether, and to the work of Eilenberg and Mac Lane, the fact has clearly emerged that mathematical structure is determined by a system of objects and their mappings, rather than by any specific features of mathematical objects viewed in isolation. »

S. Awodey, Structure in Mathematics and Logic: A Categorical Perspective, *Philosophia Mathematica*, 3, 209-237, 1996.

## Category-Theoretic Context Principle (III)

« The category-theoretic way of working and thinking […] is best characterized by an adherence to a category-theoretic 'context principle' according to which one never asks for the meaning of a mathematical concept in isolation from, but always in the context of a category. »

E. Landry and J.-P. Marquis, Categories in Context: Historical, Foundational, and Philosophical, *Philosophia Mathematica* (III)13, 1-43, 2005, p.7.

## Category-Theoretic Context Principle (IV)

« A category is a system of related objects. The objects do not live in isolation: here is some notion of map between objects, binding them together. »

T. Leinster, *Basic Category Theory*, p.9.

## Informal Characterization of a Category

Informally, a ***category*** $\mathscr{C}$ is a ***network of relations***, that is:

a collection of ***objects*** $a, b, c \ldots$

related by arrows or ***morphisms*** between them

$$f : a \to b$$

— including loops and many arrows between any two objects —

fulfilling certain conditions.

## Categories Defined as *Data-Structure-Laws* Entities

- **Data**: objects and relationships (called *morphisms*).
- **Structure**: identities and compositions.
- **Laws**: identity laws and associativity law for the composition.

## Formal Definition of Categories as *Data-Structure-Laws* Entities

### I. Data.

A ***category*** $\mathscr{C}$ consists of the following data:

- A collection of ***objects*** $\mathrm{Ob}(\mathscr{C})$.
- For every pair of objects $a, b \in \mathrm{Ob}(\mathscr{C})$, a set denoted

$$\mathrm{Hom}_{\mathscr{C}}(a, b)$$

whose elements are called ***morphisms*** from $a$ to $b$. Note that in general

$$\mathrm{Hom}_{\mathscr{C}}(a, b) \neq \mathrm{Hom}_{\mathscr{C}}(b, a)$$

Given a morphism $f : a \to b$,

the object $a$ is called ***domain*** *or* ***source*** of $f$

and the object $b$ ***codomain*** *or* ***target*** of $f$.

## Formal Definition of Categories as *Data-Structure-Laws* Entities

## II. Structure.

- For every triple of objects $a, b, c$, a ***composition function***

$$\circ \ : \ \mathrm{Hom}_{\mathscr{C}}(a,b) \times \mathrm{Hom}_{\mathscr{C}}(b,c) \longrightarrow \mathrm{Hom}_{\mathscr{C}}(a,c),$$

written $(f, g) \mapsto g \circ f$.

Note: Composition can be understood as a *non-commutative partially defined binary operation* on the set of morphisms of the category.

- For every object $a$, there is a putative ***identity morphism***

$$id_a \in \mathrm{Hom}_{\mathscr{C}}(a,a)$$

Note: for the moment we have not given any defining property of $id_a$ that distinguishes it from other morphisms in $\mathrm{Hom}_{\mathscr{C}}(a,a)$.

## Formal Definition of Categories as *Data-Structure-Laws* Entities

### III. Laws.

♥ **Associativity** law:

$$h \circ (g \circ f) = (h \circ g) \circ f$$

The associativity law eliminates the ambiguity that arises

when one composes a string of more than two arrows.

♣ **Identity laws**:

$$id_b \circ f = f \qquad \text{and} \qquad f \circ id_a = f$$

with $f : a \to b$.

Roughly speaking, identities (like 0 for + and 1 for ×) do not do anything,

i.e. they act as neutral elements for the binary partially defined operation ∘.

The identity laws are the conditions that define the identity morphisms.

## Category of Sets (I)

The objects of the category *Set* are ***sets*** $X, Y, \ldots$

and the morphisms are ***functions*** between sets

$$f : X \to Y,$$

i.e. correspondences that assign to each and every element $x$ of $X$

a unique element $f(x)$ of $Y$.

## Category of Sets (II)

For each set $X$, there is an *identity function*

$$id_X : X \to X$$

such that

$$id_X(x) = x \text{ for } x \in X$$

Functions

$$g : X \to Y \text{ and } f : Y \to Z$$

— where the source of $f$ is the target of $g$ —

can be composed into a function

$$f \circ g : X \to Z,$$

given by

$$(f \circ g)(x) = f(g(x)).$$

The fact that the definition of a category

makes use of set-theoretic notions such as

*set*, *function* and *Cartesian product*

seems to undermine the thesis

that category theory provides an alternative

to set theory as « foundations of mathematics ».

**Clue for an answer**

We shall see later that in fact a category

— considered up to an identity criterium called *equivalence of categories* —

does not have underlying sets of objets and morphisms.

## Comments on the Definition of a Category

## I. On the Nature of Objects

« The objects need not have ‘elements’, nor need the morphisms be ‘functions’, although this is the case in some motivating examples. »

S. Awodey, *Structure in Mathematics and Logic: A Categorical Perspective*.

## Comments on the Definition of a Category

## II. On the Universal Nature of the Identity Morphisms

Among the different kinds of relationships

that objects can entertain,

the identity morphisms $id_a : a \to a$ encode the fact

that there is a universal relation

that any object $a$ necessarily entertains with itself,

that of being the same as itself.

## Comments on the Definition of a Category

## III. On Identity Morphisms and Endomorphisms

• Between two objects there can be many morphisms.

• In particular, there might be many « endomorphisms » of the form $a \to a$ besides $id_a$.

« We will be careful to observe that there may be any number of arrows between $a$ and $a$ which *are not identity arrows*. What characterizes identity arrows, as always in categorial thinking, is not its position (it goes from $a$ to $a$), but its behavior within the compositions of actions or arrows (in this case its 'nullity'), and therefore the *relationship* that it maintains with anything other than itself. »

A. Badiou, *Mathematics of the Transcendental*, p.19.

## Comments on the Definition of a Category

### IV. Alternative Definition of a Category Uniquely in Terms of Morphisms.

Since the objects are in bijective correspondence with the identity morphisms,

it is possible to reformulate the definition of a category entirely in terms of morphisms.

More precisely, a category can be defined

- as a collection of morphisms $\mathscr{M}$;

- equipped with two unary operations:

$$source : \mathscr{M} \to \mathscr{M} \text{ and } target : \mathscr{M} \to \mathscr{M}$$

- and one binary operation encoding the composition:

$$\circ : \mathscr{M} \times \mathscr{M} \to \mathscr{M}$$

satisfying certain axioms encoding a) that the images of the unary maps act as neutral elements for the composition (i.e. that they are the identity morphisms), and c) the associativity of composition.

(cf. S. MacLane, *Categories for the Working Mathematician*, p.9; P.J. Freyd and A. Scedrov, *Categories, Allegories*, p.3)

Morphisms $a \to b$

will be sometimes described as

***a*-probes of** $b$,

***a*-generalized elements of** $b$,

or ***a*-shaped elements of** $b$

as if the morphism were an arrow with a particular shape defined by $a$

shot to « probe » the object $b$.

## Caveat on the image of « morphisms as arrows »

Categorical morphisms can be schematized as « arrows »

— in the sense of being an oriented relation between a source and a target —

but not in the same sense as ***vectors*** can be schematized as « arrows »

— in the sense of defining an oriented direction in a space —.

## Commutative Diagrams

A diagram is a piece of a category

containing or exhibiting some of its objects

and some of the arrows between them.

A diagram is said to be ***commutative***

when all possible directed paths in the diagram from an initial object to a final one

compose to the same morphism.

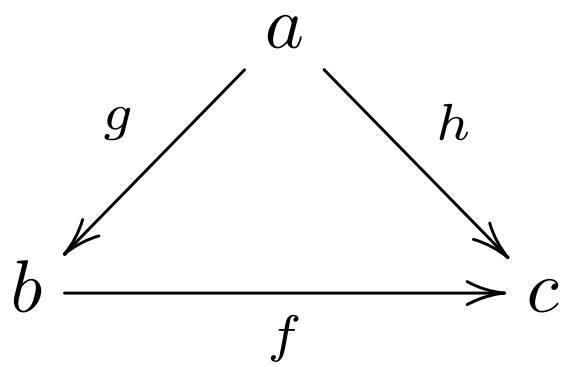
« A *diagram* in a category is a collection of objects and arrows from the category, possibly not all of them. A diagram is said to *commute* if any two composable strings with the same endpoints produce the same composite. »

E. Cheng, *The Joy of Abstraction,* p. 107.

## Equations Between Morphisms *qua* Commutative Diagrams

By encoding the fact that a given sequence of composable morphisms coincides with another sequence, commutative diagrams provide a ***diagrammatic transcriptions of equations between morphisms***.

For instance, the commutativity of the following diagram:




encodes the equation

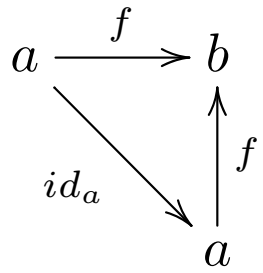
$$f \circ g = h.$$

## Identity Arrows (I)

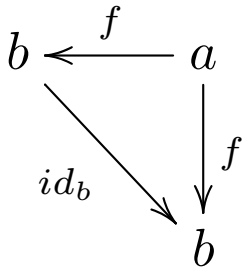
• In set theory, the identity function $id_A : A \to A$ is defined content-wise as a function such that $id_A(x) = x$, for every $x \in A$.

• Note that composition with the identity function have no effect. This is the property used to define identity morphisms categorically.

• The external-structural-diagramatic definition states that the following diagrams commute:




$f \circ id_a = f$

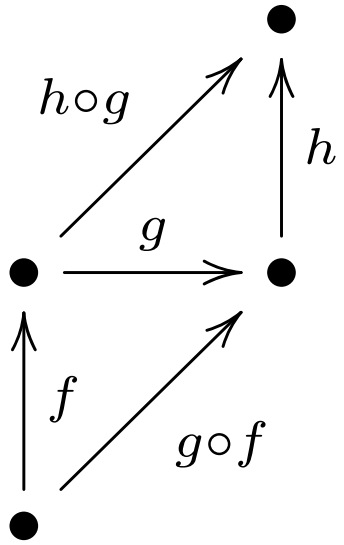



$id_b \circ f = f$

Identity morphisms act as neutral elements

for the « operation » given by composition.

# Associativity of the Composition as a Commutative Diagram




$$(h \circ g) \circ f = h \circ (g \circ f)$$

## Kinds of Categories

1) Categories defined from scratch by specifying a *directed graph* made of « points » and « arrows » (allowing for loops and for multiple arrows between the same endpoints) enriched with the corresponding compositions and identity arrows.

2) Categories associated to single mathematical objects. Examples: the category $\mathscr{C}_X$ associated to a particular set $X$, the category $\mathscr{C}_G$ associated to a particular group $G$, etc.

3) Categories of mathematical objects of the same kind. Examples: the category *Set* of sets, the category *Top* of topological spaces.

Whereas one the one hand a set $X$ defines a category $\mathscr{C}_X$,

on the other the set $X$ is itself an object of the category *Set*.

# Examples of Categories Defined as Directed Graphs

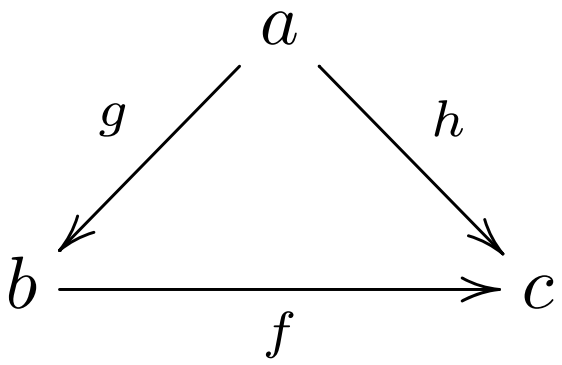
- Empty category (no objects & no morphisms):
- Category **1**:

$$a$$

- Category **2**:

$$a \to b$$

- Category **3**:



# Sets as Categories

Any ***set*** $X$ can be turned into a category $\mathscr{C}_X$

whose objects are the elements of $X$

and whose unique morphisms

are the identities $id_a$ for each $a \in X$.

In order to add « relations » to this « set of objects,

let's start by considering the set-theoretic notion of ***relation***.

A ***relation*** $R$ ***on a set*** $X$

is a set of ordered pairs

$\{(x_1, x_2)\}$

of elements of $X$,

i.e. a subset of the Cartesian product $R \subset X \times X$

## General Relations on a Set do not Define Categories

Given a set $X$ equipped with a ***relation*** $R \subset X \times X$,

we could try to define a category

where the objects would be the elements of $X$

and where there would be a morphism $a \to b$ iff $(a, b) \in R$.

However, the resulting category would lack identities (if $R$ is not reflexive)

and compositions (if $R$ is not transitive).

## Preorders as Categories (I)

A ***preorder*** $(A, \leqslant)$ is a relation that is

.**reflexive**: $a \leqslant a$.

.**transitive**: $(a \leqslant b)\ \&\ (b \leqslant c) \Rightarrow a \leqslant c$,

thereby providing the minimal structure (i.e. identities and compositions)

to define a category.

A preorder $(A, \leqslant)$ defines a category $\mathscr{C}_{(A,\leqslant)}$

whose objects are the elements of $A$

and where there is a morphism $a \to b$ if and only if $a \leqslant b$.

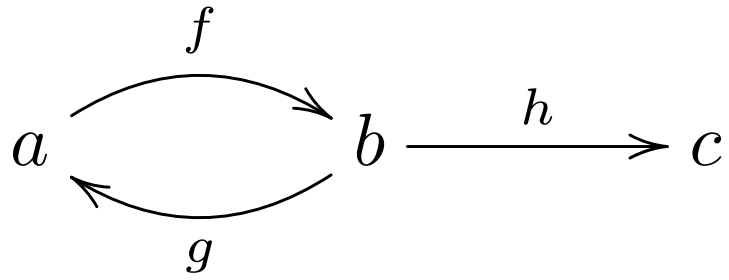
# Preorders as Categories (II)

Preorders do not necessarily establish

an *order* between its elements (i.e. they are *pre*orders)

since we can have both $a \leqslant b$ and $b \leqslant a$,

like in the following diagram:



Preorders can be further specified into

- ***Partially ordered sets*** (aka *posets*)**,** by adding an ***anti-symmetry condition***.

- ***Equivalence relations,*** by adding a ***symmetry condition***.

## Partially Ordered Sets as Categories

***Partially ordered sets*** (posets) are relations on a set $X$ that are

.reflexive: $a \leqslant a$,

.transitive: $(a \leqslant b)$ & $(b \leqslant c) \Rightarrow a \leqslant c$.

.anti-symmetric: $(a \leqslant b)$ & $(b \leqslant a) \Rightarrow a = b$).

Note 1: the last condition is called *anti-symmetry* because it prevents symmetric relations between two different objects, the absence of which is a defining property of an order.

A partial order is a relation on a set that *partially* orders its members (where the term *partial* means that two terms might be incomparable, i.e. there may be pairs for which neither element precedes the other).

**Examples of posets:**

.Open sets of a topological space.

.Logical propositions ordered by implication.

## Equivalence Relations as Categories

An ***equivalence relation*** $(X, \sim)$ is a relation that is

.reflexive $a \sim a$,

.transitive: $(a \sim b)\ \&\ (b \sim c) \Rightarrow a \sim c$

.symmetric: $a \sim b \Rightarrow b \sim a$

that is, a *symmetric preorder*.

An e*quivalence relation* $(X, \sim)$ defines a category $\mathscr{C}_{(X,\sim)}$

whose objects are the elements of $X$

and such there is an arrow $a \to b$ if and only if $a \sim b$.

Equivalence relations

are the formal tool to encode a fundamental operation

of rational thinking, namely ***abstraction***.

To abstract means

to ***bracket*** or ***« forget » certain differences***

that can be considered irrelevant

in a given context or for certain purposes.

By « forgetting » certain differences,

we can stress ***similarities***

between things that are strictly speaking different.

By classifying things according to such similarities,

we can introduce new ***abstract concepts***.

**« Loss of memory as the transcendental condition of science.**

**All reification is forgetting »**

**M. Horkheimer and T.W. Adorno, *Dialectic of Enlightenment*, p. 191.**

## (Fregean) Abstraction *qua* Equivalence Relation

- We start with a ***set of individuals*** $X$.

- We specify a ***criterium of equivalence*** according to which two individuals $a, b \in X$ will be considered similar or equivalent:

$$a \sim b$$

in spite of their differences.

- Formally, we have enriched the set $X$ with an equivalence relation $\sim$.

- We can then introduce a partition of the original set $X$ in classes of equivalent individuals called ***equivalence classes***.

- Each equivalence class is an instance of the ***abstract concept*** defined by the corresponding criterium of equivalence $\sim$.

## Quotient Set of Equivalence Classes

- An equivalence relation $\sim$ induces a partition of $X$ into subgroups called ***equivalence classes***, where $x$ and $y$ are in the same equivalence class if $x \sim y$.

- The ***set of equivalence classes*** is called ***quotient set*** and denoted

$$X/\sim$$

- The quotient or classifying function

$$q : X \to X/\sim$$

sends each $x \in X$ to its equivalence class $[x]$ such that

$$x \sim y \Rightarrow [x] = [y]$$

| | Set of individuals | Equivalence relation | Abstract concept |
|---|---|---|---|
| **Set-theoretic example** | Sets | Having the same number of elements | Cardinal |
| **Frege's example** | Straight lines | Being parallel | Direction |
| **Musical example** | Sound frequencies | Being *n* octaves apart | Pitch class |
| **Marx's example** | Commodities | Being produced by the same amount of « abstract general labour » | Exchange value |
| **Saussure's example** | Trains | Connecting the same cities and departing at the same time of the day | The « abstract train » from Paris to Geneva at 8h |

Categorically, *preorders*

(and in particular *partially ordered sets* and *equivalent relations*)

are ***thin categories***,

i.e. categories with at most one arrow between any two elements.

The notion of *relationship/morphism* in a category

*constructively* generalizes that of *relation* in set theory

in the sense that whereas the latter only encodes

whether two objects are related or not (existence or not of a relation),

the morphisms in a category encode

the multiple concrete manners through which they are related.

## Categories of Mathematical Objects

Certain categories can be understood

as ontological contexts

for a particular kind of mathematical objects.

Examples:

- Category *Set* of sets and functions.
- Category *Top* of topological spaces and continuous functions.

« Every variety of mathematical object belongs to its own category »

E. Riehl, *Infinity Category Theory Offers a Bird's-Eye View of Mathematics.*

In the case of categories of structured sets

(e.g. groups, topological spaces),

the morphisms are given by functions

between the underlying sets that preserve the structure.

E.g., a morphism of groups $f : (G, \bullet_G, e_G) \to (H, \bullet_H, e_H)$ in the category of groups

is a function between the underlying sets $G \to H$ that respects

the groups structure, i.e. such that

$$f(g \bullet_G g') = f(g) \bullet_H f(g')$$

From this property we deduce that

$$f(e_G) = e_H \qquad \text{and} \qquad f(g^{-1}) = f(g)^{-1}$$

# Kinds of Morphisms

## Kinds of Functions in *Set*

In *Set*, functions can be classified as :

- **Injective**:

$$f(x) = f(y) \Rightarrow x = y$$

- **Surjective**: for every $y \in Y$ there is an element $x \in X$ such that $f(x) = y$.

- **Bijective**: both injective and surjective.

Bijective functions $f : X \to Y$ have an inverse $f^{-1} : Y \to X$ such that

$$f^{-1} \circ f = id_X \qquad \text{and} \qquad f \circ f^{-1} = id_Y.$$

In category theory,

these three set-theoretic properties of functions

— namely, ***injectivity***, ***surjectivity*** and ***bijectivity***,

which are defined in terms of the extensional content of sets —

are reexpressed in a purely external or relational manner,

thereby allowing for the generalization of these definitions

to arbitrary categories.

# Injectivity

A function $f : X \to Y$

is ***injective*** if

$$f(x) = f(y) \Rightarrow x = y.$$

This definition is expressed

in terms of the elements of both sets.

**Arrow-theoretical recasting of the notion of element of a set**

An element of a set $x \in X$

can be redefined arrow-theoretically

by using the so-called ***characteristic function*** of $x$

$$\psi_x : \{ \star \} \to X$$

defined by $\psi_x( \star ) = x$.

Conceptually, the element $x \in X$ is individualized

by using an $\{ \star \}$-shaped arrow that « points » towards $x$.

« [E]lement is a special case of mapping. »

F.W. Lawvere and R. Rosebrugh, *Sets for Mathematics*, p.6.

« ‘Point’ stems from latin ‘punctum’, the past participle of ‘pungere’, to prick. The latter is defined by the circular image ‘to pierce slightly with a sharp point’. The important fact is that a point is a result of a movement with an arrow-like, acute (sharpened) instrument, its identification is established by a pointing action—the English expression ‘to point at’ is essential: a point is what we point at. It is the result of a pointing gesture which in German reads ‘zeigen’, to show. »

G. Mazzola, *The Topos of Music. Geometric Logic of Concepts, Theory, and Performance*, 2013, p.178.

We can now replace the elements in the definition of injectivity

$$f(x) = f(y) \Rightarrow x = y$$

by the corresponding characteristic functions:

$$f(\psi_x(\star)) = f(\psi_y(\star)) \Rightarrow \psi_x(\star) = \psi_y(\star)$$

or

$$f \circ \psi_x = f \circ \psi_y \Rightarrow \psi_x = \psi_y$$

for the diagram

$$\{\star\} \underset{\psi_y}{\overset{\psi_x}{\rightrightarrows}} A \xrightarrow{f} B$$

In other terms, the injective function $f : A \to B$ is ***left-cancellable***

with respect to the characteristic functions of elements of $A$.

A general category does not necessarily

have an object analogous to $\{ \star \}$

— called ***terminal object*** —.

Hence, an object $a$ in a category $\mathscr{C}$

does not necessarily have set-theoretical elements

obtained by « probing » or « testing » $a$ with $\{ \star \}$-shaped arrows.

But we can always « probe » $a$ with $b$-shaped arrows

for any object $b$ in $\mathscr{C}$.

# Monomorphisms (I)

*Set-theoretic* ***injective functions***

(extensionally defined in terms of the action

of the function on the elements)

translate into

*category-theoretic* ***left-cancellable morphisms***

(structurally defined in terms of morphisms).

# Monomorphisms (II)

A morphism $f : A \to B$

is a *monomorphism*

if given the following diagram

$$X \underset{g_2}{\overset{g_1}{\rightrightarrows}} A \xrightarrow{f} B$$

one has

$$f \circ g_1 = f \circ g_2 \Rightarrow g_1 = g_2,$$

i.e. if $f$ is left-cancellable.

## Monomorphisms (III)

In *Set* a function $f : A \to B$ is a monomorphism/injective if

$$f \circ \psi_x = f \circ \psi_y \Rightarrow \psi_x = \psi_y,$$

where

$$\{\star\} \underset{\psi_y}{\overset{\psi_x}{\rightrightarrows}} A$$

are the characteristic functions of the elements $x, y \in A$.

In a general category, a morphism $f : A \to B$ is a monomorphism

if it fulfills an analogous condition

$$X \underset{g_2}{\overset{g_1}{\rightrightarrows}} A$$

for every pair of « generalized elements » of $A$.

# Monomorphisms (IV)

« A monomorphism is an action that carries difference as such: it cannot install identity there where the preceding actions had instated difference. »

A. Badiou, *Mathematics of the Transcendental*, p.36.

# Epimorphisms (I)

*Set-theoretic* ***surjective functions***

(extensionally defined in terms of the action

of the function on the elements)

translate into

*category-theoretic* ***right-cancellable morphisms***

(structurally defined in terms of the morphisms).

## Epimorphisms (II)

In *Set*, a function $f : A \to B$

is ***surjective*** if given the following diagram

$$A \xrightarrow{f} B \underset{h_2}{\overset{h_1}{\rightrightarrows}} Y$$

one has $h_1 \circ f = h_2 \circ f \Rightarrow h_1 = h_2$

i.e. if $f$ is right-cancellable.

If $f$ were not surjective, $h_1 \circ f = h_2 \circ f$ would not imply $h_1 = h_2$

since $h_1$ and $h_2$ could still differ on the elements of $B$ that are not in the image of $f$.

A non-surjective $f$ entails the existence of elements in $B$

that could be the site of a potential difference between $h_1$ and $h_2$.

# Epimorphisms (III)

By generalizing this diagrammatic definition of surjectivity,

a morphism $f : A \to B$

is a ***epimorphism*** if given the following diagram

$$A \xrightarrow{f} B \underset{h_2}{\overset{h_1}{\rightrightarrows}} Y$$

one has $h_1 \circ f = h_2 \circ f \Rightarrow h_1 = h_2$

i.e. if $f$ is right-cancellable.

## Epimorphisms (IV)

« Just as a monomorphism cannot introduce identity as it preserves differences, likewise an epimorphism cannot introduce difference […]. »

A. Badiou, *Mathematics of the Transcendental*, p.40.

## Opposite or Dual Category

Given a category $\mathscr{C}$, the so-called ***opposite*** or ***dual category*** $\mathscr{C}^{op}$

is a category with the same objects as $\mathscr{C}$

and whose morphisms are obtained

by *formally* reversing the direction of all the morphisms in $\mathscr{C}$,

i.e. by exchanging the sources and targets of all the morphisms in $\mathscr{C}$.

For instance, a morphism $f : X \to Y$ in $Set^{op}$

is not a function from $X$ to $Y$,

but rather a label for a function from $Y$ to $X$ in $Set$.

Monomorphisms and epimorphisms

are ***dual notions***

in the sense that

a monomorphism in $\mathscr{C}$ becomes an epimorphism in $\mathscr{C}^{op}$

and

an epimorphism in $\mathscr{C}$ becomes a monomorphism in $\mathscr{C}^{op}$.

## References on Categorical Duality


• D. Corfield, Duality as a Category-Theoretic Concept, *Studies in History and Philosophy of Modern Physics*, 59, pp.55-61, 2017.

• J.-P. Marquis, An Historical Perspective on Duality and Category Theory: Hom is where the heart is, in R. Krömer and E. Haffner (eds.), *Duality in 19th and 20th Century Mathematical Thinking*, 2024.

***Set-theoretic bijections***

(extensionally defined in terms of the action

of the function on the elements)

translate into

***category-theoretic isomorphisms***

(structurally defined in terms of the morphisms).

## Isomorphisms

An ***isomorphism*** is a morphism

$$f : a \to b$$

such that there exists another morphism

$$f^{-1} : b \to a$$

satisfying

$$f^{-1} \circ f = id_a \qquad \text{and} \qquad f \circ f^{-1} = id_b$$

An isomorphism provides a kind of internal decomposition or factorization

of the trivial self-identity $id_a$ that passes through an external object $b$.

Note: the identity arrows $id_a$ are always isomorphisms (being their own inverses).

## On the Relation Between Isomorphisms, Monomorphism and Epimorphisms

Isomorphisms are always monic (cf. Cheng p. 196) and epic (cf. Cheng p.199).

In *Set* (and more generally in any topos),

a monic and epic morphism

is necessarily an isomorphism…

but this is not valid in general.

Counterexamples:

• In $\mathscr{C}_{(\mathbb{N},+,0)}$, every morphism is monic and epic, but the unique isomorphism is the identity morphism $id_\star = 0$.

• In a poset, every arrow is monic and epic (since there is at most one morphism between two objects), but (as a consequence of the symmetry) the only isos are the identities. More generally, in any *thin category* (categories with oat most one morphism between any two objects) every morphism is mono and epic.

## Examples of Isomorphisms

• **Isomorphisms** $f : X \to Y$ **in the category of sets**:

Bijective (or one-to-one) functions.

• **Isomorphisms** $f : (G, \bullet_G, e_G) \to (H, \bullet_H, e_H)$ **in the category of groups** :

Bijective functions btw the underlying sets $G \to H$ that respects the groups structure, i.e. such that

$$f(g \bullet_G g') = f(g) \bullet_H f(g')$$

• **Isomorphisms** $f : (X, \tau) \to (Y, \eta)$ **in the category of topological spaces (called *homeomorphisms*)** :

Bijective functions btw the underlying sets $f : X \to Y$ that respect the topological structure, i.e. such that $f$ and $f^{-1}$ are continuous.

# Groups,

# Symmetries

# and

# Expanded Identity

## Groups as Categories

A group $G \doteq (X, \bullet, e)$ defines a category $\mathscr{C}_G$ with a single object $\star$

such that every $g \in X$ defines an ***invertible*** morphism $f_g : \star \to \star$ (i.e. an isomorphism)

whose inverse is $f_{g^{-1}}$,

where $e \in X$ defines the identity morphism $id_\star$ (that is $f_e = id_\star$)

and the composition $f_g \circ f_h$ is given by $f_{g \bullet h}$

(the associativity of compositions is guaranteed

by the associativity of the group operation $\bullet$ in $G$).

By doing so, the group is categorically presented as the set of ***automorphisms*** of the object $\star$.

## Terminological Distinctions

- An ***endomorphism*** is a morphism of the form $a \to a$, i.e. such that the source and the target coincide.

- An ***automorphism*** is an invertible endomorphism.

- An ***identity morphism*** $id_a$ is an automorphism fulfilling the identity laws.

Every identity morphism $id_a$ is an automorphism

(where $id_a$ is its own inverse)

and every automorphism is an endomorphism

but the converses are not true.

## Symmetries

A ***symmetry*** is a transformation of an object

such that the final state cannot be distinguished from the initial one,

a non-trivial transformation that produces a trivial effect.

## Symmetries Form in a Group

- The transformation that consists in not doing anything to the object is a symmetry transformation, the neutral element of the group.

- Composing two symmetry transformations is a symmetry transformation.

- Every symmetry transformation has an inverse.

The notion of ***group***

encodes a ***constructivist enhancement***

of the ontological notion of ***identity***.

Wheres the classical principle of identity

$$a = a$$

states that any entity $a$ is self-identical,

the symmetry group of $a$ encodes all the transformations

by means of which an entity can be self-identified.

# On the Constructivist Notion of Proposition

From a constructivist standpoint, a proposition is not just

a statement that is either true or false (classical definition)

but rather the collection of all its proofs.

Now, how can we prove a mathematical equality

of the form

$$a = b?$$

## Proofs of a Geometric Equality are Given by Congruences

« Things which coincide with one another [when superposed] are equal to one another »

Euclid's *Elements*, Notion 4.

« Two figures are congruent if one can be placed on the other without changing anything in the two figures except their place. »

G.W. Leibniz, *Letter to Christian Huygens* (1679).

« If indeed one tries to clarify the notion of *equality*, which is introduced right at the beginning of Geometry, one is led to say that two figures are equal when one can go from one to the other by a specific geometric operation, called a motion. »

E. Cartan, *Notice sur les travaux scientifiques* (1974).

Two geometric figures $a$ and $b$

in a space $X$ are equal if there is a…




… that allows us to superpose $a$ with $b$

(if we cannot do that, $a = b$ would be false).

Now, how can we prove the particular case of an identity

$$a = a?$$

By actively identifying $a$ with itself,

i.e. by performing a transformation

such that the initial and the final state

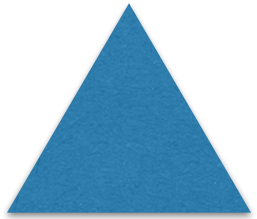
are indistinguishable.

By definition, such transformations are symmetries of $a$.

**The symmetries of an entity $a$ are proofs of its self-identity.**

For instance,

is the equilateral triangle



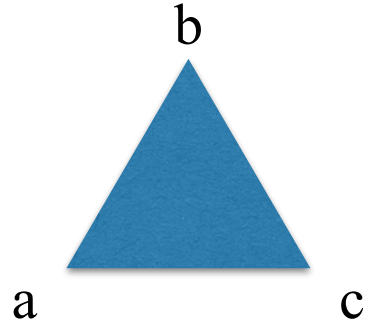
identical to itself?

How can we prove it?

Constructivist answer:

by exhibiting its self-identifications, i.e. its symmetries.

The self-identifications or symmetries of the equilateral triangle




are the following:

• the 3 counterclockwise rotations $R_0$, $R_{120}$ and $R_{240}$ in 0, 120 and 240 degrees.

• the 3 reflections $S_a$, $S_b$ and $S_c$ around the three principal axes passing by the vertices $a$, $b$ and $c$.

Note: the rotation $R_0$ in 0 degrees is in fact a transformation that does not do anything.

The set $\{R_0, R_{120}, R_{240}, S_a, S_b, S_c\}$ of these 6 transformations

is the underlying set of the ***group of symmetries*** of the equilateral triangle

(called *dihedral group of order* 6).

where $R_0$ is the neutral element,

all the transformations have an inverse in the set,

and the set of transformations is closed under compositions

(see Cayley table in the next slide).

## Cayley Table of the Group of Symmetries of the Equilateral Triangle

| | | First operation | | | | | |
|---|---|---|---|---|---|---|---|
| | $\circ$ | $R_0$ | $R_{120}$ | $R_{240}$ | $S_a$ | $S_b$ | $S_c$ |
| Second operation | $R_0$ | $R_0$ | $R_{120}$ | $R_{240}$ | $S_a$ | $S_b$ | $S_c$ |
| | $R_{120}$ | $R_{120}$ | $R_{240}$ | $R_0$ | $S_c$ | $S_a$ | $S_b$ |
| | $R_{240}$ | $R_{240}$ | $R_0$ | $R_{120}$ | $S_b$ | $S_c$ | $S_a$ |
| | $S_a$ | $S_a$ | $S_c$ | $S_b$ | $R_0$ | $R_{120}$ | $R_{240}$ |
| | $S_b$ | $S_b$ | $S_a$ | $S_c$ | $R_{240}$ | $R_0$ | $R_{120}$ |
| | $S_c$ | $S_c$ | $S_b$ | $S_a$ | $R_{120}$ | $R_{240}$ | $R_0$ |

- Note that the group is not commutative (e.g. $S_a \circ S_b = R_{240}$ and $S_b \circ S_a = R_{120}$).

***An object $a$ with symmetries has a non-trivial identity,***

in the sense that there are many proofs of the principle of identity

$$a = a.$$

The symmetry group of an object defines its expanded identity.

Even if an object $a$ has no (nontrivial) symmetries at all,

the transformation that does not anything to $a$

is a (trivial) symmetry transformation.

Hence, there is always at least the trivial proof

of the fact that an object $a$ is self-identical.

Hence, the proposition $a = a$ is always true.

« Formally we can take any shape and make a category of its symmetries. It has only one object, which we could regard as being the shape itself. The arrows are then the symmetries […] The identity arrow is the ‘trivial symmetry’ […] In normal life you’d say ‘this thing has no symmetry’ rather than ‘this thing has one symmetry, the trivial symmetry’. »

E. Cheng, *The Joy of Abstraction*, p.112.

Category theory presents groups

as groups of symmetries,

i.e. as sets of invertible transformations of a single object.

By doing so, this category theory makes clear that such symmetries

generalize the

— ontologically and categorically required —

trivial identity $id_a$ of an object $a$.

to non-trivial proofs of its self-identity.

The classical identity principle

$$a = a$$

- is ***universally true***, i.e. it is valid for any $a$:

**« No entity without identity »**

(W.V.O. Quine, *Speaking of Objects*)

- it is ***evident***:

**It is a truth that needs no proof.**

- it is ***empty***, i.e. it conveys no information about $a$:

**It is an irrelevant truth.**

## Constructivist Principle of Identity $a = a$

- it is also universally true, i.e. it is valid for any $a$, since any $a$ has at least the trivial group $G = \{e\}$ as group of symmetries:

*No entity without identity* (Quine)

- it might have non-trivial proofs given by the symmetries of $a$.

- it conveys information about $a$.

« Among the different ways of being identical to oneself, there is ‘inert’ identity, that is, the null action $id_\star$. What a group indicates is that this identity is but the degree zero of identity, its immobile figure and at rest. The other arrows of the group are dynamic identities. They are the active ways of being identical to oneself. »

A. Badiou, *Group, Category, Subject*.

« Sameness implies the relation of *with*, that is, a mediation, a connection, a synthesis: the unification into a unity. [...] But that unity is by no means the stale emptiness of that which, in itself without relation, persists in monotony. However, to get to the point where the relationship of the same with itself [...] emerges as this mediation [...] within identity, Western thought required more than two thousand years. For it is only the philosophy of speculative Idealism, prepared by Leibniz and Kant, that through Fichte, Schelling, and Hegel established an abode for the essence, in itself synthetic, of identity. [...] Just one thing we must keep in mind: since the era of speculative Idealism, it is no longer possible for thinking to represent the unity of identity as mere sameness, and to disregard the mediation that prevails in unity. Wherever this is done, identity is represented only in an abstract manner. »

M. Heidegger, *Identity and Difference*, p. 25.

## Towards Groupoids

According to their category-theoretical presentations,

***groups*** and ***equivalence relations*** are « orthogonal notions »:

- **Groups**: a single object with many isomorphisms.

- **Equivalence relations**: many objects with a unique isomorphism (if any) between any two of them.

The notion of ***groupoid*** combines

these two notions into a single one.

## Groupoids

A **groupoid** is a category

with eventually many objects (like equivalence relations)

and eventually many isomorphisms between them (like groups).

Shortly, a **groupoid** is a category

in which every morphism is an isomorphism.

*Sets*, *equivalence relations* and *groups* are all,

— when defined as categories —

particular cases of groupoids.

# On the Structuralist Understanding of Category Theory

What are the precise mathematical foundations of the structuralist reading of category theory according to which an object is determined by the relations it has with other objects?

This structuralist interpretation of category theory

is grounded on two essential components

of category theory:

♥ The category-theoretical « stretching »

of the notions of mathematical *equality* $a = b$

and, *a fortiori*, of mathematical *identity* $a = a$

into *isomorphisms*.

♣ The Yoneda lemma.

# The Paradox of Equality

Mathematical equalities

$$a = b$$

might seem paradoxical since they state

that two ostensibly different terms

$a$ and $b$

are in fact *equal*.

# The Triviality of Identity

In turn, the ***principle of identity***

stating that a thing $a$ is equal to itself

$$a = a$$

seems to be tautological or trivial.

Knowing that $a = a$ conveys

no information whatsoever about $a$.

« […] to say of two things that they are identical is nonsense,

and to say of one thing that it is identical with itself is to say

nothing at all. »

L. Wittgenstein, *Tractatus Logico-Philosophicus* (1921), §5.5303.

## The Paradox and the Truism

According to Whitehead, an equality $a = b$

is neither contradictory nor trivial

to the extent that it combines

a difference (***paradox***) and an identity (**truism**).

« It is to be observed that the proposition asserted by the equation, $a = b$, consists of two elements ; which for the sake of distinctness we will name […] the *truism* and the *paradox*. The *truism* is the partial identity of both $a$ and $b$ [...] The *paradox* is the distinction between $a$ and $b$, so that $a$ is one thing and $b$ is another thing […] »

A.N. Whitehead, *A Treatise on Universal Algebra* (1898).

According to Whitehead's description,

an interpretation of mathematical equalities $a = b$ should specify:

- where lies the ***paradox*** (the difference between $a$ and $b$) that makes equalities non trivial;

- what lies the ***truism*** (the partial identity between $a$ and $b$), i.e. the predicable content that is affirmed by the proposition.

The interpretation of equalities

$$a = b$$

proposed by Frege (*Begriffsschrift*) and Quine (*Methods of Logic*)

is that the equal sign '='

encodes a relation of ***synonymy***

according to which *a* and *b* are

just different names (*paradox*) of one and the same thing (*truism*).

## Equality as encoding synonymy according to Frege

« Identity of content differs from conditionality and negation in that it applies to names and not to contents. Whereas in other contexts signs are merely representatives of their content so that every combination into which they enter expresses only a relation between their respective contents, they suddenly display their own selves when they are combined by means of the sign for identity of content; for it expresses the circumstance that the two names have the same content. Hence, the introduction of a sign for identity of content necessarily produces a bifurcation in the meaning of all signs: they stand at times for their content, at times for themselves. »

G. Frege, *Begriffsschrift*,§8, 1879.

## Equality as synonymy according to Quine

« Identity is such a simple and fundamental idea that it is hard to explain otherwise than through mere synonyms. To say that *x* and *y* are identical is to say that they are the same thing. Everything is identical with itself and with nothing else. But despite its simplicity, identity invites confusion. E.g., it may be asked: Of what use is the notion of identity if identifying an object with itself is trivial and identifying it with anything else is false? [...] For truth of a statement of identity it is necessary only that '=' appear between names of the same object; the names may, and in useful cases will, themselves be different. For it is not the names that are affirmed to be identical, it is the things named. [...] Still, since the useful statements of identity are those in which the named objects are the same and the names are different, it is only because of a peculiarity of language that the notion of identity is needed. If our language were so perfect a copy of its subject matter that each thing had but one name, then statements of identity would indeed be useless.[...] The utility of language lies partly in its very failure to copy reality in any one-thing-one-name fashion. The notion of identity is then needed to take up the slack. [...] what are identical are the objects with themselves and not the names with one another; the names stand in the statement of identity, but it is the named objects that are identified.[...] Statements of identity consisting of '=' flanked by singular terms are needed, we saw, because language includes a redundancy of names. »

W.V.O. Quine, *Methods of Logic*, § 35.

« [...] the statements of identity that are true and not idle consist of unlike singular terms that refer to the same thing. »

W.V.O. Quine, *Word and Object*, p. 107.

## Equality as synonymy according to Wittgenstein

Mathematical equalities are nothing

but a formal device thanks to which we can keep under control

the excess of names over things

and the notion of equality can be in principle removed

by using a single name for each thing.

« Identity of object I express by identity of sign, and not by using a sign for identity. Difference of objects I express by difference of signs. […] identity is not a relation between objects. »

L. Wittgenstein, *Tractatus Logico-Philosophicus*, §5.53, 1921.

Now, the relation of synonymy

cannot exhaust the meaning of mathematical equalities,

like equalities between different mathematical expressions:

$$3 \times 2 = 5 + 1.$$

Even if the expressions $3 \times 2$ and $5 + 1$

yield (or denote) the same numerical value,

they encode different computational processes.

## Kant's Constructivist Interpretation of Equalities

« I can form a concept of one and the same magnitude by means of several different kinds of composition and separation [...]. Objectively, the concept I form is indeed identical (as in every equation). But subjectively, depending on the type of composition that I think, in order to arrive at that concept, the concepts are very different. [...] Thus I can arrive at a single determination of a magnitude $= 8$ by means of $3 + 5$, or $2 \times 4$, namely 8. But my thought '$3 + 5$' did not include the thought '$2 \times 4$'. Just as little did it include the concept '8,' which is equal in value to both of these. »

I. Kant, Letter to Schultz, November 25th, 1788.

## Girard's Fregean Rephrasing of Kant's Stance

« This equality $3 \times 2 = 5 + 1$makes sense in the mainstream of mathematics by saying that the two sides denote the same integer [...]. This is the denotational aspect, which is undoubtedly correct, but it misses the essential point: There is a finite *computation* process which shows that the denotations are equal. It is an abuse [...] to say that $3 \times 2$ *equals* $5 + 1$ […] The two expressions have different *senses* and we must *do* something (make a proof or a calculation [...]) to show that these two *senses* have the same *denotation*. »

J.-Y Girard, *Proofs and Types*, 1990.

**Equality in Frege's *Sinn und Bedeutung* (1892)**

Mathematical objects *qua* referents of mathematical discourse

are given in particular « modes of presentation ».

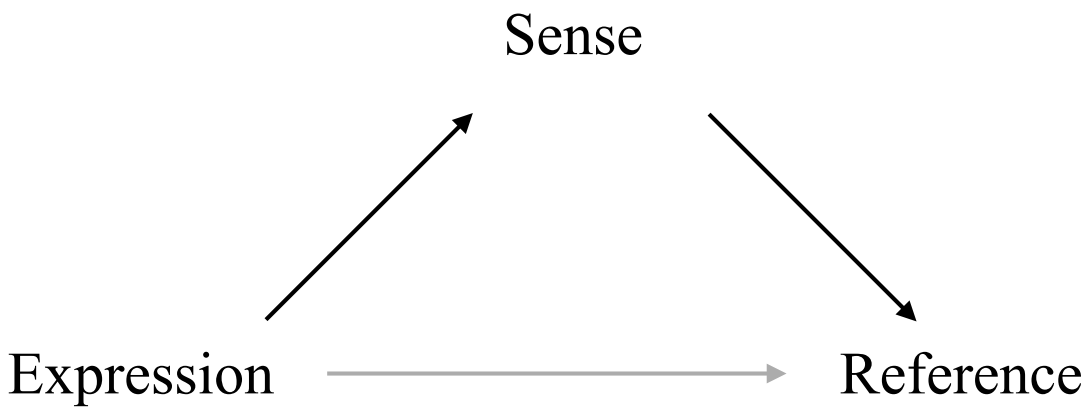
A propositional equality of the form

$$a = b$$

asserts that the signs $a$ and $b$ stand

for different presentations of the same referent.

# Frege's *Uber Sinn und Bedeutung* according to Martin-Löf




A mathematical expression

***expresses* a *sense***

(a presentation, a computation, a construction, a proof)

and ***denotes* a *reference***

(the result of the computation, the construction, or the proof).

Cf. Martin-Löf, P. (2021), The Sense/Reference Distinction in Constructive Semantics, *The Bulletin of Symbolic Logic*.

« The heart and soul of much mathematics consists of the fact that the ‘same’ object can be presented to us in different ways. »

Barry Mazur, *When is one thing equal to some other thing?*

The mathematical referent as such

— i.e. the mathematical object

*modulo* the particular mode of presentation in which it is given —

is defined through *abstraction*

in the sense that the referent (the object as such)

can be reached by « forgetting » its modes of presentation.

This is why the problem of abstraction

touches upon the very foundations of mathematics.

## Equality in Set Theory: the Axiom of Extensionality

« Two » sets $A$ and $B$ are one and the same set

noted

$$A = B$$

if $A$ and $B$ have the same elements.

In Leibniz's parlance, set-theoretic equality

encodes ***numerical identity***,

i.e. the relation of being one and the same entity.

In category theory, *strict equalities*

$$a = b$$

(meaning that $a$ and $b$ denote one and the same entity)

are ***stretched*** to *isomorphisms* (meaning « *equal form* »)

$$a \simeq b$$

In category theory, two numerically different entities are said to be « equal »

if they are iso-morphic, i.e. if they have the same « form » or « structure ».

This *form* or *structure* plays the role of the Fregean *referent*

and two isomorphic objects are understood as different presentations

of the same *form-structure-referent*.

In Whitehead's terms,

the fact that entities $a$ and $b$ verifying

$$a = b$$

denote the ***same referent***

constitute the ***truism*** of the equality proposition

and the fact that $a$ and $b$

express ***different modes of presentation*** of such a referent

constitute the ***paradox***.

Now, whereas strict equalities encode numerical identity

— being one and the same entity —,

stretched equalities encode… what?

« One philosophical reason for categorification is that it refines our concept of ‘sameness’ by allowing us to distinguish between isomorphism and [numerical] equality. In a set, two elements are either the same or different. In a category, two objects can be ‘the same in a way’ while still being different. In other words, they can be isomorphic but not [numerically] equal. Even more importantly, two objects can be the same in more than one way, since there can be different isomorphisms between them. This gives rise to the notion of the ‘symmetry group’ of an object: its group of automorphisms. »

J.C. Baez and J. Dolan, *Categorification*

## Intrisinc (set-theoretic) vs extrinsic (category-theoretic) identity

« In set theory the identification of a presented multiple (therefore of *a* being) is clear: a set is defined by its elements and two sets that have the same elements are identical (are the 'same' set). This is prescribed by the axiom of extensionality. We see that this axiom rests on an *intrinsic* (immanent) determination of the pure multiple. In category theory this is generally not the case. Because 'two' objects, while counting-for-two 'in itself' (in a set-theoretical sense), can obviously be the source and target of the same kinds of actions (or relations) in the category in which they appear. In which case their *extrinsic* identity is the same. It is however this identity, and not the pure difference of position (which is in fact a literal difference: one of these objects is called *a*, the other *b*), that is essential for the categorician. In this case we will say that the two objects are isomorphic, and will regard them as effectively 'the same'. There is therefore in category theory an essential ambiguity to identity. Caught up in a network of relations and similar external actions, 'two' objects are indiscernible, except as pure empty letters. This is inevitable since identity is not an immanent mark, but rather the effect of actions, or arrows, which operate in a categorial universe. »

A. Badiou, *Mathematics of the Transcendental*, p.14.

« A category provides a context in which to answer the question ‘When is one thing the same as another thing?’ »

E. Riehl, *Category Theory in Context*, p.7.

But what does it mean that numerically different objects

might be considered « structurally equal »

in the corresponding categorical context?

In (first-order) logic (with equality),

the equality relation $=$ is defined by two axioms

- **Reflexivity:**

$$a = a.$$

- **The Principle of Indiscernibility of Equals (PIE):** If $a$ and $b$ are « equal », then they should have the same properties

$$P(a) \text{ and } a = b \Rightarrow P(b)$$

This principle imposes a condition on the admissible properties,

that of being invariant across equalities.

In set theory,

the ***Principle of Indiscernibility of Equals***

becomes

the ***Principle of Indiscernibility of Sets with the Same Elements***

and the admissible properties

are the ***extensional properties***.

In category theory,

the ***Principle of Indiscernibility of Equals***

becomes

the ***Principle of Indiscernibility of Isomorphic Objects***

and the admissible properties

— that we shall call ***categorical*** or ***structural properties*** —

are given by the properties

that are invariant under isomorphisms.

« […] the only properties which a given object in a given category may have, *qua* object in that category, are structural ones. »

S. Awodey, Structure in Mathematics and Logic: A Categorical Perspective, *Philosophia Mathematica*, 3, 209-37, 1996.

« [...] two isomorphic structures are, essentially, the same. [...] (Principle of Isomorphism) all grammatically correct properties of objects of a fixed category are to be invariant under isomorphism. Here, a property $P(Y)$ of the variable object $Y$ in a given category is invariant under isomorphism if $P(Y)$ and $Z = Y$ imply $P(Z)$.The just-stated principle is the analog, formulated for isomorphism in place of ordinary equality, of Leibniz's indiscernibility of identicals, the usual axiom of equality that allows to substitute any thing by any other thing equal to it. [...] The basic character of the Principle of Isomorphism is that of a constraint on the language of Abstract Mathematics; a welcome one, since it provides for the separation of sense from nonsense. »

M. Makkai, *Towards a Categorical Foundation of Mathematics*.

## Extensional vs Structural Properties

The property

$P(X) =$ « The element $a$ belongs to the set $X$ »

is *extensional* but not *structural*.

The property

$Q(X) =$ « The set $X$ has 2 elements »

is *structural*.

Category theory adopts a **structural**

rather than **element-based** view of sets.

In category theory, although the objects of the category *Set* are ordinary ZF sets, they are studied « up to isomorphism »: two sets with the same cardinality are considered to be categorically or structurally indistinguishable.

In set theory, the sets {0,1} and {2,3} are (according to the extensionality axiom) different sets since they have different elements.

In category theory, they are « structurally » equal (i.e. different presentations of the same structure) since they are bijective (i.e. isomorphic in *Set*).

Having the same form or structure in *Set* means having the same number of elements.

Note: there are two isomorphisms between them, two proofs of the fact that the sets {0,1} and {2,3} are structurally equal.

In category theory, the « relations determine the objects »

in the precise sense that what an object is

— its very determinacy, that is the intrinsic properties it has —

depends on the contextual relations that it entertains

with other objects of the same category.

## On Objectivity and Automorphisms

The intrinsic (or objective) properties of an object still have to satisfy the

**Principle of the Indiscernibility of**

**Isomorphic (in this case Automorphic) Objects.**

« Objectivity means invariance with respect to the group of automorphisms. »

H. Weyl, *Symmetry*, 1952, p. 132.

The identity of an object imposes a constraint on its determinacy:

if an object has non-trivial automorphisms

its properties have to be invariant under its group of automorphisms

## Example: Minkowski space-time

In Einstein's special theory of relativity,

(Minkowski) space-time is a highly symmetric object,

being its group of symmetries the so-called ***Poincaré group***.

Hence, only the invariants under the action of the Poincaré group

define intrinsic properties of Minkowski space-time

(e.g. the temporal lapse between two events is not invariant).

**The Referents of Category-Theoretical Discourse**

The relevant « entities » of category theory

— the referents of category-theoretical discourse —

are not given by the particular objects of a category,

but rather by the *abstracta* given by their isomorphism classes.

These *abstracta*

— which are instantiated by all the objects in a the corresponding isomorphism class —

are called *structures*.

Category theory is a *structuralist theory*

in the sense that its referents are abstract « structures »

and not the particular objects that instantiate or present these structures.

« The subject matter of pure mathematics is **invariant form** […] »

S. Awodey, Structure in mathematics and logic: A categorical perspective, *Philosophia Mathematics* (3), 4, 1996, pp. 209-237.

« In the mathematical development of recent decades one sees clearly the rise of the conviction that the relevant properties of mathematical objects are those which can be stated in terms of their **abstract structure** rather than in terms of the elements which the objects were thought to be made of. »

F. W. Lawvere, *The Category of Categories as a Foundation of Mathematics*, Proc. Conference Categorical Algebra (LaJolla 1965), Springer, N.Y. 1-20, 1966.

## Structure vs kind of structure (I)

It is important to distinguish between

$a$ and $b$ having the same ***kind of form***, ***kind of structure*** or ***structure type***

— which is the case when $a$ and $b$ are objects of the same category —

and $a$ and $b$ having the same form or structure

— which is the case when $a$ and $b$ are are isomorphic ($a \simeq b$) —

For instance, two groups have the same *structure type*

(the structure of a group),

but they do not have necessarily the same *structure*,

that is they are not necessarily isomorphic.

## Structure vs kind of structure (II)

« Intuitively, two things with the same kind of structure may indeed have the same structure. The categorical notion of isomorphism may now serve as a *definition* of 'having the same structure of a given kind'. »

S. Awodey, Structure in Mathematics and Logic: A Categorical Perspective, *Philosophia Mathematica*, 3, 209-37, 1996.

« Group homomorphism and similar general maps between structures of other types are colloquially called structure-preserving. This is somewhat misleading because if such maps preserve anything at all it is a type of structure but not a particular structure [preserved under isomorphisms]. »

A. Rodin, Categories without Structures, *Philosophia Mathematica* (III) 19, p.28, 2011.

# Towards Higher Categories

## The Pitfalls of Category Theory

• Category theory stretches the notion of strict equality $=$ ***between objects*** to the notion of isomorphism $\simeq$.

• However, strict equalities are still used in the very definition of a category in the form of strict equalities ***between morphisms***:

.the associativity of composition

$$(h \circ g) \circ f = h \circ (g \circ f)$$

.the identity laws verified by the trivial identities:

$$id_b \circ f = f \qquad \text{and} \qquad f \circ id_a = f$$

$$\text{with } f : a \to b.$$

• Moreover, the very definition of isomorphisms also uses strict equalities between morphisms:

$$f^{-1} \circ f = id_a \qquad \text{and} \qquad f \circ f^{-1} = id_b$$

$$\text{for} \quad f : a \to b \quad \text{and} \quad f^{-1} : b \to a.$$

Category theory is based on a ***two-sorted ontology***

composed of ***objects*** and ***morphisms***.

Whereas morphisms are used

to weaken the strict equalities between objects,

the equalities between morphisms remain strict.

Whereas objects are just instantitions of a structure,

morphisms are treated non-structurally as full-fledged individuals.

Category theory does not implement

a full-fledged stretching of strict equalities into isomorphisms.



**Category theory remains partially strict after all.**

## Stretching all the way up

• In order to solve this problem, we should stretch the strict equalities in the definition of a category:

$$(h \circ g) \circ f = h \circ (f \circ g)$$

$$id_b \circ f = f \qquad f \circ id_a = f$$

and in the definition of an isomorphism:

$$f^{-1} \circ f = id_a \qquad f \circ f^{-1} = id_b.$$

into isomorphisms $\simeq$.

• Now, the entities related by such equalities (e.g. $f^{-1} \circ f$ and $id_a$) are already morphisms.

• So the new isomorphisms should be isomorphisms between morphisms !!!

A full-fledged or maximally stretched replacement of equalities with isomorphisms

can only be performed if we make the leap towards ***higher categories***,

i.e. towards a theory of ***categories with higher morphisms***,

i.e., with morphisms between morphisms between morphisms,…

## Second Argument Towards Higher Categories (I)

« In basic category theory our insight begins by saying we should study relationships between objects, not just the objects themselves. But what about the relationships? If we consider those to be new mathematical objects, shouldn't we also study relationships between those? This gives us one more dimension. »

E. Cheng, *The Joy of Abstraction*, p. 21.

« But now, if we believe in studying relationships between things, what about relationships between morphisms? What about amore nuanced notion of sameness between morphisms? What about a way of studying various kinds of abstract structure on morphisms? If we're interested in categories rather than sets, how about categories of morphisms instead of sets of them? »

E. Cheng, *The Joy of Abstraction*, p. 21.

## Second Argument Towards Higher Categories (II)

If relationships are also mathematical objects, they should also be objects in a category.

Hence, each $Hom_{\mathscr{C}}(a, b)$ should not be a set but rather a category.

If we do so, the objects in the category $Hom_{\mathscr{C}}(a, b)$ are 1-morphisms in $\mathscr{C}$

and the morphisms in $Hom_{\mathscr{C}}(a, b)$ are 2-morphisms in $\mathscr{C}$, i.e.

2-morphisms between 1-morphisms.

The same argument can be applied to $Hom_{Hom_{\mathscr{C}}(a,b)}(f, g)$ where $f, g : a \to b$,

which should also be considered as a category, which requires to introduce 3-morphisms…

If we iterate this argument, we shall have $n$-morphisms all the way up.

## Higher Category Theory

An $n$-category $\mathscr{C}$ is given by:

.a collection of *objects* $a, b, c \ldots$

.a collection of 1-morphisms $f : a \to b$ between objects.

.a collection of 2-morphisms $H : f \Rightarrow g$ between 1-morphisms $f$ and $g$ with the same source $a$ and target $b$.

.a collection of 3-morphisms $\eta : H \Rrightarrow K$ between 2-morphisms $H$ and $K$ between 1-morphisms $f$ and $g$ with the same source $a$ and target $b$.

- And so on and so forth all the way up to $n$-morphisms.

- In particular:

.0-categories: sets.

.1-categories: ordinary categories.

## Higher Compositions

- In 1-category theory, two 1-morphisms can be composed when they meet at an object (i.e. at an entity one dimension less).

- In higher category theory, $k$-morphisms can meet along morphisms of any dimension from 0 (objects) to $k-1$.

- Hence, there are $p = 0,...,k-1$ possible kinds of composition $\circ_p^k$ between $k$-morphisms.

- For instance, 2-morphisms can meet at objects and at 1-morphisms.

- It follows that 2-morphisms can be composed

  .along objects (called ***horizontal composition*** and denoted $\circ_0^2$);

  .along 1-morphisms (called ***vertical composition*** and denoted $\circ_1^2$).




Animation by Manuel Eguía




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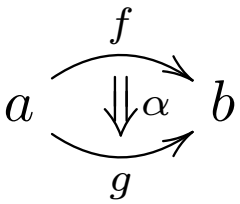
# From Globular Sets to Higher Categories
# I. Data

- **Globular Set**:

.Collection of 0-morphisms, 1-morphisms, 2-morphisms, …, $n$-morphisms, where each $k$-morphism has a source and a target of dimension $k-1$.

.Satisfying certain « globularity conditions » stating that an $n$-morphism relates two $n-1$-morphisms that have the same source and target.

Thanks to the globularity conditions a 2-cell has the following globular shape




rather than the form

$$\begin{array}{ccc} a & \xrightarrow{f} & b \\ & \Downarrow\alpha & \\ c & \xrightarrow[g]{} & d \end{array}$$

## Globular Sets Formally

A ***globular set*** is a diagram in *Set*

$$C_n \underset{t_n}{\overset{s_n}{\rightrightarrows}} C_{n-1} \underset{t_{n-1}}{\overset{s_{n-1}}{\rightrightarrows}} \cdots \underset{t_2}{\overset{s_2}{\rightrightarrows}} C_1 \underset{t_1}{\overset{s_1}{\rightrightarrows}} C_0$$

satisfying the ***globularity conditions*** stating that a $k$-morphism in $C_k$ has to satisfy that

$$s_{k-1}s_k = s_{k-1}t_k$$

— saying that a $k$-morphism relates two $(k-1)$-morphisms that have the same source —

and

$$t_{k-1}s_k = t_{k-1}t_k$$

— saying that a $k$-morphism relates two $(k-1)$-morphisms that have the same target —.

## From Globular Sets to Higher Categories
## II. Structure

• **Reflexive Globular Set**: A globular set in which each $k$-morphism $\alpha$ is equipped with a putative $(k+1)$-identity morphism $id_\alpha : \alpha \to \alpha$.

• **Magma**: a reflexive globular set enriched with all possible binary compositions.

**Magma = globular set (data) + identities and compositions (structure).**

## Compositions of Multi-Composable Diagrams

• The composition of certain multi-composable diagrams can be performed according to different parsing orders.

• 1-categorical example:

.In the multi-composable diagram

$$a \xrightarrow{f} b \xrightarrow{g} c \xrightarrow{h} d$$

there are two possible manners to perform the composition, namely

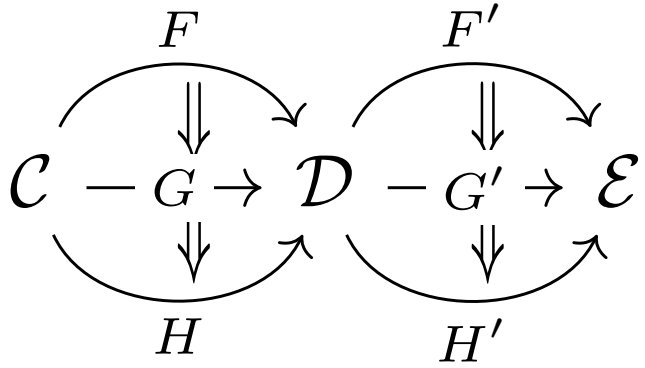
$$h \circ (g \circ f) \qquad \text{and} \qquad (h \circ g) \circ f$$

The associativity law in the definition of a 1-category states

that these two parsing orders are equal.

• 2-categorical example: interchange law.

# Interchange Law



## Strict vs. Weak Higher Categories

« In a *strict n*-category, we can take a pasting diagram of cells and compose them in any order we like|the result will be the same. To make a *weak n*-category, we are going to 'stretch out' a strict *n* -category a bit, so that there is a bit of 'distance' in between these composites done in different orders — but not *too* much. And it shouldn't be empty space in between; they must be connected by a suitable mediating cell. »

E. Cheng and A. Lauda, *Higher-Dimensional Categories: An Illustrated Guide Book*, p.9.

## Magmas are Incoherent

**A *magma***

**— i.e. a reflexive globular set with all possible binary compositions —**

**is « incoherent »**

**in the sense that there are no laws**

**fixing the relations between different compositions**

**of the same multi-composable diagram.**

Whereas in 1-category theory,

the data+structure layers

are supplemented

by the *strict* associativity and identity laws,

in higher category theory

we want to supplement the data+structure layer (the magmas)

with (weakly invertible) higher morphisms

guaranteeing the coherence

of the multicomposable diagrams

(including as a particular case the neutrality of the putative identities).

Rather than forcing

the different possible compositions of a given diagram

to yield the same result

by imposing laws given by strict equalities

between the corresponding morphisms,

we shall add (weakly invertible) morphisms one dimension higher

in order to guarantee that different parsing orders

of a multi-composable diagram

are *structurally* equal.

Rather than endowing the structure

with strict juridical coherence,

the structure self-regulates its own coherence

by flowing upwards.

## Immanent Coherence through « Contraction »

The « space » of different parsing orders

of the same multicomposable diagram of $k$-morphisms

should be « contractible »

in the sense that any two « points » in that « space »

— i.e. any two different parsing orders

of the same multicomposable diagram —

should be related by a mediating weakly invertible $(k + 1)$-morphism.

« Historical experience suggests, however, that the most exotic mathematics of today will eventually be thought of as easy enough to teach to mathematics undergraduates in the future. It is fun to speculate, as a researcher in $\infty$-category theory, about how this subject could be simplified. In this case, there is a linguistic trick […] that could make $\infty$-categories as easy for late 21st-century undergraduates to think about as ordinary categories are today. The key axiom in an ordinary category is the existence of a unique composite transformation $g \circ f : X \to Z$ for each composable pair of transformations $f : X \to Y$ and $g : Y \to Z$, chosen from all the elements of the set of transformations from $X$ to $Z$. In contrast, in an $\infty$-category, there is a space of arrows leading from $X$ to $Z$, which in the fundamental $\infty$-groupoid can be understood as a kind of 'path space.' The correct analogue of the uniqueness of composites in an ordinary category is the assertion that in an $\infty$-category, the space of composites is 'contractible,' meaning that each of its points can be continuously collapsed via a reverse big bang to a single point of origin. »

E. Riehl, *Infinity Category Theory Offers a Bird's-Eye View of Mathematics*

The addition of these mediating morphisms

guaranteeing coherence

requires to add their own putative identities

and the new mediating morphisms

required by the new multicompossable diagrams

in which they enter.

Hence, the structure *does* flow upwards.

## Stretching or Straightening

- **Straightening (or juridical) option:**

We can impose ***coherence laws*** according to which

the different possibles compositions of a multi-composable diagram

should be strictly equal.

This yields ***Strict Higher Categories***

- **Stretching (or anarchist) option**:

The different possibles compositions

of a multi-composable diagram

should be related by morphisms one level higher (***coherators***).

This yields ***Weak Higher Categories***

# Examples of (what we shall call) « Coherators »

- Associators:

$$h \circ (g \circ f) = (h \circ g) \circ f \rightsquigarrow assoc_{h,g,f} : h \circ (g \circ f) \longrightarrow (h \circ g) \circ f)$$

- Unitors:

$$f \circ id_a = f \rightsquigarrow unitor_f : f \circ id_a \longrightarrow f$$

## Category-Theoretical Anarchist Credo

**We can stretch *strict laws* by adding higher *coherators*.**

(e.g. associators, unitors, pentagonators, federators,

mutualisers, mediators, kidtors, cancelators, etc.)

In Hegelian terms,

the *static laws* that guarantee coherence

are replaced by *mediating processes*.

**Category-Theoretical « Background Independence »**

Given the Data-Structure-Laws trichotomy,

the category theoretical anarchist tendency

is to substitute the laws by higher dimensional structure.

The problem is that we need to introduce

further laws to rule this new structure.

Is there a fixed point to this process

in which the structure becomes fully self-regulated in a lawless manner?

## Revisiting isomorphisms

In 1-categories, the definition of an isomorphism

$$f : a \to b$$

is given by strict equalities

$$f^{-1} \circ f = id_a \quad \text{and} \quad f \circ f^{-1} = id_b.$$

In higher category theory,

this strict equalities can be stretched.

## Stretching Isomorphisms Further and Further

**The correct notion of « sameness » in a higher category**

**can be obtained by maximally stretching**

**the 1-categorical notion of isomorphism.**

**Two objects $a$ and $b$ in a higher category are « equal »**

**if there is a morphism $f : a \to b$**

**that is *invertible in the weakest possible sense* of the term.**

## Towards Maximally Stretched Equalities

• In 0-categories (sets), strict equalities are the *maximally stretched notion of sameness* available (since there are no 1-morphisms to stretch them).

• In 1-categories, isomorphisms are the *maximally stretched notion of sameness* available.

• Indeed, the definition of isos

$$f^{-1} \circ f = id_a$$

uses a strict equality $=$ between $f^{-1} \circ f$ and $id_a$ because we do not have 2-isos to stretch further these strict equalities in something like

$$f^{-1} \circ f \simeq id_a$$

• But in higher categories we do have higher isomorphisms to stretch further.

**The higher the category,**

**the more higher « ligaments » we have**

**to stretch equalities and identities**

**more and more.**

## Maximally Stretched Equalities in 2-Categories

- In a 2-category, we now have a notion of 2-isos to stretch the strict equalities

$$f^{-1} \circ f = id_a$$

used to define 1-isos.

- Hence, a maximally stretched equality between two objects in a 2-category is given by a 1-morphism

$$f : a \to b$$

such that there exists a 1-morphism

$$f^{-1} : b \to a$$

such that there exists a 2-iso

$$\alpha : f^{-1} \circ f \to id_a$$

with inverse

$$\alpha^{-1} : id_a \to f^{-1} \circ f$$

such that

$$\alpha^{-1} \circ \alpha = id_{f^{-1} \circ f} \text{ and } \alpha \circ \alpha^{-1} = id_{id_a}$$

**Note**: these last strict equalities between 2-morphism cannot be stretched since in order to do that we would need 3 -morphisms.

**Higher Principle of the Indiscernibility**

**of Maximally Stretched Equals**

The structural properties of an object in a higher category

are the properties that are invariant under

the maximally stretched notion of equality available.

# The Conceptual Kernel of Category Theory

- The ***Stretching Equality***:




- ***Principle of the Indiscernibility of Isomorphic Objects***:

The structural properties of an object

are the properties that are invariant under isomorphisms.

## The Categorical Imperative

I. Think relationally, i.e. in terms of morphisms.

II. Consider the referents of mathematical discourse up to isomorphism.

> II.a In particular, their admissible properties are the structural or isomorphism-invariant properties.

# Universal Constructions

# Through

# Limits and Colimits

Mathematical formalization

of fundamental affordances and categories of rational thinking

**Order structures $\Longleftrightarrow$ Partially Ordered Sets**

**Abstraction $\Longleftrightarrow$ Equivalence Relations**

**Identity $\Longleftrightarrow$ Groups**

**Determination (attribution of properties) $\Longleftrightarrow$ Invariance under isomorphisms**

We shall now enrich the list

of affordances and categories of rational thinking

two introducing two further operations:

♣ Restrictions (« cutting out ») through the imposition of compatibility conditions:

**Limits** (of a diagram in a category)

♥ Constructions of new objects by « gluing » existing ones together:

**Colimits (of a diagram in a category)**

Depending of the diagram being considered,

the dual notions of limit and colimit

take the form of the following dual constructions:

- **Terminal and initial objects.**
- **Products and coproducts.**
- **Domains and codomains.**
- **Pullbacks and pushouts.**
- **Equalizers and coequalizers.**

Roughly speaking,

***limits*** can be understood as ***generalized products***

and ***colimits*** as ***generalized additions***.

Hence, we shall explore the varieties

of ***categorical multiplication***

and ***categorical addition***.

## Concrete Universals in Category Theory

In category theory, we are interested in objects

satisfying a given diagrammatic property.

A ***concrete universal*** for that property satisfies the following conditions:

- it is an object of the category that has the property at stake (it is a *concrete* universal),

- every other object having the property « participates » of the universal object through a morphism to or from it (it is a concrete *universal*).

## Abstract vs Concrete Universals

In pictorial terms, we could say that an object is white

either because it « participates » in the abstract universal called *whiteness*

(« vertical » abstract universality)

or because it is related to a paradigmatic white object,

a concrete universal that canonically incarnates whiteness

(« horizontal » concrete universality).

« […] there are two types of universals, the non-self-participating, vertical, or 'abstract' universals of set theory and the self-participating, horizontal, or 'concrete' universals which are given by the universal mapping properties of category theory […]. The abstract or non-self-participating universals of set theory collect together instances of a property but involve no machinery about the determination of the instances having the property. In contrast, a self-participating universal has the property itself and every other instance of the property has it by participating in (e.g., uniquely factoring through) the universal. The determination that the other instances have the property 'flows through' the universal to the instances. »

D. Ellerman, *A Theory of Adjoint Functors — with some Thoughts about their Philosophical Significance*, 2005.

« For the concrete universals of category theory, the *participation relation* is the *uniquely-factors-through* relation »

D.P. Ellerman, *Category Theory and Concrete Universals*, Erkenntnis (1975-), Vol. 28, No. 3 (May, 1988), pp. 409-429.

In the present case,

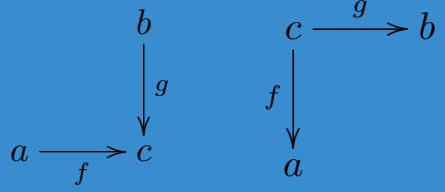
***limits*** and ***colimits***

are objects in a category that entertain

a universal relation with respect to a given diagram,

that of being a ***universal cone*** over the diagram (limits)

or a ***universal cocone*** under the diagram (colimis).

| | Limits | Colimits |
|---|---|---|
| **Empty diagram** | Terminal Objects | Initial Objects |
| $a \qquad b$ | Products | Coproducts |
| $a \to b$ | Domain | Codomain |
|  | Pullbacks | Pushouts |
| $A \rightrightarrows B$ | Equalizers | Coequalizers |

Up to now, we have considered relations

— given by morphisms —

between objects.

We shall now consider relations

between a single object on the one hand

and a whole diagram $D$ of the same category on the other.

If the diagram $D$ is just a collection of objects $a_i$ with no arrows

— besides the trivial identities $id_{a_i}$—,

then a « relation » between an object $c$ and the whole diagram $D$

is just a collection of arrows

$$c \to a_i$$

to all the objects of the diagram $D$.

If the diagram $D$ has more arrows

besides the trivial identities $id_{a_i}$,

a relation between $c$ and the whole diagram $D$

will have to be compatible with

the relations between the objects $a_i$ in $D$.

## Cones

Given a diagram $D$ in a category $\mathscr{C}$,

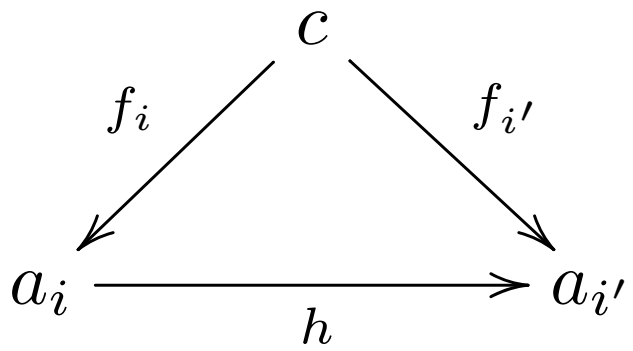
a ***cone*** over the diagram $D$ is an object $c$ in $\mathscr{C}$

(called *vertex* of the cone)

equipped with morphisms $f_i : c \to a_i$ to every object $a_i$ in $D$

(called *legs* of the cone)

such that the following triangles commute




for every arrow $a_i \to a_{i'}$ in the diagram $D$.

## Limits

Now, there might be many different objects

that entertain such relations to a diagram *D*.

A ***universal cone*** — called the ***limit*** of the diagram *D* and denoted *lim*(*D*) —

is a cone from which

« we see the diagram as closely as possible » (A. Badiou, *op.cit.*, p. 27)

in the sense that every other cone over to *D*

will be factored through or mediated by the universal one.

The limit of a diagram *D* is the ***concrete universal***

for the property « being a cone over the diagram *D* ».

## Limits Formally Defined

A ***limit*** of a diagram $D$ in a category $\mathscr{C}$ is a ***universal cone*** over $D$,

that is a cone over $D$ with vertex denoted $lim(D)$

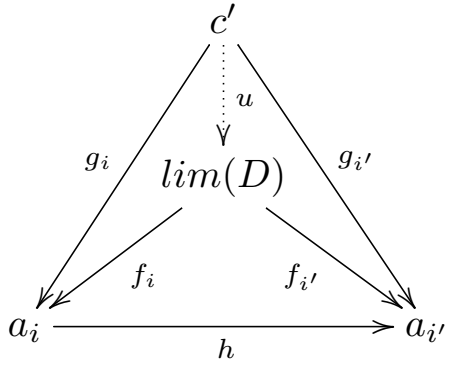
such that for any other cone over $D$ with vertex $c'$

there is a ***unique classifying*** or ***mediating morphism*** $u : c' \to lim(D)$

such that the legs $g_i : c' \to a_i$ in the cone over $D$ with vertex $c'$

factorize uniquely through $u : c' \to lim(D)$

and the legs $f_i : lim(D) \to a_i$ in the cone with vertex $c$:




$$g_i = f_i \circ u$$

## Cocones

Analogously, a ***cocone*** under a diagram $D$

is an object $c$ in $\mathscr{C}$ that is the target of a « relation »

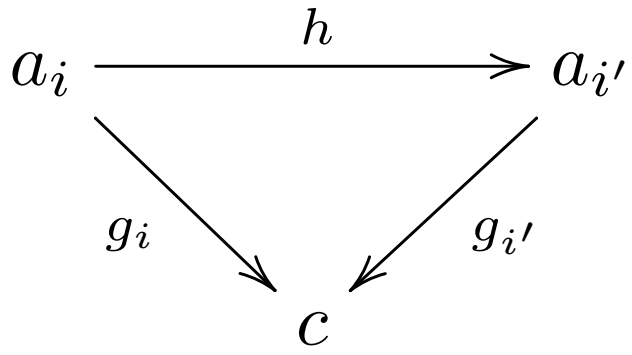
whose source is a whole diagram $D$.

Given a diagram $D$ in a category $\mathscr{C}$,

a ***cocone*** under the diagram $D$ is an object $c$ in $\mathscr{C}$ (called the *nadir* of the cone)

equipped with morphisms $g_i : a_i \to c$ from every object $a_i$ in the diagram $D$

such that the following triangles commute for every morphism $h : a_i \to a_{i'}$ in the diagram



# Colimits

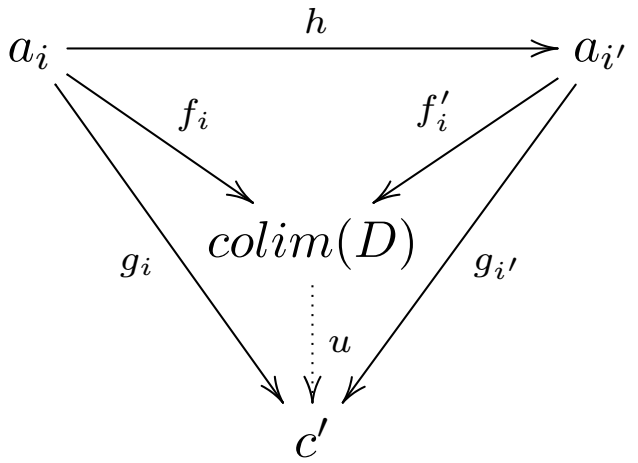
A ***colimit*** of a diagram *D*

is a ***universal cocone*** under *D*

i.e. a cocone with nadir denoted *colim*(*D*)

that has the universal property

of mediating or classifying co-cones under the diagram *D*:




$$g_i = u \circ f_i$$

# Initial and Terminal Objects

## Terminal Objects as Limits Over the Empty Diagram

The limit $lim(D_{\varnothing})$ of the empty diagram $D_{\varnothing}$ in a category $\mathscr{C}$

— called ***terminal object*** and denoted 1 —

is a universal cone over $D_{\varnothing}$,

that is an object equipped with commuting morphisms to all the objects in $D_{\varnothing}$

(that is, equipped with no morphisms at all since there are no objects in $D_{\varnothing}$)

such that any other object $c'$ in $\mathscr{C}$ satisfying the same condition (i.e. any other object in $\mathscr{C}$)

is equipped with a unique morphism $c' \to 1$ to 1.

In other terms, an object 1 in a category $\mathscr{C}$ is a ***terminal object***

if for every other object $c'$ in $\mathscr{C}$ there is a *unique* morphism $c' \to 1$.

**The terminal object $1$ of a category**

**— if it exists —**

**is unique up to (a unique) isomorphism.**

Demonstration:

If $1$ and $1'$ are terminal objects, we have unique morphisms

$f : 1 \to 1'$ (since $1'$ is terminal)

and

$g : 1' \to 1$ (since $1$ is terminal).

Since the unique endomorphisms of $1$ and $1'$ are their trivial identities (because they are both terminal),

we have

$$g \circ f = id_1 \text{ and } f \circ g = id_{1'}$$

that is, $f$ and $g$ are inverses and $1$ and $1'$ are isomorphic.

## Examples of Terminal Objects

• In *Set*, every singleton $\{ \star \}$ (they are all isomorphic) is a terminal object.

• In a preorder $(A, \leqslant)$, every maximal element, i.e. every element satisfying $a \leqslant 1$ for every object $a$, is a terminal object (note that there can be many of them).

• In a poset $(A, \leqslant)$, a maximal element $0 \leqslant a$ is necessarily unique, the *maximum*. The poset $(\mathbb{Z}, \leqslant)$ has no terminal object.

In *Set*, a singleton $\{ \star \}$ has no internal structure

that could be used to define different functions

$$X \to \{ \star \}.$$

Reciprocally, since $\{ \star \}$ has no internal structure,

each function of the form

$$\{ \star \} \to X$$

Selects an element of $X$,

i.e. a part of $X$ with no internal structure.

In *Set*, an element $x \in X$ can be described

by means of a « characteristic morphism »

$$\psi_x : \{ \star \} \to X$$

— whose source is the terminal object $1 = \{ \star \}$ —,

defined by

$$\psi_x( \star ) = x.$$

Conceptually, the element $x \in X$ is individuated

by « pointing » to it by means

of the $\{ \star \}$-shaped « arrow » $\psi_x$.

The existence of a bijection of sets

$$X \simeq Hom_{Set}(1,X)$$

means that a set $X$ can be completely described

by « probing » or « testing » it uniquely

with { ★ }-shaped « arrows ».

## Terminal Objects Provide Pointed Arrows

In a general category $\mathscr{C}$ with a terminal object 1,

morphisms of the form

$$1 \to a$$

provide the category-theoretical generalization

of the set-theoretic notion

of « element of a set ».

## Initial Objects as Colimits Over the Empty Diagram

The colimit of the empty diagram $D_\varnothing$ in a category $\mathscr{C}$

— called ***initial object*** and denoted $0$ —

is a universal cocone under $D_\varnothing$,

that is an object equipped with commuting morphisms from all the objects in $D_\varnothing$

(i.e. equipped with no morphisms at all since there are no objects in $D_\varnothing$)

such that any other object $c'$ in $\mathscr{C}$ satisfying the same condition (i.e. any other object in $\mathscr{C}$)

is equipped with a unique morphism $0 \to c'$ to $0$.

In other terms, an object $0$ in a category $\mathscr{C}$ is an ***initial object***

if for every object $a$ in $\mathscr{C}$ there is a *unique* morphism $0 \to a$.

## Examples of Initial Objects

- In *Set*, the empty set $\emptyset$ is an initial object.

- In a preorder $(A, \leqslant)$, every minimal element $0 \leqslant a$ for every object $a$ is an initial object (note that there can be many of them).

- In a partially ordered set $(A, \leqslant)$, a minimal element $0 \leqslant a$ is necessarily unique, the *minimum*.

# Products and Coproducts

***Limits*** and ***colimits*** can be understood

as generalized ***Cartesian products*** and ***disjoint sums*** of sets

submitted to further constraints

encoded in the morphisms of the corresponding diagram.

These further conditions

select subsets of the Cartesian product for limits

and define quotients of the disjoint union for colimits.

The particular cases given by diagrams without morphisms

yield the categorical notions of *products* and *coproducts*.

# Cartesian Product of Sets

The Cartesian product of two sets $A$ and $B$

*pairies* elements of $A$ with elements of $B$.

For example, the product $A \times B$

of the sets $A = \{1,2,3,4\}$ and $B = \{\clubsuit, \heartsuit\}$

is the set of ordered pairs :

$$A \times B = \{(1,\clubsuit), (1,\heartsuit), (2,\clubsuit), (2,\heartsuit), (3,\clubsuit), (3,\heartsuit), (4,\clubsuit), (4,\heartsuit)\}$$

## Canonical Projections of the Cartesian Product

There are two canonical projections

$$\pi_A : A \times B \to A$$

$$(a, b) \mapsto a$$

— defining $A \times B$ as a « fibration » over the base $A$ with fiber $B$ —

and

$$\pi_B : A \times B \to B$$

$$(a, b) \mapsto b$$

— defining $A \times B$ as a « fibration » over the base B with fiber $A$ —.

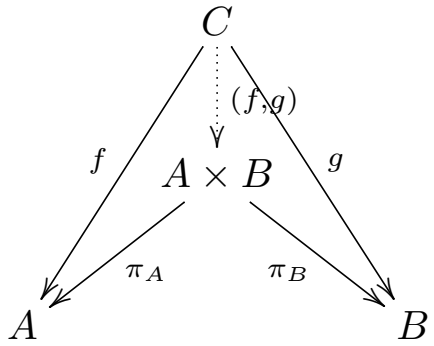
**Universal Property of the Cartesian Product**

Given a third set $C$ equipped with two functions

$$f : C \to A \text{ and } g : C \to B \text{ to } A \text{ and } B,$$

there is a unique factorization

through the Cartesian product $A \times B$




where the mediating morphism $(f, g) : C \to A \times B$ is given by $c \mapsto (f(c), g(c))$

and $(\pi_A \circ (f, g))(c) = \pi_A(f(c), g(c)) = f(c) \in A$ for $c \in C$.

# Categorical Products

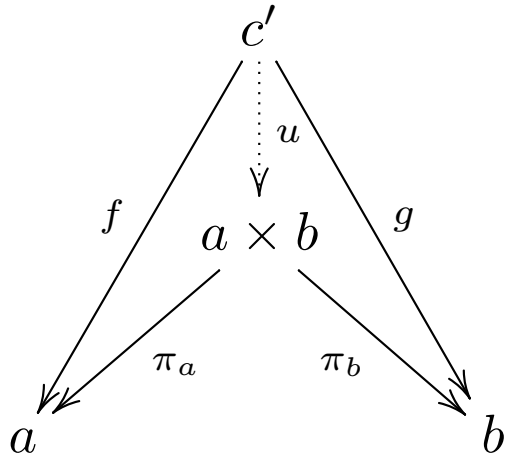
In a category $\mathscr{C}$ with products,

the product of two objects $a$ and $b$ is the limit

— denoted $a \times b$ —

of the diagram given by two objects $a$ and $b$

with no morphisms (besides the trivial identities).



A category does not necessarily have products.

Example: the discrete two-object category:



## Examples of Categorical Products

**Products $p \times q$ in a categorical preorder** $\mathscr{C}_{(A,\leqslant)}$ are given

by the ***greatest lower bound*** of $p$ and $q$,

i.e. $p \times q \leqslant p$, $p \times q \leqslant q$, and $c \leqslant p \times q$ if $c \leqslant a, b$.

**Products in a categorical deductive system**

— where $p \to q$ is a proof

that the proposition $p$ implies the proposition $q$ (i.e. a deduction) —

are given by the **logical conjunction « and »**.

## The Power of Categorical Abstraction

***Cartesian products*** of sets,

***greatest lower bound*** in a preorder,

and the ***logical conjunction « and »***

are all particular cases of the category-theoretical notion

of limit over a diagram given by two unrelated objects.

# Disjoint Unions of Sets

Given the sets $A = \{1,2,3\}$ and $B = \{3,4,5\}$,

their ***disjoint union*** is the set

$$A \coprod B = \{1,2,3_A,3_B,4,5\}$$

obtained by « adding » the two sets $A$ and $B$ « side by side »

(without conflating common elements if any).

## Canonical Injections into the Disjoint Union of Sets

Whereas a Cartesian product of sets $A \times B$ is equipped

with two canonical projections

$$\pi_A : A \times B \to A \quad \text{and} \quad \pi_B : A \times B \to B$$

the disjoint union of sets $A \coprod B$ is equipped

with two canonical injections

$$i_a : A \hookrightarrow A \coprod B$$

$$a \mapsto a_A$$

$$i_B : B \hookrightarrow A \coprod B$$

$$b \mapsto b_B$$

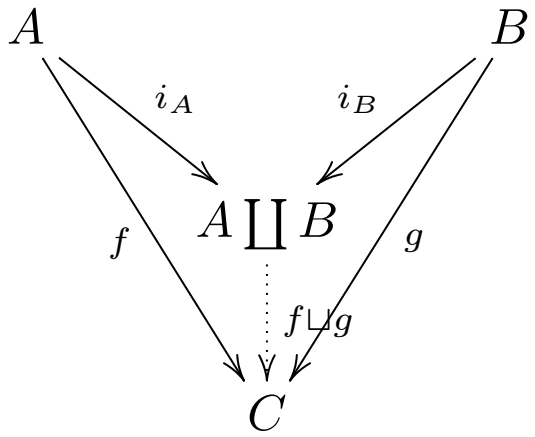
## Universal Property of the Disjoint Union

Given a set $C$ with functions $f : A \to C$ and $g : B \to C$,

there is a unique mediating function

$$f \sqcup g : A \coprod B \to C$$

such that the following diagrams commutes:




where $(f \sqcup g)(x) = f(x)$ if $x \in A$ and $(f \sqcup g)(x) = g(x)$ if $x \in B$.

# Categorical Coproducts as Colimits

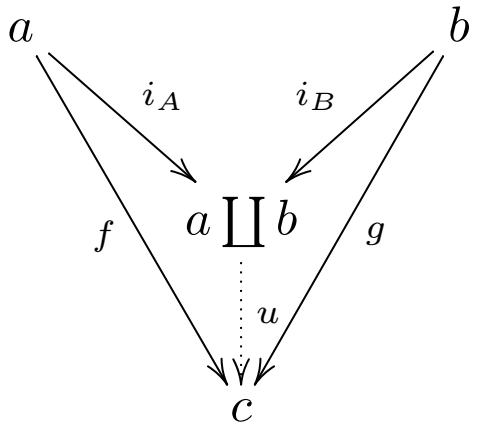
In a category $\mathscr{C}$ with coproducts,

the coproduct of two objects $a$ and $b$

is an object denoted $a \coprod b$

given by the colimit of the diagram

defined by two unrelated objects $a$ and $b$



## Examples of Categorical Coproducts

**Coproducts in a categorical preorder** $\mathscr{C}_{(A,\leqslant)}$

are given by the **least upper bound** of the corresponding objects.

**Coproducts in a categorical deductive system**

are given by the **logical disjunction « or »**.

In general, a ***limit*** can be understood

as a ***constrained product***,

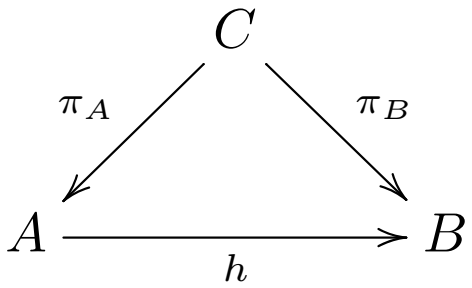
that is, as a product between the objects in the diagram

constrained by the conditions

encoded in the morphisms in the diagram.

## Domains as Limits

The limit in *Set* of the diagram $h : A \to B$ is a universal cone of the form




The elements of the set $C$ are pairs $(x, y)$ with $x \in A$ and $y \in B$

subjected to the condition $h(\pi_A(x, y)) = h(x) = \pi_B(x, y) = y$,

i.e. by pairs $(x, h(x))$, which (given the function $h$) are completely determined by $x \in A$.

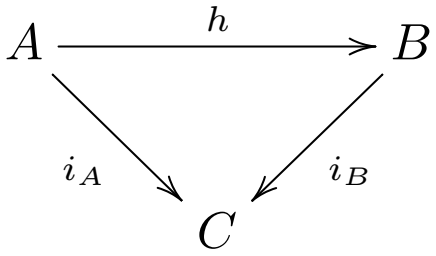
Hence, $C \simeq A$, i.e. the limit of the diagram $h : A \to B$ is the domain $A$ of the function $h$.

Conceptually, the subset of the Cartesian product $A \times B$

given by the graph of the function $h : A \to B$ is in bijection with the domain $A$.

## Codomains as Colimits

The colimit in *Set* of the diagram $h : A \to B$ is a universal cocone of the form




The elements of $C$ are given by the disjoint union of $A$ and $B$

constrained by the condition $i_B(h(x)) = h(x) = i_A(x) = x$, i.e. quotiened by the identification $x \sim h(x)$.

Hence, $C$ is the disjoint union of $A$ and $B$ where each element $x \in A$ is identified with its image $h(x)$.

Since $h$ might not be injective, different elements in $A$

might be sent (and be identified with) to the same element in $B$.

After all the elements in $A$ are identified with their images in $B$, we are left just with a copy of $B$.

All in all, $C \simeq B$,

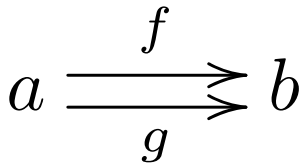
i.e. the colimit of the diagram $h : A \to B$ is the codomain $B$ of $h$.

# Equalizers and Coequalizers

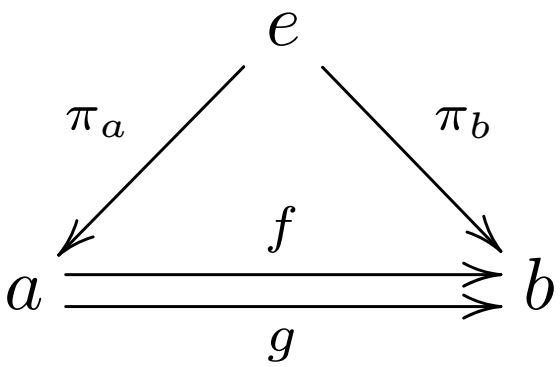
# Equalizers as Limits

An ***equalizer*** is the limit

of a diagram of the form




that is a universal cone of the form



## Equalizers in Set

In *Set*, the equalizer of a diagram

$$A \underset{g}{\overset{f}{\rightrightarrows}} B$$

is a set $E = \{(x, y)\}$ equipped

with maps $\pi_A : E \to A$ and $\pi_B : E \to B$ constrained by the conditions

- $f(\pi_A(x, y)) = f(x) = \pi_B(x, y) = y$,
- $g(\pi_A(x, y)) = g(x) = \pi_B(x, y) = y$,

which implies $y = f(x) = g(x)$.

Hence, the second component $y$ is completely fixed by the first one $x$

and the first one is restricted to the elements $x \in A$ such that $f(x) = g(x)$,

i.e. to the elements that « equalize » $f$ and $g$.

The equalizer $E$ of the parallel morphisms

$$A \underset{g}{\overset{f}{\rightrightarrows}} B$$

is the subset $E \subset A$ containing the elements on which $f$ and $g$ yield the same value.

This means that $\pi_A : E \to A$ is an inclusion, i.e. a monomorphism.

In a general category, every equalizer is a monomorphism,

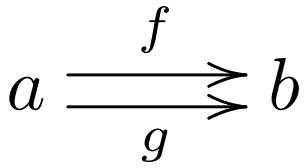
that is a kind of generalized subset

(cf. Awodey, *op.cit.*, Prop. 3.15, p.56).

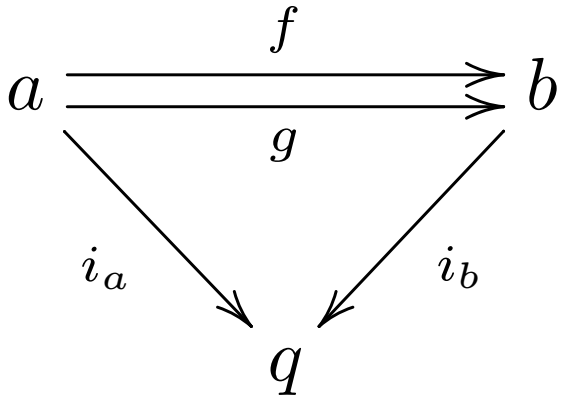
## Coequalizers as Colimits

A ***coequalizer*** is the colimit

of a diagram of the form




that is, a universal cocone of the form

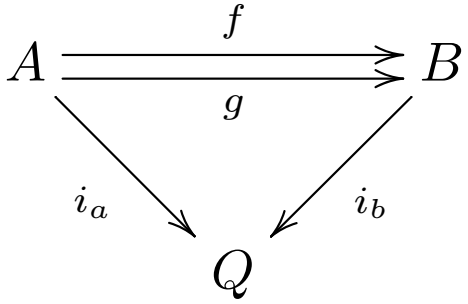


# Coequalizers in Set

In *Set*, the coequalizer of a diagram

$$A \underset{g}{\overset{f}{\rightrightarrows}} B$$

is given by a universal cocone of the form




where the set $Q$ is given by the disjoint union of $A$ and $B$ subjected to the conditions:

$$i_A(x) = x = i_B(f(x)) = f(x) = i_B(g(x)) = g(x).$$

This yields the codomain B (as it was the case for the colimit of a single function $A \to B$ )

but now quotiened by the equivalence relation $f(x) \sim g(x)$ for each $x \in A$:

Conceptually, $Q = B/\sim$ is the quotient set of $B$ obtained by identifying/gluing

the images $f(x)$ and $g(x)$ for each $x \in A$.

A coequalizer

of a pair of parallel arrows

is an epimorphism.

(cf. Awodey, *op.cit.*, Prop. 3.18, p.58)

# Pullbacks and Pushouts

## Pullbacks or Fiber Products of Sets

The limit of the following diagram in *Set*

$$\begin{array}{ccc} & & B \\ & & \downarrow{\scriptstyle g} \\ A & \xrightarrow[f]{} & C \end{array}$$

is given by a set denoted $A \times_C B = \{(x, y)\}$ with $x \in A$ and $y \in B$

that is the source of the following morphisms

$$\begin{array}{ccc} A \times_C B & \xrightarrow{\pi_B} & B \\ \downarrow{\scriptstyle \pi_A} & \searrow^{\pi_C} & \downarrow{\scriptstyle g} \\ A & \xrightarrow[f]{} & C \end{array}$$

subjected to the conditions $g(\pi_B(x, y)) = g(y) = \pi_C(x, y)$ and$f(\pi_A(x, y)) = f(x) = \pi_C(x, y)$,

which implies $f(x) = g(y) \in C$. All in all,

$$A \times_C B = \{(x, y)/f(x) = g(y) \in C\}$$

## Why pullbacks are called « pullbacks »?

If $g : B \to C$ is understood as a ***fiber bundle*** over the base $C$,

where the fiber over $c \in C$ is given by $g^{-1}(c) \subset B$,

$A \times_C B \to A$ can be understood as a fiber bundle over the base $A$

obtained by « pulling back »

the fibration $g : B \to C$ over the base $C$ « along » $f$:

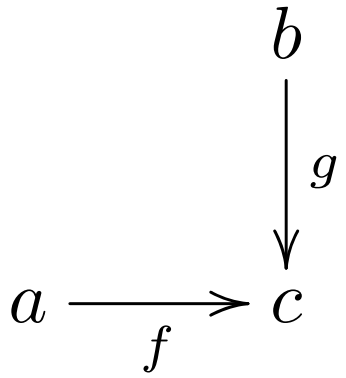
$$\begin{array}{ccc} A \times_C B & \xrightarrow{\pi_B} & B \\ {\scriptstyle \pi_A}\downarrow & & \downarrow{\scriptstyle g} \\ A & \xrightarrow[f]{} & C \end{array}$$

This means that the fiber of $A \times_C B \to A$ over $x \in A$

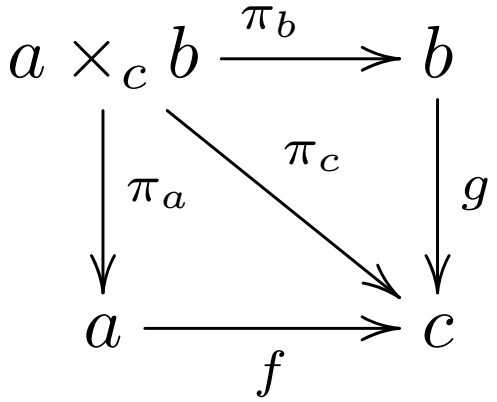
is isomorphic to the fiber of $g : B \to C$ over $f(x) \in C$, i.e. by $g^{-1}(f(x))$.

# Pullbacks or Fiber Products

In a general category (with pullbacks), the limit of the diagram

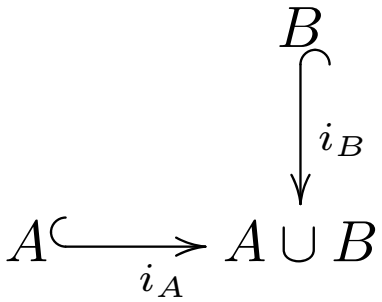



is given by a universal cone of the form

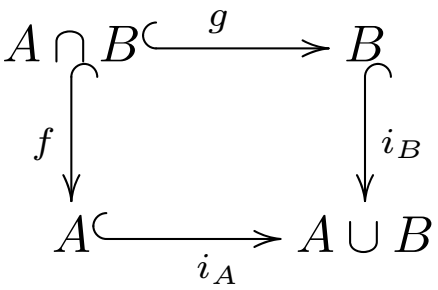


**Intersections as Pullbacks**

The pullback of the diagram




is the intersection $A \cap B$



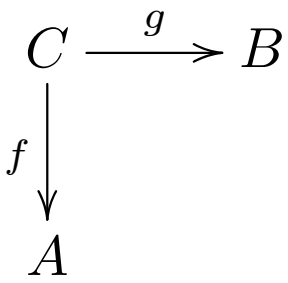

**The intersection $A \cap B = A \times_{A \cup B} B$ results from the Cartesian product $A \times B$ by**

**only keeping the ordered pairs of elements in $A$ and $B$**

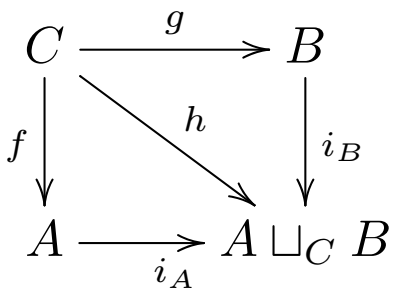
**that are injected into the same element of $A \cup B$.**

**Pushouts or Amalgamated Sums in *Set***

The colimit of the following diagram in *Set*




is given by a set denoted $A \sqcup_C B$ that is the target of the following morphisms




The condition that $A \sqcup_C B$ is a cocone means that

$(i_B \circ g)(c) = h(c)$ and $(i_A \circ f)(c) = h(c)$, which imply $(i_B \circ g)(c) = (i_A \circ f)(c)$,

Hence, the set $A \sqcup_C B$ can be understood as the disjoint union of $A$ and $B$

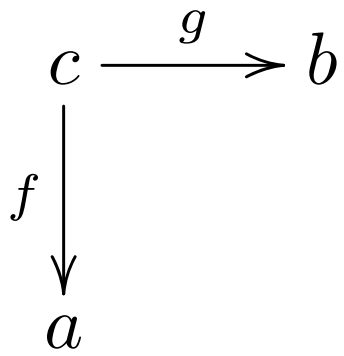
where $a \in A$ and $b \in B$ are *identified*, *amalgamated* or *glued*

if they are in the image of the same element in $C$ through $f$ and $g$, that is if $a = f(c)$ and $b = g(c)$.

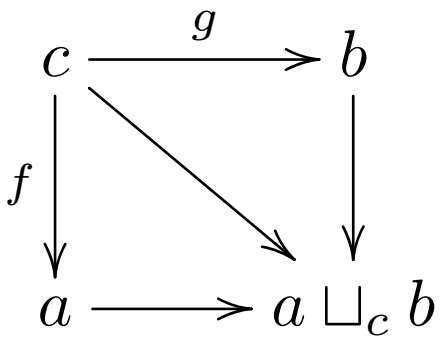
## Pushouts or Amalgamated Sums

In a genera category (with upshots),

the colimit of the following diagram

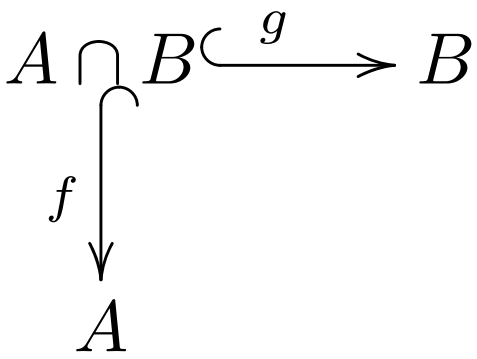



is given by a universal cocone of the form

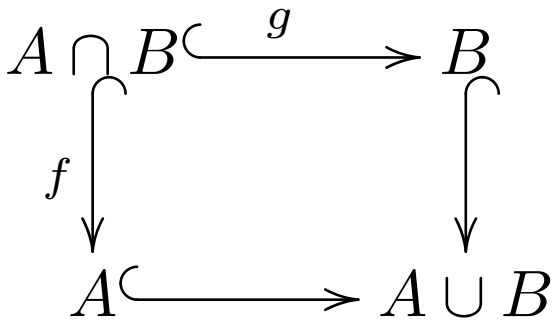


## Unions as Pushouts

The pushout of the diagram




is



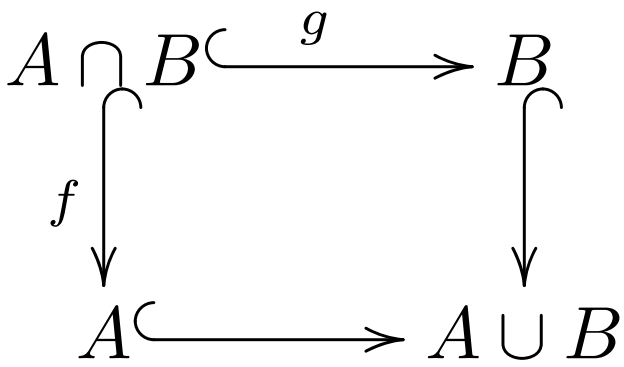

**The union $A \cup B = A \sqcup_{A \cap B} B$ results from the disjoint union $A \sqcup B$ by**

**« gluing » elements of $A$ and $B$**

**when they correspond to the same elements in $A \cap B$.**

## An Exact Square

The square




is both a pullback $A \cap B = A \times_{A \cup B} B$

and a pushout $A \cup B = A \sqcup_{A \cap B} B$.

Such squares are called ***exact***.

The projection from $A$ and $B$ to the « quotient set » $A \cup B$

is obtained by gluing $A$ and $B$ *exactly* along their overlap, i.e. $A \cap B$.

# Functors

## Why Functors? To consider Categories as Mathematical Objects

According to the contextual principle,

any mathematical object lives in category

understood as a relational context.

If we want to understand categories themselves as mathematical objects,

they must also be objects of a category.

In order for this ***category of categories*** to exist

we need morphisms between categories.

This is given by the notion of ***functor***

(Note: Mac Lane took the term *functor* from Carnap).

« One of the lessons of category theory is that whenever we meet a new type of mathematical object, we should always ask whether there is a sensible notion of ‘map’ between such objects. We can ask this about categories themselves. The answer is yes, and a map between categories is called a functor. »

T. Leinster, *Basic Category Theory*, p.17.

## Functors Translate Between Different Ontological Contexts

Functors can be understood

as *translators* or *bridges* between ontological contexts.

These bridges allow us to recontextualize mathematical objects,

that is, to represent certain mathematical objects

by objects of a different ontological kind.

This is in fact the historical origin of category theory:

establishing translation functors

between geometric objects (typically topological spaces)

and algebraic objects (typically homotopy groups).

« Mathematics is full of constructions that translate mathematical objects of one kind into objects of another kind. A construction that converts the objects in one category into objects in another category is *functorial* if it can be extended to a mapping on morphisms in such a way that composites and identity morphisms are preserved. Such constructions define morphisms between categories, called *functors*. »

E. Riehl, *Category Theory in Context*, p.xii.

## Definition of *Covariant* Functors

A ***covariant functor*** $F : \mathscr{C} \to \mathscr{D}$ is a mapping between categories that associates

- to each object $a$ in $\mathscr{C}$ an object

$$F(a) \text{ in } \mathscr{D},$$

- to each morphism $f : a \to b$ in $\mathscr{C}$ a morphism

$$F(f) : F(a) \to F(b) \text{ in } \mathscr{D}$$

such that

♠ Identities are preserved:

$$F(id_a) = id_{F(a)} \text{ for every object } a$$

♥ Compositions are preserved:

$$F(g \circ f) = F(g) \circ F(f)$$

## Definition of *Contravariant* (or arrow-reversing) Functors

A ***contravariant functor*** $F : \mathscr{C} \to \mathscr{D}$ is a mapping that associates

- to each object $a$ in $\mathscr{C}$ an object

$$F(a) \text{ in } \mathscr{D},$$

- to each morphism $f : a \to b$ in $\mathscr{C}$ a morphism

$$F(f) : F(b) \to F(a) \text{ in } \mathscr{D}$$

such that

♠ Identities are preserved:

$$F(id_a) = id_{F(a)} \text{ for every object } a$$

♥ Compositions are preserved:

$$F(g \circ f) = F(f) \circ F(g)$$

for compostable morphisms $f : a \to b$ and $g : b \to c$ in $\mathscr{C}$.

**Note**

Contravariant functors

$$F : \mathscr{C} \to \mathscr{D}$$

are sometimes described as covariant functors

$$F : \mathscr{C}^{op} \to \mathscr{D}$$

where $\mathscr{C}^{op}$ is the category dual to $\mathscr{C}$,

i.e. the category obtained by *formally* inverting the morphisms in $\mathscr{C}$.

## Properties of Functors

♦ Functors preserve commutativity of diagrams.

♣ Functors preserve isomorphisms:

If

$$f : a \to b$$

is an isomorphism in $\mathscr{C}$, then

$$F(f) : F(a) \to F(b)$$

is an isomorphism in $\mathscr{D}$.

Note: the converse is not necessarily true, $F(a) \simeq F(b)$ does not necessarily imply $a \simeq b$.

# Examples of Functors

## Functors between preorders

Given two preorders

$(X, \leqslant)$ and $(Y, \leqslant)$

a covariant functor

$$F : \mathscr{C}_{(X,\leqslant)} \to \mathscr{C}_{(Y,\leqslant)}$$

is an order preserving function

$$x \leqslant x' \Rightarrow F(x) \leqslant F(x')$$

## Covariant Hom-Functors

Given an object $a$ of a category $\mathscr{C}$,

we can define the following covariant ***hom-functor***

$$Hom_{\mathscr{C}}(a, -) : \mathscr{C} \to Set$$

where

$$b \mapsto Hom_{\mathscr{C}}(a, b)$$

and

$$(g : b \to b') \mapsto ((h : a \to b) \mapsto g \circ h : a \to b')$$

## Contravariant Hom-Functors

Given an object $a$ of a category $\mathscr{C}$,

we can define the following contravariant ***hom-functor***

— also known as ***functor of points*** of $a$ —

$$Hom_{\mathscr{C}}(-, a) : \mathscr{C}^{op} \to Set$$

where

$$b \mapsto Hom_{\mathscr{C}}(b, a)$$

— i.e. the functor of points assigns to each $b$ the set of ***generalized $b$-points*** of $a$ —

and

$$(g : b \to b') \mapsto ((h : b' \to a) \mapsto h \circ g : b \to a).$$

## Hom-(bi)Functors

The previous two hom-functors

can be combined in a single covariant hom-functor

$$Hom_{\mathscr{C}}(-,-) : \mathscr{C}^{op} \times \mathscr{C} \to Set$$

where

$$(a, b) \mapsto Hom_{\mathscr{C}}(a, b)$$

and

$$(f^{op} : a \to a', g : b \to b') \mapsto ((h : a \to b) \mapsto g \circ h \circ f : a' \to b').$$

## Diagonal Functor

The covariant diagonal functor

$$\Delta(-) : \mathscr{C} \to \mathscr{C} \times \mathscr{C}$$

is given on objects by

$$\Delta(a) = (a, a)$$

and on morphisms by

$$(f : a \to b) \mapsto ((f, f) : (a, a) \to (b, b))$$

# Diagrams as Functors

A covariant functor

$$D : \mathscr{I} \to \mathscr{C}$$

from an « indexing category » $\mathscr{I}$ to $\mathscr{C}$

— encoding an abstract diagrammatic shape —

can be understood as a ***$\mathscr{I}$-shaped diagram***

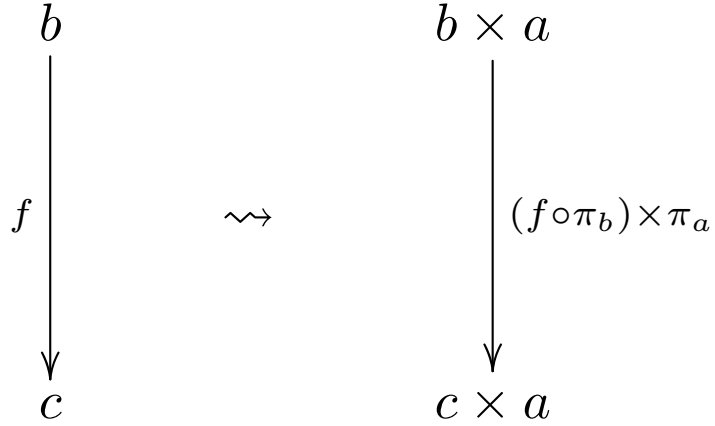
or a ***diagram of type*** $\mathscr{I}$ in $\mathscr{C}$.

## Right Product Functor

In a category $\mathscr{C}$ with products, each object $a$ in $\mathscr{C}$ defines a covariant ***right product functor***

$$(-) \times a : \mathscr{C} \to \mathscr{C}$$

is given on objects by $b \mapsto b \times a$ and on morphisms




where the morphism $(f \circ \pi_b) \times \pi_a$ is defined

by using the universal property of the product $c \times a$.

Note: Similarly, each object $a$ in $\mathscr{C}$ defines

a covariant ***left product functor*** $a \times (-) : \mathscr{C} \to \mathscr{C}$.

## Forgetful Functors

If $\mathscr{C}$ is a category of structured sets

— like *Grp* or *Top* —,

there is a (covariant) ***forgetful functor***

$$U_{\mathscr{C}} : \mathscr{C} \to Set$$

that sends each structured set in $\mathscr{C}$ to its underlying set

and each morphism between structured sets

(i.e. each structure-preserving function)

to the corresponding function between the underlying sets.

For instance,

$$U_{Grp} : Grp \to Set$$

is given by

$$(X, \bullet, e) \mapsto X$$

Since a group morphism $(X, \bullet, e) \to (Y, \bullet', e)$

is given by a function $f : X \to Y$

between the underlying sets that preserves the group structure,

the corresponding function in $Set$ is just $f$

(which explains that $U_{Grp}$ is covariant).

## Free Functors

While a ***forgetful functor*** acts on a structured set

by « forgetting » the structure,

a ***free functor*** takes a set

and constructs the most general (« free » or unconstrained)

structure of the corresponding type

(e.g. groups, rings, vector spaces, etc.).

See Example 1.2.4 in Leinster, *op.cit.*, p.19.

## Example of a Free Functor

The covariant free functor $F : Set \to Grp$ sends a set $X = \{x, y, z, \dots\}$

to the group $F(X)$ freely generated out of $X$,

where the elements of $F(X)$ are the reduced finite « words »

obtained by superposing the « letters » $x, y, z, \dots$ in $X$ and its formal inverses $x^{-1}, y^{-1}, z^{-1}, \dots,$

and then removing all sequences of the form $xx^{-1}$ and $x^{-1}x$

(e.g. the word $x^2y^{-1}yz^{-1}$ can be reduced to $x^2z^{-1}$)

The group operation is the concatenation of words and the neutral element is the empty word.

Given a function $f : X = \{x, y, z, \dots\} \to \{u, v, w, \dots\}$

the corresponding group morphism $F(f) : F(X) \to F(Y)$

is defined by substituting each letter in $X$ in a word in $F(X)$ by the letter $f(x)$ in $Y$,

thereby obtaining a word in $F(Y)$ (hence, $F$ is covariant).

## Actions of Monoids and Groups as Functors

Given a *monoid* $\mathscr{M} = \{M, \bullet, e\}$ (and less generally a *group*)

a functor $F : \mathscr{C}_{\mathscr{M}} \to Set$

— where $\mathscr{C}_{\mathscr{M}}$ is the one object category defined by $\mathscr{M}$ —

is a **left $\mathscr{M}$-set**,

i.e. a set equipped with a ***left $\mathscr{M}$-action***

(the *left-character* of the action results from the fact that $g \circ f \in M$ acts on the set $F(\bullet)$ by acting first with the function $F(f)$ and then with $F(g)$, as if — symbolically — $g \circ f$ acted on $M$ « from the left »)

Analogously, a functor $F : \mathscr{M}^{op} \to Set$ is a *right $\mathscr{M}$-action*.

Cf. Leinster, *op.cit.*, pp. 22 & 24.

# Fundamental Group Functor

The ***fundamental group functor***

$$\pi_1 : Top_* \to Grp$$

sends *pointed topological spaces* $(X, p)$

to the fundamental $\pi_1(X, p)$ group of $X$ based at $p$,

i.e. to the group of loops based at $p$ modulo homotopy.

Cf. Leinster, *op.cit.*, Example 1.2.5(a), p.21.

# Presheaves

A *Set*-valued contravariant functor

$$F : \mathscr{C}^{op} \to Set$$

is usually called a ***presheaf*** when the category $\mathscr{C}$ is

the poset $\mathcal{O}(X)$ of open subsets of a topological space $X$,

ordered by inclusion (i.e. $V \to U$ if $V \subseteq U$).

A topical example is given by

$$F(U) = \{\text{continuous functions } U \to \mathbb{R}\}$$

and $F(U) \to F(V)$ is given by the *restriction*.

# Natural Transformations

Are functors mathematical objects

in the category-theoretical sense of the term,

that is, are themselves objects of a category?

## Morphisms Between Functors

Given two categories $\mathscr{C}$ and $\mathscr{D}$,

we can define morphisms between functors

$$F, G : \mathscr{C} \to \mathscr{D}$$

known as ***natural transformations***.

Natural transformations

are bridges

between bridges

between ontological contexts.

« Categories are themselves mathematical objects, and with that in mind, it is unsurprising that there is a good notion of ‘map between categories’. Such maps are called functors. More surprising, perhaps, is the existence of a third level: we can talk about maps between functors, which are called natural transformations. These, then, are maps between maps between categories. »

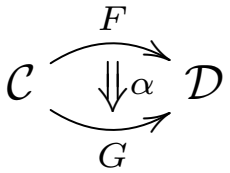
T. Leinster, *Basic Category Theory*, p.9.

**Natural Transformations Between Functors**

Given two functors between the same categories

$$F, G : \mathscr{C} \to \mathscr{D},$$

a ***natural transformation*** between them




is given by a collection of morphisms in $\mathscr{D}$

$$\eta_a : F(a) \to G(a)$$

(called the *components* of $\eta$)

indexed by the objects $a$ of $\mathscr{C}$ such that for every arrow $f : a \to b$ in $\mathscr{C}$

the following square in $\mathscr{D}$ commutes:

$$\begin{array}{ccc} a & & F(a) \xrightarrow{\eta_a} G(a) \\ f \downarrow & \qquad & F(f) \downarrow \qquad \downarrow G(f) \\ b & & F(b) \xrightarrow{\eta_b} G(b) \end{array}$$

Note that the definition of these 2-morphisms

— natural transformations—

between 1-morphisms

— functors —

between categories

essentially uses the fact that there are morphisms

within the categories.

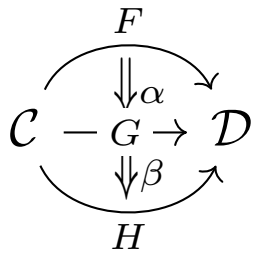
**Functors Between Two Fixed Categories Form a Category**

Composable natural transformations

$$\alpha : F \Rightarrow G \text{ and } \beta : G \Rightarrow H$$

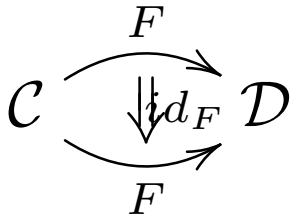
between functors between the same categories « vertically » compose to

$$\beta \circ \alpha : F \Rightarrow H$$




defined componentwise by $(\beta \circ \alpha)_a = \beta_a \circ \alpha_a$

with identity functors $id_F : F \Rightarrow F$




defined componentwise by $(id_F)_a = id_{F(a)}$.

Hence, the set of functors between two categories $\mathscr{C}$ and $\mathscr{D}$

$$\mathscr{C} \to \mathscr{D}$$

forms a $1-$category

named ***functor category***

— denoted $[\mathscr{C}, \mathscr{D}]$ —

with functors $F : \mathscr{C} \to \mathscr{D}$ as objects

and natural transformations between such functors

as morphisms.

When two functors

$$F, G : \mathscr{C} \to \mathscr{D}$$

are categorically « equal »?

## Natural Isomorphisms

Since $[\mathscr{C}, \mathscr{D}]$ is a 1-category,

the weakest notion of « structural sameness » available

is given by isomorphisms in $[\mathscr{C}, \mathscr{D}]$, which are called ***natural isomorphisms***.

The functors $F, G : \mathscr{C} \to \mathscr{D}$ are categorically equal

if there there are natural transformations (i.e. 1-morphisms in $[\mathscr{C}, \mathscr{D}]$)

$$\alpha : F \Rightarrow G \text{ and } \alpha^{-1} : G \Rightarrow F$$

such that

$$\alpha^{-1} \circ \alpha = id_F \text{ and } \alpha \circ \alpha^{-1} = id_G.$$

Since $(id_F)_a = id_{F(a)}$, the components $\alpha_a$ of the isomorphism $\alpha$

will be isomorphisms

$$\alpha_a : F(a) \to G(a).$$

# Limits Functorially
# I. Diagrams as Functors

A covariant functor

$$D : \mathscr{I} \to \mathscr{C}$$

from an « indexing category » $\mathscr{I}$ to $\mathscr{C}$

— encoding an abstract diagrammatic shape —

can be understood as a ***$\mathscr{I}$-shaped diagram***

or a ***diagram of type*** $\mathscr{I}$ in $\mathscr{C}$.

The functor category $[\mathscr{I}, \mathscr{C}]$

contains all the diagrams of shape $\mathscr{I}$ in $\mathscr{C}$.

**Limits Functorially**
**II. Constant Diagrams**

For any object $c$ in $\mathscr{C}$

we can define the ***constant diagram*** of shape $\mathscr{I}$

as the constant functor

$$\Delta_c : \mathscr{I} \to \mathscr{C}$$

given by

$$\Delta_c(i) = c$$

for every object $i$ in $\mathscr{I}$

and

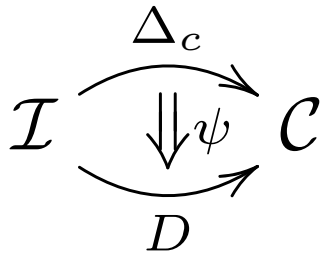
$$\Delta_c(f) = id_c$$

for every morphism $f : i \to i'$ in $\mathscr{I}$.

## Limits Functorially
## III. Cones as Natural Transformations

A ***cone*** over a diagram $D : \mathscr{I} \to \mathscr{C}$ of shape $\mathscr{I}$ in $\mathscr{C}$ with summit $c$

is a natural transformation $\Delta_c \Rightarrow D$:




i.e. a morphism in the functor category $[\mathscr{I}, \mathscr{C}]$.

**Note**: the commutative squares in the definition of a natural transformation guarantee that all the triangles from $c$ to pairs of objects in the diagram $D$ do commute.

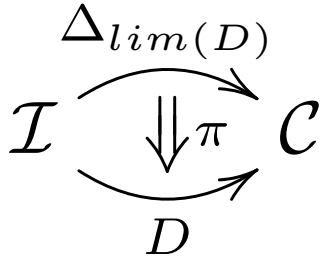
If $Cones(c, D)$ denotes the set of cones on $D$ with vertex $c$, we have

$$Cones(c, D) \simeq Hom_{[\mathscr{I},\mathscr{C}]}(\Delta(c), D).$$

# Limits Functorially
# IV. Limits as Natural Transformations

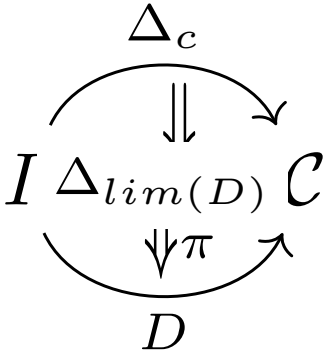
A limit for a diagram $D : \mathscr{I} \to \mathscr{C}$ is a universal cone,

that is, a natural transformation




for an object $lim(D)$ in $\mathscr{C}$ such that any other cone $\psi : \Delta_c \to D$

factorizes uniquely through $\pi$:



# Category of Categories

Are categories themselves

mathematical objects,

that is, objects of a category?

If we understand category theory

as a « *mathematical* meta-mathematics »,

then categories themselves should be understood

as mathematical objects, that is as objects in a category.

Since

***functors compose***

and

there is an ***identity functor***

$$id_{\mathscr{C}} : \mathscr{C} \to \mathscr{C}$$

for each category $\mathscr{C}$,

it seems that categories themselves

form a category *Cat* with functors as morphisms.

When two categories

are structurally equal?

What is the structural criterium of identity

for categories?

# Isomorphism of Categories

We could guess that two categories $\mathscr{C}$ and $\mathscr{D}$

are structurally « equal »

if they are isomorphic in *Cat*, i.e. if there are functors

$$F : \mathscr{C} \to \mathscr{D} \qquad \text{and} \qquad F^{-1} : \mathscr{D} \to \mathscr{C}$$

such that

$$F^{-1} \circ F = id_{\mathscr{C}} \qquad \text{and} \qquad F \circ F^{-1} = id_{\mathscr{D}}$$

On the objects of $\mathscr{C}$ and $\mathscr{D}$, the definition of isomorphism of categories

takes the form

$$(F^{-1} \circ F)(c) = c \qquad \text{for } c \text{ an object in } \mathscr{C}$$

$$(F \circ F^{-1})(d) = d \qquad \text{for } d \text{ an object in } \mathscr{D}.$$

The notion of *isomorphism of categories*

does not provide a good structural criterium of equality

for categories

since it relies on a strict equality for objects.

In other terms, the notion of *isomorphism of categories* is too strict

since it does not allow for the possibility that $(F^{-1} \circ F)(c)$ is merely isomorphic to $c$,

rather than strictly equal to it.

« […] the concept of an isomorphism between two categories is clearly seen to involve the idea of equality of objects […] Therefore, categories cannot be isomorphic; as a consequence, they cannot be objects of a category. »

M. Makkai, *Towards a Categorical Foundations of Mathematics*, 1998, pp. 162-3.

But this objection seems to contradict

the thesis according to which

the general structural criterium of equality

for objects in a category

— in the present case, the category of categories —

is given by isomorphisms.

The way out of this conundrum is to realize that

the $Hom_{Cat}(\mathscr{C}, \mathscr{D})$'s in *Cat* are not sets

as it is the case for 1-categories,

but rather 1-categories

with functors as objects and natural transformations as morphisms.

This means that *Cat* is not a 1-category,

but rather a 2-category

with objects (categories), 1-morphisms (functors)

and 2-morphisms (natural transformations).

Now, in 2-categories the weakest or maximally stretched

structural criterium of equality is not given by isomorphisms.

## Morphisms in *Cat*

- In general, the category of $n$-categories is a $(n + 1)$-category.

- In particular, the category *Cat* of 1-categories is a 2-category:

  .Objects: 1-categories.

  .1-morphisms: ***functors*** between 1-categories.

  .2-morphisms: ***natural equivalences*** between functors.

## Natural Isomorphisms

The maximally stretched notion of equality between categories in *Cat*

can be obtained by substituting the strict equalities $=$ between functors

in the definition of the notion of isomorphism of categories

$$F^{-1} \circ F = id_{\mathscr{C}} \quad \text{and} \quad F \circ F^{-1} = id_{\mathscr{D}}$$

with natural isomorphisms $\simeq$ (i.e. invertible natural transformations):

$$F^{<-1>} \circ F \simeq id_{\mathscr{C}} \quad \text{and} \quad F \circ F^{<-1>} \simeq id_{\mathscr{D}}$$

(where the notation $F^{<-1>}$ is used to stress

that this functor is not a strict inverse of $F$ but rather a *pseudo-inverse*).

**Equivalence of Categories**
**(Grothendieck, 1957)**

Two categories $\mathscr{C}$ and $\mathscr{D}$ are ***equivalent categories***

if there exist functors

$$F : \mathscr{C} \to \mathscr{D} \text{ and } F^{<-1>} : \mathscr{D} \to \mathscr{C}$$

— called ***pseudo-inverses*** of each other —

such that there are natural isomorphisms

$$\eta : id_{\mathscr{C}} \to F^{<-1>} \circ F \text{ and } \epsilon : F \circ F^{<-1>} \to id_{\mathscr{D}}.$$

« One can think of equivalence of categories as 'isomorphism up to isomorphism.' »

S. Awodey, *Category Theory*, p.148.

The natural transformation

$$\eta : id_{\mathscr{C}} \to F^{<-1>} \circ F$$

is a natural isomorphism if it has a strict inverse

$$\eta^{-1} : F^{<-1>} \circ F \to id_{\mathscr{C}}$$

such that

$$\eta^{-1} \circ \eta = id_{id_{\mathscr{C}}} \quad \text{and} \quad \eta \circ \eta^{-1} = id_{F^{<-1>} \circ F}$$

These last strict equalities

between natural transformations (2-morphisms in *Cat*)

cannot be stretched further

since there are no 3-morphisms in *Cat*.

The components $\eta_a$ of the natural isomorphism

$$\eta : id_{\mathscr{C}} \to F^{<-1>} \circ F$$

are given by isomorphisms

$$\eta_a : id_{\mathscr{C}}(a) = a \to (F^{<-1>} \circ F)(a).$$

These isomorphisms « stretch » (as we wanted) the strict equalities

$$(F^{-1} \circ F)(a) = a$$

that implement the (too strict) notion of isomorphism of categories.

This means that the composition of $F$ and its pseudo-inverse $F^{<-1>}$

lands in an object isomorphic to the initial one.

## Objection to the Foundational Pretensions of Category Theory

Category theory cannot provide new structural foundations

since it presupposes set-theoretic notions

such as *set* and *function*.

Indeed, the very definition of a category $\mathscr{C}$ proceeds by asserting

- the existence of a set (or class) of objects;
- the existence, for each pair of objects $a$ and $b$, of a set $Hom_{\mathscr{C}}(a,b)$ of morphisms;
- for every triple of objects $a, b, c$, a ***composition function***:
  - $\circ : \mathrm{Hom}_{\mathscr{C}}(a,b) \times \mathrm{Hom}_{\mathscr{C}}(b,c) \longrightarrow \mathrm{Hom}_{\mathscr{C}}(a,c).$

« […] when explaining the general notion of structure and of particular kinds of structures such as groups, rings, categories, etc. we implicitly presume as understood the ideas of operation and collection; e.g. we say that a group consists of a collection of objects together with a binary operation satisfying such and such conditions. Next, when explaining the notion of homomorphism for groups or functor for categories, etc., we must again understand the concept of operation. […] Thus at each step we must make use of the unstructured notions of operation and collection to explain the structural notions to be studied. The logical and psychological priority if not primacy of the notions of operation and collection is thus evident. »

S. Feferman, Categorical foundations and foundations of category theory', in R. E. Butts and J. Hintikka, eds., *Logic, Foundations of Mathematics and Computability Theory*. Dordrecht: Reidel, 1977, pp. 149-169.

## Categories are not Structured Sets

- Two categories with a different number of objects might still be equivalent.

- For instance,

.a category $\mathscr{C}$ with a unique object and trivial self-identity;

.a category $\mathscr{D}$ with two isomorphic objects;

are equivalent, even if they have different underlying sets of objects.

If we add objects isomorphic to some objects of category $\mathscr{C}$,

the resulting extended category is equivalent to $\mathscr{C}$.

If we remove some of the objects

in each equivalence class of isomorphic objects $\mathscr{C}$,

the resulting reduced category is still equivalent to $\mathscr{C}$.

Every category is equivalent

to a ***skeletal category***,

that is, to a category in which isomorphic objects

$$a \simeq b$$

are strictly equal

$$a = b.$$

In other terms, every category has a skeleton.

Hence, a category $\mathscr{C}$ cannot be understood

as a *structured set*

since the underlying set of objects

is not a structural property of $\mathscr{C}$,

but just an artifact of the presentation

of the corresponding categorical « structure ».

In *Set*, the criterium of equality

given by isomorphisms can be described

• either in terms of morphisms

(existence of an inverse such that the compositions give the identities)

• or element-wise as ***bijections***

— i.e. ***injective*** and ***surjective functions*** —

between sets.

Can we provide a similar description

of the functors that establish

the equivalence between two categories?

In the case of categories, a functor $F : \mathscr{C} \to \mathscr{D}$

acts on both the objects and the morphisms of $\mathscr{C}$.

Hence, we can consider the injectivity and the surjectivity

of the action of $F$ on both objects and morphisms.

We then have to consider fours notions:

- Injectivity on objects;
- Surjectivity on objects;
- Injectivity on morphisms;
- Surjectivity on morphisms.

Since the main reference of categorical discourse

are *objects up to isomorphism*,

the strict injectivity and strict surjectivity of a functor

$$F : \mathscr{C} \to \mathscr{D}$$

on the objects of $\mathscr{C}$ is not structurally relevant

in order to asses

whether $F$ defines an equivalence of categories or not.

In the example of the equivalence of categories

between the one-object category

$$\bullet$$
$$a$$

and a category with two isomorphic objects

$$x \rightleftarrows y,$$

a functor

$$F : (a) \to (x \rightleftarrows y)$$

cannot be surjective on objects

and the pseudo-inverse functor

$$F^{<-1>} : (x \rightleftarrows y) \to (a)$$

is not injective on objects.

However, the two categories are equivalent.

Hence, we have to weaken the conditions of strict injectivity and strict surjectivity on objects to (what we shall call)

***structural injectivity*** and ***structural surjectivity***.

**Essential (we would rather call is Structural) Surjectivity on Objects**

A functor $F : \mathscr{C} \to \mathscr{D}$

is ***essentially surjective***

if it is surjective on objects *up to isomorphism*,

that is, if for every object $b$ of $\mathscr{D}$,

there exists an object $a$ of $\mathscr{C}$

such that $F(a) \simeq b$ is an isomorphism in $\mathscr{D}$.

## « Essential » or Structural Injectivity on Objects

Functors $F : \mathscr{C} \to \mathscr{D}$ necessarily *preserve* isomorphisms

$$(a \simeq b) \Rightarrow (F(a) \simeq F(b))$$

but do not necessarily « reflect » them back:

$$(F(a) \simeq F(b)) \Rightarrow (a \simeq b)$$

This last non-trivial property

is the structurally weakened notion of injectivity on objects.

By contrast to the case of objects, in the framework of 1-categories,

morphisms are treated as « individuals »

since there are no 2-morphisms that could be used

to stretch the strict equalities between morphisms.

Hence, the strict injectivity and the strict surjectivity of the action of a functor

$$F : \mathscr{C} \to \mathscr{D}$$

on the morphisms in $\mathscr{C}$

is structurally relevant in order to asses

whether $\mathscr{C}$ and $\mathscr{D}$ are equivalent or not

(i.e. it cannot be « stretched » further).

## Full and Faithful Functors

A functor $F : \mathscr{C} \to \mathscr{D}$ induces a function

$$F_{X,Y} : Hom_{\mathscr{C}}(a, b) \to Hom_{\mathscr{D}}(F(a), F(b))$$

for every pair of objects $a$ and $b$ in $\mathscr{C}$.

The functor $F$ is said to be

- ***full*** if $F_{X,Y}$ is surjective.
- ***faithful*** if $F_{X,Y}$ is injective.
- ***fully faithful*** (= full and faithful) if $F_{X,Y}$ is bijective.

for each $a$ and $b$ in $\mathscr{C}$.

**Terminological note**: we would rather call these three properties of functors ***strict surjectivity/injectivity/bijectivity on morphisms***.

Full and faithful functors reflect isomorphisms

$$(F(a) \simeq F(b)) \Rightarrow (a \simeq b),$$

which is the weakened or stretched version

of the set-theoretic notion of injectivity.

**Demonstration.** Let's suppose that $f : a \to b$ is a morphism in $\mathscr{C}$ such that $F(f)$ is an iso in $\mathscr{D}$. Since $F(f)$ is an iso, it has an inverse $F(f)^{-1} : F(b) \to F(a)$. Since $F$ is full there is a morphism $g : b \to a$ in $\mathscr{C}$ such that $F(f)^{-1} = F(g)$. We have

$$F(g \circ f) = F(g) \circ F(f) = F(f)^{-1} \circ F(f) = id_{F(a)} = F(id_a),$$

that is $F(g \circ f) = F(id_a)$. Since $F$ is injective on arrows (faithful), then $F(g \circ f) = F(id_a)$ implies $g \circ f = id_a$, which means that $g$ is an inverse of $f$, i.e. that $f$ is also an isomorphism.

Hence, it is not necessary to require from a functor

$$F : \mathscr{C} \to \mathscr{D}$$

to be structurally injective on the objects of $\mathscr{C}$

as an independent condition

for $F$ to define an equivalence of categories

between $\mathscr{C}$ and $\mathscr{D}$.

## (Full) Subcategories

A ***subcategory*** $\mathcal{S}$ of a category $\mathcal{C}$

is a subclass of the objects of $\mathcal{C}$

together with, for each pairs of objects $a$ and $b$ of $\mathcal{C}$,

a subset of $Hom_{\mathcal{C}}(a, b)$ that contains identities and compositions.

A subcategory $\mathcal{S}$ of a category $\mathcal{C}$ is a ***full subcategory*** if

$$Hom_{\mathcal{S}}(a, b) = Hom_{\mathcal{C}}(a, b)$$

for all pairs of objects of $\mathcal{S}$.

Hence, a full subcategory of $\mathcal{C}$ is a collection of objects of $\mathcal{C}$

together with *all* the morphisms between them.

If

$$F : \mathscr{C} \to \mathscr{D}$$

is a full and faithful functor

(i.e. structurally bijective on morphisms)

then $\mathscr{C}$ is equivalent

to the full subcategory of $\mathscr{D}$

whose objects are of the form $F(a)$ for some $a$ in $\mathscr{C}$.

Cf. Leinster, *op.cit.*, Corollary 1.3.19, p.35.

If in addition to being ***full*** and ***faithful***

(i.e. structurally bijective on morphisms),

the functor

$$F : \mathscr{C} \to \mathscr{D}$$

is ***essentially surjective***

(i.e. structurally surjective on objects)

then $\mathscr{C}$ and $\mathscr{D}$ are ***equivalent categories*** (and viceversa).

## Compositions in *Cat*

## I. Horizontal Compositions

For each triplet of categories $(\mathscr{C}, \mathscr{D}, \mathscr{E})$,

the 2-category *Cat* is equipped with a ***composition functor***

(rather than a function as it is the case for 1-categories where the *Hom*'s are sets)

$$\circ : Hom_{Cat}(\mathscr{C}, \mathscr{D}) \times Hom_{Cat}(\mathscr{D}, \mathscr{E}) \to Hom_{Cat}(\mathscr{C}, \mathscr{E})$$

where $Hom_{Cat}(\mathscr{C}, \mathscr{D}) \times Hom_{Cat}(\mathscr{D}, \mathscr{E})$

is the *product category* with objects given by pairs of functors

$$(F : \mathscr{C} \to \mathscr{D}, G : \mathscr{D} \to \mathscr{E})$$

and morphisms $(F, G) \Rightarrow (F', G')$ given by pairs of natural transformations

$$(\alpha : F \Rightarrow F', \beta : G \Rightarrow G').$$

Acting on objects, the composition functor

$$\circ : Hom_{Cat}(\mathscr{C}, \mathscr{D}) \times Hom_{Cat}(\mathscr{D}, \mathscr{E}) \to Hom_{Cat}(\mathscr{C}, \mathscr{E})$$

sends

$$(F : \mathscr{C} \to \mathscr{D}, G : \mathscr{D} \to \mathscr{E}) \mapsto (G \circ F : \mathscr{C} \to \mathscr{E})$$

Acting on morphisms, the composition functor sends

$$(\alpha : F \Rightarrow F', \beta : G \Rightarrow G') \mapsto (\beta \circ \alpha : G \circ F \Rightarrow G' \circ F)$$

which can be understood

as the « horizontal composition » of the natural transformations

$$\alpha : F \Rightarrow F' \text{ and } \beta : G \Rightarrow G'$$

## Compositions in *Cat*

## II. Vertical Compositions

In *Cat*, every $Hom_{Cat}(\mathscr{C}, \mathscr{D})$ is not a set but a category.

Hence, given the functors $F, G, H : \mathscr{C} \to \mathscr{D}$,

two composable morphisms (natural transformations)

$$\eta : F \Rightarrow G \text{ and } \theta : G \Rightarrow H$$

in the category $Hom_{Cat}(\mathscr{C}, \mathscr{D})$

can be « vertically » composed to

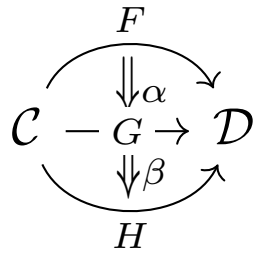
$$\theta \circ \eta : F \Rightarrow H.$$

**Functor Categories**

Composable natural transformations

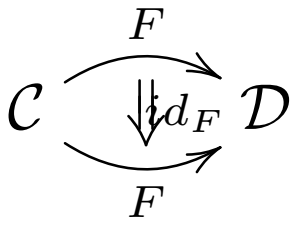
$$\alpha : F \Rightarrow G \text{ and } \beta : G \Rightarrow H$$

between functors between the same categories « vertically » compose to $\beta \circ \alpha : F \Rightarrow H$




defined componentwise $(\beta \circ \alpha)_a = \beta_a \circ \alpha_a$

with identity functors $id_F : F \Rightarrow F$




defined componentwise $(id_F)_a = id_{F(a)}$

Since $Hom_{Cat}[\mathscr{C}, \mathscr{D}]$ is a category, the associativity of the composition and the unity laws hold strictly.

## *Cat* as a strict 2-category

The category of categories *Cat*

— which can be understood as a category enriched over categories, i.e. as a category in which each *Hom* is not a set but a category —

is a strict 2-category,

which means that

vertical compositions of functors are strictly associative and unital (since each $Hom_{Cat}(\mathscr{C}, \mathscr{D})$ is now a category, the vertical composition of functors is strictly associative and unital)

and the interchange law holds strictly.

## Partial Summary

Category theory can be understood

as a ***mathematical meta-mathematics***

that organizes mathematical objects

in ***relational ontological contexts***

called ***categories***.

## Partial Summary

Categories are ***relational ontological contexts***

given by

***self-identical objects***

and

***morphisms***

that can be composed in an associative manner.

## Partial Summary

By using the self-identities of the objects

and the presence of morphisms,

it is possible to « stretch » the strict notion of equality

into ***isomorphisms***.

Isomorphisms provide a

category-dependent criterium of equality

for mathematical objects.

## Partial Summary

The definition of isomorphisms

as categorical criterium of identity

implies that the referents of mathematical discourse

are ***objects of a category up to isomorphism***,

sometimes called ***structures.***

## Partial Summary

This characterization of the ontological content of category theory

has two central ontological categories

— ***determinacy*** and ***identity*** —

are redefined.

### Structural Determinacy

Only structural (or isomorphism-invariant) properties

are admissible properties.

### Structural Identity

Objects are endowed

with an expanded self-identity

given by theirs groups of automorphisms.

# Partial Summary

The « categorical imperative »

according to which the strict notion of equality

has to be stretched to isomorphisms

elicits the upgrading of category theory

to ***higher category theory.***

Higher morphisms are required in order to « stretch »

the strict equalities between morphisms in category theory

(e.g. associativity laws, identity laws, definition of isomorphism).

## Partial Summary

A ***higher category*** is an enrichment of the notion of category

in which we add 2-morphisms between 1-morphisms,

3-morphisms between 2-morphisms,

and so on and so forth.

An entity at level $k$ is equipped

with a trivial self-identity at level $k+1$.

$k$-morphisms can meet

— and thereby be compose— along morphisms

of any dimension from 0 (objects) to $k-1$.

# Partial Summary

For instance, in the case of a 2-category,

2-morphisms can be composed

when they meet at an object

— ***horizontal composition*** —

and when they meet at a morphism

— ***vertical composition*** —.

**Partial Summary**

All the equations between morphisms

guaranteeing the coherence of the structure,

— i.e. specifying that the different possible compositions

of a multi-composable diagram are « equal »

(e.g. associativity, identity laws, interchange law, etc.) —

are replaced by morphisms

one level higher.

# Summary

Rather than endowing the structure

with strict juridical coherence (laws),

the structure self-regulates its own coherence

by flowing upwards.

## Partial Summary

The introduction of functors

$$F : \mathscr{C} \to \mathscr{D}$$

*qua* inter-categorical bridges

allows us to treat categories themselves

as objects of a category *Cat*,

and to meaningfully ask when two categories

ought to be regarded as categorically equal.

## Partial Summary

If categories are objects of a category of categories *Cat*,

then two categories $\mathscr{C}$ and $\mathscr{D}$ should be considered categorically equal

when they are related by a functor

$$F : \mathscr{C} \to \mathscr{D}$$

that is an isomorphism, i.e. that has an inverse $F^{-1} : \mathscr{C} \to \mathscr{D}$ such that

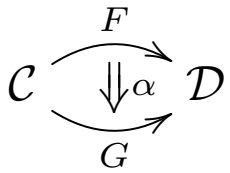
$$F^{-1} \circ F = id_{\mathscr{C}} \quad \text{and} \quad F \circ F^{-1} = id_{\mathscr{D}}.$$

## Partial Summary

This criterium of equality for categories does not take into account

the existence of a notion of morphism between functors

called ***natural transformations*** (2-morphisms in *Cat*)




By using invertible natural transformations,

one can weaken the strict equalities in the definition of an isomorphism

$$F^{-1} \circ F = id_{\mathscr{C}} \quad \text{and} \quad F \circ F^{-1} = id_{\mathscr{D}}$$

to

$$F^{-1} \circ F \simeq id_{\mathscr{C}} \quad \text{and} \quad F \circ F^{-1} \simeq id_{\mathscr{D}}.$$

These last expressions define the correct criterium of equality for categories

called ***equivalence of categories***.

## Partial Summary

Whereas the correct criterium of equality for 1-categories

is given by isomorphisms,

the correct correct criterium of equality for 2-categories

is given by the notion of ***isomorphism up to 2-isomorphisms***.

The ***category of categories*** *Cat* is indeed a 2-category

with categories as objects,

functors as 1-morphisms

and natural transformations as 1-morphisms.

The notion of ***equivalence of categories***

is the corresponding notion of isomorphism up to 2-isomorphisms.

# Adjoint Functors

# (Daniel Kan, 1958)

« The notion of adjoint functor […] captures an important mathematical phenomenon that is invisible without the lens of category theory. Indeed, I will make the admittedly provocative claim that adjointness is a concept of fundamental logical and mathematical importance that is not captured elsewhere in mathematics. Many of the most striking applications of category theory involve adjoints and many important and fundamental mathematical notions are instances of adjoint functors. »

S. Awodey, *Category Theory*, p.179.

By having functors *qua* inter-categorical bridges

at our disposal,

we can also consider « ***heteromorphisms*** »

between objects belonging to different categories.

## Categorical Odyssey

Adjoint functors

$$\mathcal{D} \underset{R}{\overset{L}{\leftrightarrows}} \mathcal{C}$$

can be defined in (at least) two different manners:

1) By introducing « heteromorphisms »

between objects in different categories $\mathscr{C}$ and $\mathscr{D}$

either by « abducting » an object from $\mathscr{C}$ to $\mathscr{D}$

— where the heteromorphism takes the form $c \to R(d)$ —

or by « inducting » an object from $\mathscr{D}$ to $\mathscr{C}$.

— where the heteromorphism takes the form $L(c) \to d$ —

2) By using « sublated [abducted-&-inducted] objects »

as « universal mediators with the aliens ».

## Functorial Dialectics

Given two functors of the form

$$\mathcal{D} \underset{R}{\overset{L}{\leftrightarrows}} \mathcal{C}$$

we can use them to take an object $c$ in $\mathscr{C}$,

« abduct » it to $\mathscr{D}$ through the functor $L$

and then « induct » the alienated object $L(c)$ back to $\mathscr{C}$ through $R$.

We are interested in comparing (what we shall call)

the « sublated [abducted and inducted] object »

$R(L(c))$ in $\mathscr{C}$

with the original object $c$ in $\mathscr{C}$.

Given two functors

$$\mathcal{D} \underset{R}{\overset{L}{\rightleftarrows}} \mathcal{C}$$

the comparison between the original object $c$ and the sublated one $R(L(c))$

can take three forms:

• **Isomorphism of categories**: the sublated object $R(L(c))$ is strictly equal to the original one $c$:

$$R(L(c)) = c$$

• **Equivalence of categories**: the sublated object $R(L(c))$ is a « clone » of (i.e. isomorphic to) the original one $c$:

$$R(L(c)) \simeq c$$

• **Adjonctions**: the sublated object $R(L(c))$ is a « universal mediator for relations with aliens ».

If the functors

$$\mathcal{D} \underset{R}{\overset{L}{\rightleftarrows}} \mathcal{C}$$

define an ***isomorphism of categories***,

$R$ is the ***strict inverse*** of $L$ (that we have denoted $L^{-1}$ before)

and the functors satisfy

$$R \circ L = id_{\mathscr{C}} \quad \text{and} \quad L \circ R = id_{\mathscr{D}}$$

This means that the sublated object

is strictly equal to the original one:

$$R(L(c)) = c.$$

If the functors

$$\mathcal{D} \underset{R}{\overset{L}{\rightleftarrows}} \mathcal{C}$$

define an ***equivalence of categories***,

$R$ is the ***pseudo-inverse or weak inverse*** of $L$

and there are ***natural isomorphisms*** (rather than strict equalities)

$$\eta : id_{\mathscr{C}} \to R \circ L \quad \text{and} \quad \epsilon : L \circ R \to id_{\mathscr{D}}$$

This means that the sublated object

is a « clone » of (i.e. isomorphic to) the original one:

$$R(L(c)) \simeq c.$$

If the functors

$$\mathcal{D} \underset{R}{\overset{L}{\leftrightarrows}} \mathcal{C}$$

define an ***adjonction***, the natural transformations

$$\eta : id_{\mathscr{C}} \to R \circ L \quad \text{and} \quad \epsilon : L \circ R \to id_{\mathscr{D}}$$

— called ***unit*** and ***counit*** of the adjunction —

are no longer required to be natural isomorphisms

(as it is the case for an equivalence of categories)

but still fulfill a ***universal mapping property***

encoding the fact that the sublated object $R(L(c))$ plays a universal role.

In the case of an adjunction,

the sublated object $R(L(c))$

is neither strictly equal

— as in an isomorphism of categories —

nor a clone of the original one $c$

— as in an equivalence of categories —

but plays the role of being

a ***universal mediator with the aliens***.

The sublated object $R(L(c))$ fulfills the role

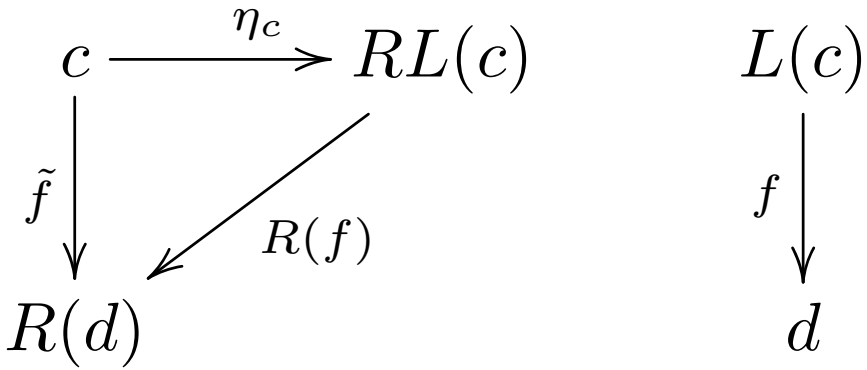
of being a « universal mediator with the aliens »,

in the sense that any relation between $c$ and an « alien » $R(d)$

is mediated or factorized by a universal morphism

— the component $\eta_c$ of the natural transformation $\eta : id_{\mathscr{C}} \to R \circ L$ —

to its « sublated self » $R(L(c))$:




where $f : L(c) \to d$ is a morphism in $\mathscr{D}$.

In pictorial terms, the « sublated object »

$$R(L(c))$$

— being related at once to the original object $c$ (that is, being familiar enough)

and to the alien world $\mathscr{D}$ (that is, being alien enough)—

functions as a universal “mediator”

between the original object $c$

and any “alien” object $R(d)$ arriving from the « world » $\mathscr{D}$

through the functor $R$.

In particular, the so-called « adjunct »

$$\widetilde{id_{L(c)}} \doteq \eta_c : c \to R(L(c)) \text{ in } \mathscr{C}$$

of the identity morphism

$$id_{L(c)} : L(c) \to L(c) \text{ in } \mathscr{D}$$

is the *component at c of natural transformation*

$$\eta : id_{\mathscr{C}} \Rightarrow R \circ L,$$

(the unit of the adjunction)

(Cf. nlab, *adjoint functor*, Proposition 2.3.2).

In pictorial terms, the universal morphism

$$\eta_c : c \to R(L(c))$$

between $c$ and its sublated avatar $R(L(c))$

is the functorial transposition to $\mathscr{C}$ of the trivial identity

$$id_{L(c)} : L(c) \to L(c)$$

of the alienated $c$.

## Adjuntions *qua* Natural Isomorphisms Between Hom-Bifunctors

Two functors

$$\mathcal{D} \underset{R}{\overset{L}{\leftrightarrows}} \mathcal{C}$$

define a pair of adjoint functors

where $L$ is called the ***left adjoint*** and $R$ the ***right adjoint***

denoted $F \dashv U$

if there is a ***natural isomorphism***

$$Hom_{\mathscr{D}}(L(-), -) \simeq Hom_{\mathscr{C}}(-, R(-))$$

where both hom-functors are functors of the form

$$\mathscr{C}^{op} \times \mathscr{D} \to Set$$

The natural isomorphism

$$Hom_{\mathscr{D}}(L(-),-) \simeq Hom_{\mathscr{C}}(-,R(-))$$

means that for every $c$ in $\mathscr{C}$ and every $d$ in $\mathscr{D}$, there is a bijection of sets

$$Hom_{\mathscr{D}}(L(c),d) \simeq Hom_{\mathscr{C}}(c,R(d))$$

that sends each morphism in $\mathscr{D}$ of the form

$$f : L(c) \to d$$

to its « adjunct » in $\mathscr{C}$

$$\tilde{f} = R(f) \circ \eta_c : c \to R(d)$$

and each morphism in $\mathscr{C}$ of the form

$$g : c \to R(d)$$

to its « adjunct » in $\mathscr{D}$

$$\tilde{g} = \epsilon_d \circ L(g) : L(c) \to d.$$

In pictorial terms, the bijection of sets

$$Hom_{\mathscr{D}}(L(c), d) \simeq Hom_{\mathscr{C}}(c, R(d))$$

with $c$ an object in $\mathscr{C}$ and $d$ an object in $\mathscr{D}$

says that an « heteromorphism » between objects $c$ and $d$ living in different « worlds »

can be equivalently defined

either by « abducting » $c$ to $\mathscr{D}$ through the functor $L$

or by « inducting » $d$ to $\mathscr{C}$ through the functor $R$.

In other terms, if $c$ wants to have an alien friend

it can either abduct itself to the world of the alien

or induct the aliens to its own world.

The fact that

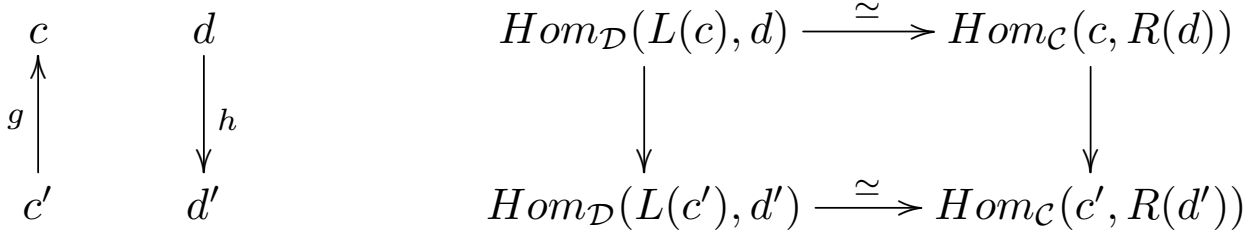
$$Hom_{\mathscr{D}}(L(-),-) \simeq Hom_{\mathscr{C}}(-,R(-))$$

is a natural isomorphism between two hom-bifunctors

— or that $Hom_{\mathscr{D}}(L(c),d) \simeq Hom_{\mathscr{C}}(c,R(d))$ is « natural » in $c$ and $d$ —

means that we have the following commutative squares

for each morphism in $\mathscr{C}^{op} \times \mathscr{D}$




where the vertical morphisms in the square are

$$(m : L(c) \to d) \mapsto (h \circ m \circ L(g) : L(c') \to d')$$

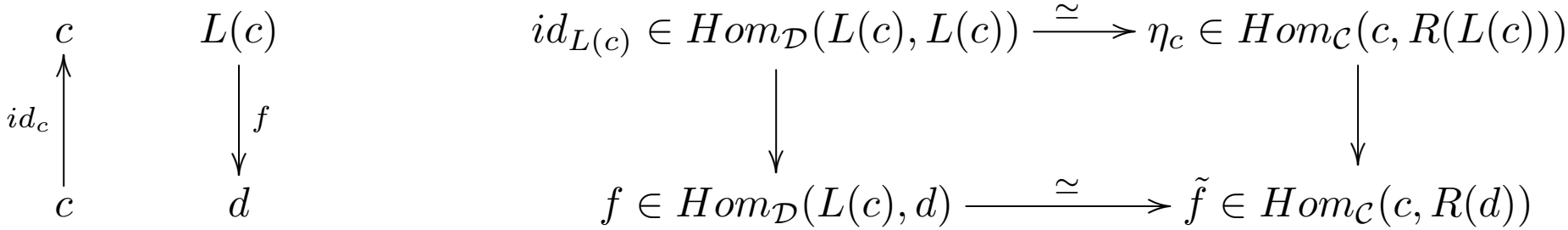
and

$$(s : c \to R(d)) \mapsto (R(h) \circ s \circ g : c' \to R(d')$$

By focusing on

$$id_{L(c)} \in Hom_{\mathscr{D}}(L(c), L(c)),$$

the commutativity of the following square diagram

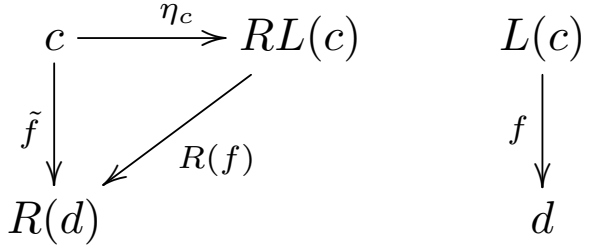



implies that the adjunct of $f : L(c) \to d$ is given by $\tilde{f} = R(f) \circ \eta_c : c \to R(d)$

In other terms, for each morphism $f : L(c) \to d$ in $\mathscr{D}$

there is a morphism in $\tilde{f} : c \to R(d)$ in $\mathscr{C}$

such that the following diagram commutes



# Examples of Adjunctions

## Adjonction Between Products & Hom-Functors
## I. Left Product Functor and Hom-Functor

In the category *Set* (which is endowed with the Cartesian product),

given a set $A$, we can define a *left product functor*

$A \times (\,-\,) : Set \to Set$ given by $B \mapsto A \times B$.

In turn, a set $A$ defines a (covariant) Hom-functor

$Hom_{Set}(A, -) : Set \to Set$ given by $C \mapsto Hom_{Set}(A, C)$.

We shall see that $A \times (\,-\,)$ and $Hom_{Set}(A, -)$ are a pair of adjoint functors

where $A \times (\,-\,)$ is the left adjoint and $Hom_{Set}(A, -)$ is the right adjoint:

$$Hom_{Set}(A \times B, C) \simeq Hom_{Set}(B, Hom_{Set}(A, C)).$$

## Adjonction Between Products & Hom-Functors
## II. Adjoint Functions

A function

$$f : A \times B \to C$$

defines an adjoint function

$$\tilde{f} : B \to Hom_{Set}(A, C)$$

where

$$\tilde{f}(b) : A \to C$$

is given by

$$\tilde{f}(b)(a) = f(a, b).$$

# Adjonction Between Products & Hom-Functors
# III. Adjoint Functions

A function

$$g : B \to Hom_{Set}(A, C)$$

defines an adjoint function

$$\tilde{g} : A \times B \to C$$

given by

$$\tilde{g}(a, b) = g(b)(a).$$

# Adjonction Between Products & Hom-Functors
# IV. Natural bijection Between Hom-Sets.

Hence,

$$Hom_{Set}(A \times (\,-\,), -\,) \simeq Hom_{Set}(\,-\,, Hom_{Set}(A, -\,))$$

or

$$Hom_{Set}(A \times B, C) \simeq Hom_{Set}(B, Hom_{Set}(A, C))$$

i.e. the hom-functor

$$Hom_{Set}(A, -\,) : Set \to Set$$

is right adjoint to the product functor

$$A \times (\,-\,) : Set \to Set.$$

# Adjonction Between Products & Hom-Functors
# V. Currying

In computer science,

the conversion of a function on two variables

$$f : A \times B \to C$$

into an adjoinct function on one variable

$$\tilde{f} : B \to \{A \to C\}$$

is called ***currying***

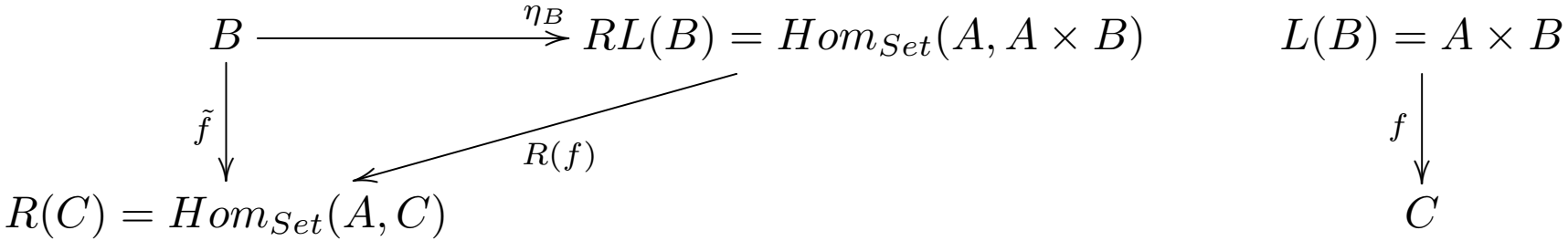
— a reference to the logician Haskell Curry —.

**Adjonction Between Products & Hom-Functors**
**VI. Unit of the adjunction**

Given the adjunct situation

$$Set \underset{R=Hom_{Set}(A,-)}{\overset{L=A\times(-)}{\rightleftarrows}} Set$$

The commutative triangle that defines the universal role of the unit is




where

$$\eta_B(b)(a) = (a, b)$$

and

$$R(f)(h : A \to A \times B) = (f \circ h : A \to C)$$

make the triangle commute.

## Products and Coproducts as Right and Left Adjoints to the Diagonal Functor (I)

Let's consider the diagonal functor

$$\Delta(-) : \mathscr{C} \to \mathscr{C} \times \mathscr{C}$$

given by $\Delta(a) = (a, a)$.

According to the definition of the product category $\mathscr{C} \times \mathscr{C}$,

a morphism $f : (a, a) \to (b, c)$ in $\mathscr{C} \times \mathscr{C}$

is given by a pair of morphisms $f_b : a \to b$ and $f_c : a \to c$.

According to the the universal property of the categorical product,

such a couple of morphisms defines a unique morphism $a \to b \times c$.

Hence, we have a bijection

$$Hom_{\mathscr{C}\times\mathscr{C}}(\Delta(a), (b, c)) \simeq Hom_{\mathscr{C}}(a, b \times c)$$

All in all, the product functor $(-) \times (-) : \mathscr{C} \times \mathscr{C} \to \mathscr{C}$

is right adjoint to the diagonal functor $\Delta(-) : \mathscr{C} \to \mathscr{C} \times \mathscr{C}$.

## Products and Coproducts as Right and Left Adjoints to the Diagonal Functor (II)

Analogously,

a morphism $f : (b, c) \to (a, a)$ in $\mathscr{C} \times \mathscr{C}$

is given by a pair of morphisms $f_b : b \to a$ and $f_c : c \to a$.

According to the the universal property of the categorical coproduct,

such a couple of morphisms defines a unique morphism $b \coprod c \to a$.

Hence, we have a bijection

$$Hom_{\mathscr{C}\times\mathscr{C}}((b, c), \Delta(a)) \simeq Hom_{\mathscr{C}}(b \coprod c, a).$$

All in all, the coproduct functor $(-) \coprod (-) : \mathscr{C} \times \mathscr{C} \to \mathscr{C}$

is left adjoint to the diagonal functor $\Delta(-) : \mathscr{C} \to \mathscr{C} \times \mathscr{C}$.

# Adjonction Between Directed Graphs and Categories

[To be completed]

## Adjunction Between the Forgetful Functor and the Free Group Functor

Let's consider the following situation

$$Set \underset{F}{\overset{U}{\leftarrow\!\!\rightarrow}} Grp$$

where the forgetful functor $U : Grp \to Set$

sends a group $G = (X_G, *, e)$ to its underlying set $X_G$

and the free functor $F : Set \to Grp$

sends a set $X$ to the group $F(X)$ of reduced « words »

freely generated out of the « letters » in $X$ and its formal inverses.

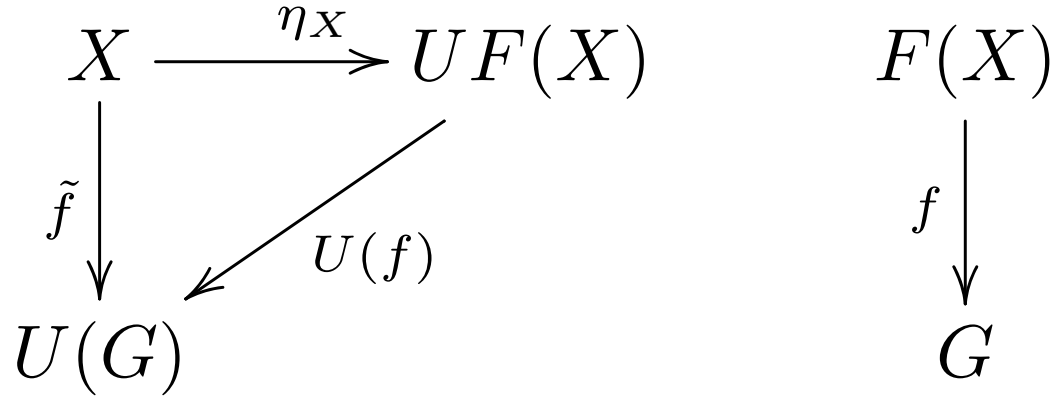
## The unit of the forgetful-free adjunction

Given a function $f$ from the free group $F(X)$ of words generated out of $X$ to a group $G$,

we can define an adjoint function $\tilde{f}$ from $X$ to the underlying set $U(G)$ of $G$

by including each letter in $X$ as a one-letter word in $UF(X)$

and then using $f$ to transform each one-letter word in $UF(X)$ into an element of $G$:




All in all, the free functor $F$ is left adjoint to the forgetful functor $U$

$$Hom_{Set}(X, U(G)) \simeq Hom_{Grp}(F(X), G).$$

**The counit of the forgetful-free adjunction**

Given a function $g : X \to U(G)$,

we can define a group homomorphism $\tilde{g} : F(X) \to G$

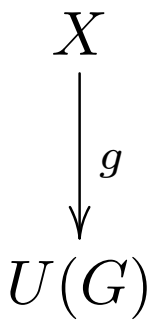
by substituting each letter $x \in X$ in the words in $F(X)$ by $g(x)$,

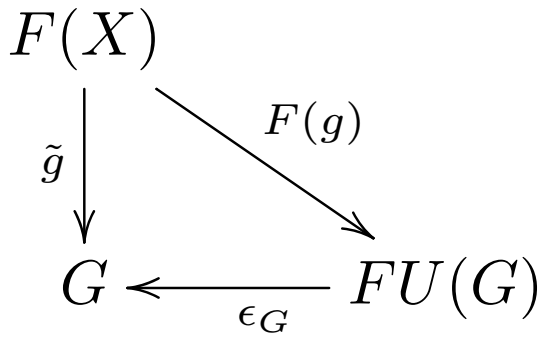
thereby obtaining a word in $FU(G)$

and then using the counit $\epsilon_G : FU(G) \to G$,

which sends a word in $FU(G)$ to a single element in $G$

by using the operation, inverses and identity in $G$.



# Galois Correspondences

[To be completed]

# Existential and Universal Quantifiers (Lawvere, 1960s)

[To be completed]

# Discrete and Indiscrete Topologies

[To be completed]

# Limits and Colimits as Adjoints

## I. Diagonal Functor

Let's consider the *diagonal functor*

$$\Delta : \mathscr{C} \to [\mathscr{I}, \mathscr{C}]$$

that sends each object $a$ in $\mathscr{C}$ to the constant diagram

$$\Delta(c) : \mathscr{I} \to \mathscr{C},$$

where

$$\Delta(c)(i) = c, \text{ for } i \in \mathscr{I}$$

and

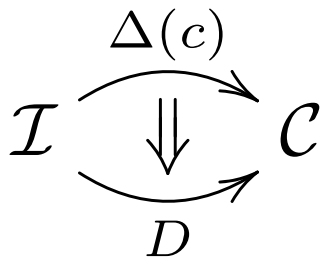
$$\Delta(c)(f) = id_c \text{ for } f \text{ a morphism in } \mathscr{I}.$$

## Limits and Colimits as Adjoints

## II. Cones as Natural Transformations

Given a diagram $D : \mathscr{I} \to \mathscr{C}$ and an object $c$ in $\mathscr{C}$,

a cone on $D$ with vertex $c$ is a natural transformation,




where the commutative squares of the natural transformation

encodes the fact that the morphisms from $c$ to the objects in the diagram

commute will all the morphisms in the diagram.

## Limits as Adjoints

If every diagram $D$ in $[\mathscr{I}, \mathscr{C}]$ has a limit $lim(D)$,

then

$$\Delta(-) : \mathscr{C} \to [\mathscr{I}, \mathscr{C}]$$

has a right adjoint

$$lim(-) : [\mathscr{I}, \mathscr{C}] \to \mathscr{C}$$

where

$$Hom_{\mathscr{C}}(a, lim(D)) \simeq Hom_{[\mathscr{I},\mathscr{C}]}(\Delta(a), D)$$

$$(a \to lim(D)) \mapsto \text{Cone to } D \text{ with vertex } a.$$

This can be rephrased by saying that the functor

$$Hom_{[\mathscr{I},\mathscr{C}]}(\Delta(-), D) : \mathscr{C} \to Set$$

is a representable functor with representing object $lim(D)$.

# The Yoneda Lemma,

# (Non-)Representable Functors,

# and Generalized Objects.

« […] the abstractions in this chapter

are one of the great joys of category theory. »

E. Cheng, *The Joy of Abstraction*, p. 351.

« By the Yoneda lemma—which, despite its innocuous statement, is arguably the most important result in category theory—»

E. Riehl, *Category Theory in Context*, p. xiii.

« Mathematics thrives on going to extremes whenever it can. Since the ‘compromise’ we sketched above has ‘mathematical objects determined by the network of relationships they enjoy with all the other objects of their species,’ perhaps we can go to extremes within this compromise, by taking the following further step. Subjugate the role of the *mathematical object* to the role of its network of relationships—or, a further extreme—simply replace the mathematical object by this network. […] one of the elegant–and surprising—accomplishments of category theory is that it performs this act, and does it with ease. […] The lights are dimmed on *mathematical objects* and beamed rather on the corresponding functors; that is, on the networks of relationships entailed by the objects. »

Barry Mazur, *When is one thing equal to some other thing?*

« People have written about Grothendieck's intense category-theoretic genius. The phrase 'category-theoretic,' as far as it goes, is correct as a very vague pointer to Grothendieck's attitude, where, for example, Yoneda's Lemma plays such an important role. Yoneda's Lemma asserts that an object *X* of a category is determined (up to unique isomorphism) by the functor that records morphisms from *X* to *each* of the objects of that category. Or, in more evocative terms, a mathematical object *X* is best thought of in the context of a category surrounding it, and is determined by the network of relations it enjoys with *all* the objects of that category. Moreover, to understand X it might be more germane to deal directly with the functor representing it. This is reminiscent of Wittgenstein's 'language game'; i.e., that the meaning of a word is—in essence—determined by, in fact is nothing more than, its relations to all the utterances in a language.'

Barry Mazur, *Thinking about Grothendieck*, 2016.

In the category *Set* we have a bijection

$$X \simeq Hom_{Set}(\{ \star \}, X),$$

which means that a set $X$

can be fully determined

by « probing » it with $\{ \star \}$-shaped arrows

that « point » to the different elements of $X$

(since a set $X$ is nothing but the collection of its elements).

In structuralist terms,

we could say that the set $X$

can be fully determined

by considering *uniquely* its relations

with a single privileged object of the category *Set*,

namely the terminal object $1 = \{ \star \}$.

« In other words, in *Set* it's enough to look at everything from the one-element set 1 — the only thing that matters about a set is its elements! »

T. Leinster, *The Yoneda Lemma: What's It All About?*

In a general category, this is not necessarily the case since:

- there might not be a terminal object 1 from which we could define such « pointed arrows », like in the poset $(\mathbb{Z}, \leqslant)$;

- even if the category has a terminal object 1, the set $Hom_{\mathscr{C}}(1,a)$ might be insufficient to determine $a$.

For example, in a category of structured sets,

the morphisms from the terminal object 1

detect only the underlying set of an object,

while its additional structure remains invisible to such probes alone.

**Example of Objects that Cannot be Distinguished by Probing Them With Pointed Arrows**

The two non-isomorphic posets

$A = \{a \leqslant b \leqslant c\}$ and $X = \{x \leqslant y; x \leqslant z\}$

in the category *Pos* of posets and order-preserving functions

cannot be distinguished by probing them with pointed arrows

$1 \to A, X$

since $A$ and $X$ are isomorphic *qua* sets, i.e. they have the same number of elements.

The point-shaped arrows can only « see » the elements of the posets,

but the structure of morphisms remain « invisible » to such arrows.

But $A$ and $X$ can be distinguished by using 2-shaped arrows $2 \to A, X$

where 2 is the poset $p \leqslant p'$ (or $p \to p'$)

Indeed, $A$ has 6 and $X$ has 5 generalized elements of shape 2 (i.e. 6 and 5 morphisms from 2).

Example taken from Awodey, *op.cit.*, pp.31-2.

## Example of $\mathscr{C} = Grp$

The terminal object in the category $Grp$ of groups is the trivial group $1 = \{e\}$.

♣ The morphisms $1 = \{e\} \to G = \{X, \bullet, e_G\}$ encode no information at all about $G$ since there is only one such morphism given by $e \mapsto e_G$.

♥ The morphisms $(\mathbb{Z}, +, 0) \to G$ encode information about the number of elements in $G$. Indeed any such morphism $\phi : (\mathbb{Z}, +, 0) \to G$ is fixed by the image of $1 \in \mathbb{Z}$. Indeed, $\phi(n) = \phi(1)^n$. Hence, there are as many morphisms $(\mathbb{Z}, +, 0) \to G$ as possible images of $1 \in \mathbb{Z}$, which is given by the number of elements in $G$.

**Note**: Morphisms from $(\mathbb{Z}, +, 0)$ are still insufficient to determine $G$, because they only encode the underlying set of elements, not the group structure.

Example taken from Leinster, *The Yoneda Lemma: What's It All About?,* p.9.

The general result that is valid in any category

— encoded in the so-called ***Yoneda lemma*** —

is that an object $a$ of a category $\mathscr{C}$ can be fully determined

by probing is with all the $b$-shaped arrows

for every object $b$ in $\mathscr{C}$,

that is by specifying its $b$-generalized elements

$Hom_{\mathscr{C}}(b, a)$ for all the objects $b$ in $\mathscr{C}$.

In order to structurally describe an object $a$ in a category $\mathscr{C}$,

we have to consider the sets $Hom_{\mathscr{C}}(b, a)$

of $b$-generalized points of $a$

for every object $b$ in $\mathscr{C}$.

Therefore, we have to consider the contravariant functor

$$Hom_{\mathscr{C}}(-, a) : \mathscr{C}^{op} \to Set$$

where the vacant place can be occupied by any object $b$ in $\mathscr{C}$.

The ***functor category*** $[\mathscr{C}^{op}, Set]$

is defined by contravariant functors

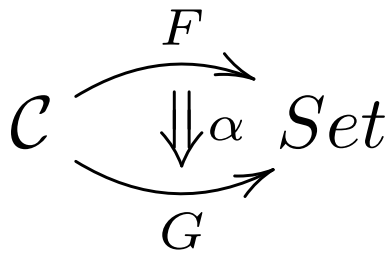
$$F : \mathscr{C}^{op} \to Set$$

as objects

and natural transformations

$$\alpha : F \Rightarrow G$$

between these functors as morphisms



The so-called ***Yoneda Embedding*** is a covariant functor

$$h_{(-)} : \mathscr{C} \to [\mathscr{C}^{op}, Set]$$

that sends each object $a$ in a category $\mathscr{C}$

to a contravariant functor

$$h_a(-) \doteq Hom_{\mathscr{C}}(-, a) : \mathscr{C}^{op} \to Set$$

in the functor category $[\mathscr{C}^{op}, Set]$

called the ***functor of points*** of $a$.

## Action of the Yoneda Embedding Functor on Morphisms

Given a morphism $f : a \to b$ in $\mathscr{C}$,

the Yoneda embedding functor

$$h_{(-)} : \mathscr{C} \to [\mathscr{C}^{op}, Set]$$

defines a natural transformation

$$h_f : Hom_{\mathscr{C}}(-, a) \Rightarrow Hom_{\mathscr{C}}(-, b)$$

whose component at an object $c$

$$(h_f)_c : Hom_{\mathscr{C}}(c, a) \Rightarrow Hom_{\mathscr{C}}(c, b)$$

is given by

$$(h : c \to a) \mapsto (f \circ h : c \to b)$$

(note that the Yoneda embedding functor $h_{(-)}$ is covariant).

The functor of points $h_a(-)$ of an object $a$ of a category $\mathscr{C}$

— i.e. the image of $a$ under the Yoneda embedding functor $h_{(-)}$ —

is a contravariant functor

$$h_a(-) \doteq Hom_{\mathscr{C}}(-, a) : \mathscr{C}^{op} \to Set$$

defined by

$$b \mapsto Hom_{\mathscr{C}}(b, a) \text{ on objects in } \mathscr{C}$$

and

$$(f : b \to c) \mapsto h_a(f) : Hom_{\mathscr{C}}(c, a) \to Hom_{\mathscr{C}}(b, a)$$

on morphisms in $\mathscr{C}$, where

$$h_a(f)(g : c \to a) = (g \circ f : b \to a).$$

Conceptually, the functor of points

$$h_a(-)$$

assigns to each object $b$ in $\mathscr{C}$,

the set $Hom_{\mathscr{C}}(b, a)$

of $b$-probes, $b$-generalized elements, $b$-shaped elements or $b$-points of $a$.

Hence, the functor of points $h_a(-)$ encodes the « variable set »

of generalized elements of the object $a$.

This variable set contains one set $Hom_{\mathscr{C}}(b, a)$ of $b$-shaped elements of $a$

for each object $b$ in $\mathscr{C}$,

where each morphism $f : b \to c$ in $\mathscr{C}$ defines a « parallax function »

$$F(f) : Hom_{\mathscr{C}}(c, a) \to Hom_{\mathscr{C}}(b, a)$$

that converts $c$-probes of $a$ into $b$-probes.

In phenomenological terms,

the functor of points

$$h_a(-)$$

encodes all the $b$-perspectives on $a$

defined by all the objects $b$ in $\mathscr{C}$

as well as the « parallax » transformations

associated to the changes in the standpoint of observation.

Each object $a$ in a category $\mathscr{C}$

has a functorial avatar

$$h_a( - )$$

in the functor category

$$[\mathscr{C}^{op}, Set].$$

This functorial avatar of $a$ encodes all the manners

in which the other objects in $\mathscr{C}$

can « point » towards $a$.

## (Non-)Representable Functors
## (Grothendieck, 1961)

There are functors in the functor category $[\mathscr{C}^{op}, Set]$

which are not in the image of the Yoneda embedding,

i.e. which are not of the form

$$h_a(-) \doteq Hom_{\mathscr{C}}(-, a)$$

for some object $a$ in $\mathscr{C}$.

The functors in $[\mathscr{C}^{op}, Set]$ that are in the image of the Yoneda embedding

are called ***representable functors***.

Not all functors in $[\mathscr{C}^{op}, Set]$ are representable.

Even if not all functors in $[\mathscr{C}^{op}, Set]$ are representable functors,

there is a fundamental relation between any functor

$$F(-) : \mathscr{C}^{op} \to Set$$

and the representable functors given by the so-called

***Yoneda lemma***

— lemma which was described,

according to the myth, by Nobuo Yoneda to Saunders Mac Lane

at the Gare du Nord train station in Paris in 1954 —.

The ***Yoneda Lemma*** states

that there is a bijection of sets

$$Hom_{[\mathscr{C}^{op},Set]}(h_a(-), F(-)) \simeq F(a)$$

that is « natural » in $a \in \mathscr{C}$ and $F \in [\mathscr{C}^{op}, Set]$.

This means that the set of natural transformations

$$h_a(-) \Rightarrow F(-)$$

— understood as morphisms in the functor category $[\mathscr{C}^{op}, Set]$ —

is in bijective correspondence

with the elements of the set $F(a)$.

For a non-representable functor

$$F(-) : \mathscr{C}^{op} \to Set,$$

the Yoneda lemma

$$Hom_{[\mathscr{C}^{op},Set]}(h_a(-), F(-)) \simeq F(a)$$

states that the elements in the set $F(a)$

can still be understood as $a$-probes, $a$-shaped elements, or $h_a$-generalized elements of $F$.

Even if we cannot point from $a$ to $F$ in $\mathscr{C}$

— since the functor $F$ is not an object in $\mathscr{C}$ —,

we can point to $F$ from the functorial avatar $h_a(-)$ of $a$.

The Yoneda lemma states that

the set-theoretic elements in the variable set defined by $F$

— i.e. all the elements $x \in F(a)$ for each $a \in \mathscr{C}$ —

defines the generalized elements of $F$

of representable shape.

Hence, the action of the functor $F(-)$ on objects

is fully determined by the $h_a$-points of $F(-)$.

What about the action of the functor $F(-)$ on morphisms?

Is the action of $F(-)$ on morphisms

also determined by the morphisms to $F(-)$

from representable functors $h_a(-)$?

**Naturality of the Yoneda Bijection of Hom-Sets**

The fact that the bijection of sets

$$Hom_{[\mathscr{C}^{op},Set]}(h_a(-), F(-)) \simeq F(a)$$

is « natural » in both $a$ in $\mathscr{C}$ and $F(-)$ in $[\mathscr{C}^{op}, Set]$

means that there is a natural isomorphism $\Phi$ between the « evaluation functor »

$$\mathscr{C}^{op} \times [\mathscr{C}^{op}, Set] \to Set$$

given by $(a, F(-)) \mapsto F(a)$

and the functor

$$\mathscr{C}^{op} \times [\mathscr{C}^{op}, Set] \to Set$$

given by

$$(a, F(-)) \mapsto Hom_{[\mathscr{C}^{op},Set]}(h_a(-), F(-))$$

— note: remember that a hom-bifuntor is contravariant in the first variable

and covariant in the second ones —.

The commutative squares of the natural isomorphism $\Phi$

associated to a morphism $f : b \to a$

fixes the morphism $F(f) : F(a) \to F(b)$:

$$\begin{array}{ccc} a & Hom_{[\mathcal{C}^{op},Set]}(h_a(-), F(-)) & \xrightarrow{\phi_{a,F}(\simeq)} F(a) \\ \uparrow f & \downarrow & \downarrow F(f) \\ b & Hom_{[\mathcal{C}^{op},Set]}(h_b(-), F(-)) & \xrightarrow{\phi_{b,F}(\simeq)} F(b) \end{array}$$

where the left vertical morphism is given by

$$(h_a \to F) \mapsto (h_b \to h_a \to F).$$

All in all, the action of $F(-)$ on morphisms is given by the naturally in $a$.

The Yoneda lemma does not only states

that the set $F(a)$ can still be interpreted

as a set of morphisms « probing » $F(-)$,

but also that $F(-)$ is fully determined

by pointing to it *uniquely* from the representable functors

$h_a(-)$ for every $a$ in $\mathscr{C}$.

Even if there are non-representable functors in $[\mathscr{C}^{op}, Set]$,

any functor (be it representable or not) can be fully determined

from the generalized points defined by the representable ones.

## Demonstration of the Yoneda Lemma (I)

In order to demonstrate that

$$Hom_{[\mathscr{C},Set]}(h_a, F) \simeq F(a)$$

we shall first fix a natural transformation

$$\eta : h_a(-) \Rightarrow F(-)$$

and use it to define an element in the set $F(a)$.

To do so, we shall apply the component of the natural transformation $\eta$ at $a$

$$\eta_a : h_a(a) \to F(a)$$

to the identity morphism $id_a \in h_a(a) = Hom_{\mathscr{C}}(a, a)$,

thereby defining an element in $F(a)$:

$$\eta_a(id_a) \in F(a).$$

## Demonstration of the Yoneda Lemma (II)

Conversely, we shall now use an element $u \in F(a)$ to define a natural transformation

$$\eta^{(u)} : h_a(-) \Rightarrow F(-)$$

To do so, we have to define each component of $\eta^{(u)}$

$$\eta^{(u)}_b : h_a(b) \to F(b),$$

which amounts to send any morphism $(f : b \to a) \in h_a(b) = Hom_{\mathscr{C}}(b, a)$

to an element in $F(b)$.

Since $F(-)$ is a contravariant functor,

it sends a morphism $f : b \to a$ in $\mathscr{C}$

to a morphism of sets $F(f) : F(a) \to F(b)$.

Hence, we can define an element in $F(b)$ out of $f$, by applying $F(f)$ to $u \in F(a)$:

$$\eta^{(u)}_b(f) = F(f)(u) \in F(b).$$

## Demonstration of the Yoneda Lemma (III)

We now have to show that the defined correspondences

between natural transformations $\eta : h_a(-) \Rightarrow F(-)$

and element in $F(a)$ are inverses of each other.

Let's start with an element $u \in F(a)$

and let's apply to $id_a \in h_a(a)$ the component at $a$

$$\eta_a^{(u)} : h_a(a) \to F(a)$$

of the induced natural transformation

$$\eta^{(u)} : h_a(-) \Rightarrow F(-)$$

This gives:

$$\eta_a^{(u)}(id_a) = F(id_a)(u) = id_{F(a)}(u) = u.$$

**Demonstration of the Yoneda Lemma (IV)**

Let's start now with a natural transformation $\eta : h_a(-) \Rightarrow F(-)$,

then define the corresponding element $\eta_a(id_a) \in F(a)$

and define a natural transformation $\eta^{\eta_a(id_a)}$ out of this element to see if we reobtain $\eta$.

These two natural transformations will be equal

if their components coincide on every $f \in Hom_{\mathscr{C}}(b, a)$.

On the one hand we have $\eta^{\eta_a(id_a)}(f) = F(f)(\eta_a(id_a)) \in F(b)$

Hence we need to show that $F(f)(\eta_a(id_a)) = \eta_b(f)$.

This is given by the commutativity of the following square associated to the natural transformation $\eta$

$$\begin{array}{ccc} a & id_a \in Hom_{\mathcal{C}}(a,a) & \xrightarrow{\eta_a} \eta_a(id_a) \in F(a) \\ f \uparrow & \downarrow & \downarrow F(f) \\ b & f \in Hom_{\mathcal{C}}(b,a) & \xrightarrow{\eta_b} F(b) \end{array}$$

According to the demonstration of the Yoneda lemma,

a natural transformation

$$\eta : h_a(-) \Rightarrow F(-)$$

is completely determined

by choosing an element in $F(a)$ to be the image of

$$id_a \in h_a(a) \text{ under}$$

$$\eta_a : h_a(a) \to F(a).$$

The overall behavior of $\eta$

is completely defined by $\eta_a(id_a) \in F(a)$.

Conceptually, if we think of the trivial identity $id_a$ of $a$

as the relational index of $a$ itself

— remember that we can define a category uniquely in terms of morphisms

by using $id_a$ as a relational replacement for $a$ —

the Yoneda lemma states that the specification

of how $a$ (in the form of $id_a$) is inserted into $F$ or sits within $F$

automatically defines how each generalized point of $a$

is inserted into $F$.

## The Yoneda Lemma for Representable Functors

If $F(-)$ is of the form $h_b(-)$

— i.e. if $F$ itself is representable —,

then the Yoneda lemma states

that there is a bijection of sets

$$Hom_{[\mathscr{C}^{op},Set]}(h_a(-), h_b(-)) \simeq h_b(a) = Hom_{\mathscr{C}}(a, b).$$

This means that ***the Yoneda embedding***

$$h_{(-)} : \mathscr{C} \to [\mathscr{C}^{op}, Set]$$

given by $a \mapsto h_a(-)$

***is full and faithful*** (i.e. bijective on morphisms).

Since not every functor is isomorphic

to a representable one,

the Yoneda embedding

is not it *essentially surjective*,

which means that $h_{(_)} : \mathscr{C} \to [\mathscr{C}, Set]$

does not define an equivalence of categories.

However, the category $\mathscr{C}$ is equivalent

to a ***full subcategory***

of the functor category $[\mathscr{C}^{op}, Set]$,

namely the full subcategory

composed of the representable functors

$$h_a(-) \doteq Hom_{\mathscr{C}}(-, a) : \mathscr{C}^{op} \to Set.$$

Hence, the category $\mathscr{C}$

and the subcategory of $[\mathscr{C}^{op}, Set]$ composed of the representable functors

are categorically the same.

Since the Yoneda embedding is full and faithful,

it reflects isomorphisms,

which means that

$$(h_a(-) \simeq h_b(-)) \Rightarrow (a \simeq b)$$

— i.e. the Yoneda embedding is structurally injective —.

Since every functor preserves isomorphisms, we have

$$(h_a(-) \simeq h_b(-)) \Leftarrow (a \simeq b)$$

All in all,

$$(h_a(-) \simeq h_b(-)) \Leftrightarrow (a \simeq b).$$

## Yoneda Embedding and Structuralism

This corollary of the Yoneda lemma means that

the *relations* do determine the *relata*

in the precise sense that any object $a$ in $\mathscr{C}$

can be replaced by the functor of points $h_a(-)$

that encodes all the (incoming or outgoing) relations

that $a$ entertains with the other objects in $\mathscr{C}$.

« […] Two objects are the same

if an only if they look the same from all viewpoints »,

T. Leinster, *Basic Category Theory*, p.104.

An object $a$ in a category $\mathscr{C}$ has a double existence,

as an object in $\mathscr{C}$

and as a functor of points $h_a(-)$ in $[\mathscr{C}^{op}, Set]$,

which encodes how $a$ looks like

when it is observed from all the other objects in $\mathscr{C}$.

By contrast, a non-representable object $F(-)$ in $[\mathscr{C}^{op}, Set]$

is a ***generalized object***

that *uniquely* « exists » as a « functor of points »,

i.e. as a « variable set » of representations

defined with respect to the objects in $\mathscr{C}$.

## The Yoneda Embedding as an Ontological Extension

The Yoneda embedding can be understood

as an ***ontological extension*** of the category $\mathscr{C}$

in the sense that enriches $\mathscr{C}$ with generalized objects

— non-representable functor in $[\mathscr{C}^{op}, Set]$ —-

that only have a functorial existence (i.e. that do not exist as objects in $\mathscr{C}$).

« […] the category $[\mathscr{C}^{op}, Set]$ is like an extension of $\mathscr{C}$ by 'ideal elements' […] »

S. Awodey, *Category Theory*, p.167.

The new « ideal elements »

given by the non-representable functors

can be obtained by « gluing »

— i.e. by computing a colimit of —

representable functors.

**Recipe for Concocting a Generalized Object**

The category $\mathscr{C}$ contains the basic building blocks

of the « lego » and the relations between them.

The functor $F(-) : \mathscr{C}^{op} \to Set$

defines the set of copies $F(a)$ of each building block $a$ in $\mathscr{C}$

that will compose the generalized object.

The action of $F(-)$ on the morphisms in $\mathscr{C}$

specify how the relations between the building blocks in $\mathscr{C}$

are functorially transported to functions between the sets $F(a)$,

thereby encoding the « gluing conditions »

between the different copies of the building blocks.

## The Simplex Category

Let's consider the so called ***simplex category*** $\Delta$

whose objects are thought of

as elementary geometric archetypes of dimension $n$

(called ***n-simplices***)

encoded by finite totally ordered sets like

an archetypal point $[0] = \{0\}$ (0-simplex),

an archetypal directed edge $[1] = \{0 < 1\}$ (1-simplex),

an archetypal oriented triangle $[2] = \{0 < 1 < 2\}$ (2-simplex),

an archetypal tetrahedra $[3] = \{0 < 1 < 2 < 3\}$ (3-simplex), etc.

The morphisms in the ***simplex category*** $\Delta$ are given by

non-decreasing functions

between the corresponding sets and encode the incidence relations between $n$-simplices.

It can be shown that any such morphism can be written

as a finite composition of two kinds of maps (subject to certain *simplicial relations*.):

• **Face maps**: $\delta_i^n : [n-1] \to [n]$ which are injections whose image leaves out $0 \leqslant i \leqslant n$.

For instance the three maps $\delta_0^2, \delta_1^2, \delta_2^2 : [1] \to [2]$ encode the three possible injections of the archetypal oriented edge [1] in the archetypal oriented triangle [2].

• **Degeneracy maps**: $\sigma_i^n : [n] \to [n-1]$ which are surjections $\sigma_i(i) = \sigma(i+1) = i$ that yield a $(n-1)$-simplex by identifying two vertices of a $n$-simplex.

For instance the three maps $\sigma_0^2, \sigma_1^2, \sigma_2^2 : [2] \to [1]$ encode the three manners in which identifying two vertices in an oriented triangle yields an oriented edge.

The « ideal objects » obtained by Yoneda embedding $\Delta$

into the functor category $[\Delta^{op}, Set]$

are « generalized spaces »

$$F(-) : \Delta^{op} \to Set$$

constructed by « gluing »

copies of each archetypal building block $[n]$ in $\Delta$

— the set $F([n])$ is the set of copies

of the archetypal $n$-simplex $[n]$—.

## Example of a Generalized Object

We shall describe a generalized object given by two edges $(e_1, e_2)$

with vertices $(v_0, v_1)$ and $(v_1, v_2)$ respectively,

which means that the two edges are linked at the vertex $v_1$.

This generalized object is given by a non-representable presheaf

$$F(-) : \Delta^{op} \to Set$$

whose action on objects is given by

$F([0]) = \{v_0, v_1, v_2\}$ (three copies of the archetypal vertex [0])

$F([1]) = \{e_0, e_1\}$ (two copies of the archetypal edge [1])

$F([n]) = \emptyset$ for all $n \geq 2$ (no copies of higher archetypal simplices).

Since the generalized object

only has vertices and edges

we have to consider uniquely

copies of the 0-simplex [0] (the archetypal vertex)

and copies of the 1-simplex [1] (the archetypal edge).

The maps in $\Delta$ (the non-decreasing functions between $n$-simplices) are given by

- the identities

$$id_{[0]} : [0] \to [0] \text{ and } id_{[1]} : [1] \to [1],$$

- two face maps describing the two injections of the archetypal vertex in the archetypal edge:

$$\delta_1^1 : [0] \to [1] \text{ given by } 0 \mapsto 0,$$

$$\delta_0^1 : [0] \to [1] \text{ given by } 0 \mapsto 1,$$

- a unique degeneracy map describing the coalescence of the two vertices of the archetypal edge into the archetypal vertex:

$$\sigma_0^1 : [1] \to [0].$$

The functor $F$ converts these maps into the following maps:

$$F(\delta_1^1) : F([1]) \to F([0])$$

given by $F(\delta_1^1)(e_0) = v_0$ and $F(\delta_1^1)(e_1) = v_1$

— since $\delta_1^1$ injects $[0]$ in the initial vertex of $[1]$, $F(\delta_1^1)$

sends each copy of the archetypal edge into its initial vertex —

$F(\delta_0^1) : F([1]) \to F([0])$ given by $F(\delta_0^1)(e_0) = v_1$ and $F(\delta_0^1)(e_1) = v_2$

— since $\delta_0^1$ injects $[0]$ in the final vertex of $[1]$, $F(\delta_0^1)$

sends each copy of the archetypal edge into its final vertex —.

Note that $F(\delta_1^1)(e_1) = v_1$ and $F(\delta_0^1)(e_0) = v_1$ encode the fact

that the initial vertex of $e_1$ is « identified » or « glued » with the final vertex of $e_0$.

The generalized object at stake

$$F(-) : \Delta^{op} \to Set$$

is completely determined by probing it

from the archetypal building blocks

[0] and [1] in $\Delta$.

By the Yoneda lemma,

each copy $x \in F([n])$ of the archetypal $n$-simplex $[n]$

defines a natural transformation

$$\eta^x : h_{[n]}(-) \to F(-)$$

that encodes how that copy $x$ of $[n]$

is « inserted » in the generalized object $F$.

In other terms, the natural transformation $\eta^x$

describes the copy $x$ of $[n]$ in $F$

as a particular $[n]$-probe of $F$.

As the demonstration of the Yoneda lemma explicitly shows,

the whole natural transformation

$$\eta^x : h_{[n]}(-) \to F(-)$$

is fully determined out of the condition

$$id_{[n]} \in h_{[n]}([n]) \mapsto \eta^x_{[n]}(id_{[n]}) = x \in F([n]).$$

By specifying where $[n]$ itself

— i.e. its trivial self-identity —

lands within $F$,

the action of $\eta^x$ on all the generalized points of $[n]$

is automatically determined.

The generalized object $F$ that we have constructed

does not exist as an archetypal object

in the simplex category $\Delta$,

but it can be perfectly defined by producing

sets of copies of the archetypal objects in $\Delta$

and specifying the incidence relations

between all these copies.

# Thanks for your kind attention !!!